Fpt University

TRƯỜNG ĐẠI HỌC FPT

Software Requirement Specification

Smart Cafe Management System

Project Code: SCMS

Document Code: version 01

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I. Record of Changes

Date	A* M, D	In charge	Change Description
25/09	A	VuNL	Context diagram
25/09	A	TamDHM	Usecase list, product overview
25/09	A	KhanhPG	Screen flow, screen description, Non-function requirment
25/09	A	AnhTV	Usecase list, Entity Class Diagram
25/9	A	VanNT	Usecase diagram, product overview
28/9	M	KhanhPG, VuNL, AnhVT	Context diagram, description
30/9	M	TamDHM, VanNT	Usecase diagram, usecase description
1/10	M	VuNL	Entity Class diagram
1/10	M	AnhVT	Context diagram
2/10	M	VanNT, TamDHM	Usecase list
2/10	M	KhanhPG	Screen flow
2/10	A	KhanhPG, AnhVT, TamHDM	Screen mockup
3/10	M	VanNT, VuNL	Screen description

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

1.1 Business Context

The Smart Cafe Management System (SCMS) targets expanding mid-sized cafe and F&B chains. This segment has outgrown the single owner-operated store model but is not yet large enough to build its own proprietary management system like Highlands Coffee or The Coffee House. A reference business in this segment operates around 6 to 10 branches. Over a 2-year period, the enterprise might open one additional branch, or possibly none, with daily operating hours from 06:00 to 23:00.

Each branch employs about 12 to 15 staff members divided into 2 to 3 shifts, consisting of Branch Managers, Cashiers, and Baristas. The COO centrally synchronizes menus, pricing, and promotions, and tracks financial data across the entire chain. Each branch serves approximately 300 to 500 orders daily, generating a monthly revenue of around 400 to 600 million VND. Cash flows include physical cash in the registers at each branch and digital transfers via QR code directly into the chain's central bank account.

1.2 Current Problems

Each branch operates almost independently, while the COO requires centralized control. When using disjointed POS systems combined with manual ledgers and Excel, the business faces three groups of problems:

First, inconsistent configuration. Every time prices change, new items are introduced, or promotions are launched, the COO must manually update each branch, leading to omissions and discrepancies between stores. Revenue and Cost of Goods Sold (COGS) reports must also be consolidated from multiple sources, causing reporting delays.

Second, losses at the counter and in inventory. Cashiers verify bank transfers by opening banking apps or checking transaction screenshots provided by customers, which is slow during peak hours and carries the risk of forged images. The cash drawer lacks a strict shift opening and closing process, and there are no approval thresholds for order cancellations and refunds. Inventory is manually recorded and not linked to the standard recipe/ingredients of each item, making shrinkage difficult to detect.

Third, high costs and dependency. Third-party POS software often charges per store, meaning costs scale up with the number of branches, and it is difficult to customize the software to specific proprietary workflows. Manual paper-based timesheets are also prone to errors in calculating working hours.

1.3 Solution and Value Proposition

SCMS is an independent web-based application that runs on browsers using the store's existing hardware. The system integrates multiple modules: POS order taking, printing preparation tickets to the bar, time attendance, cash drawer management, bulk-packaging inventory management,

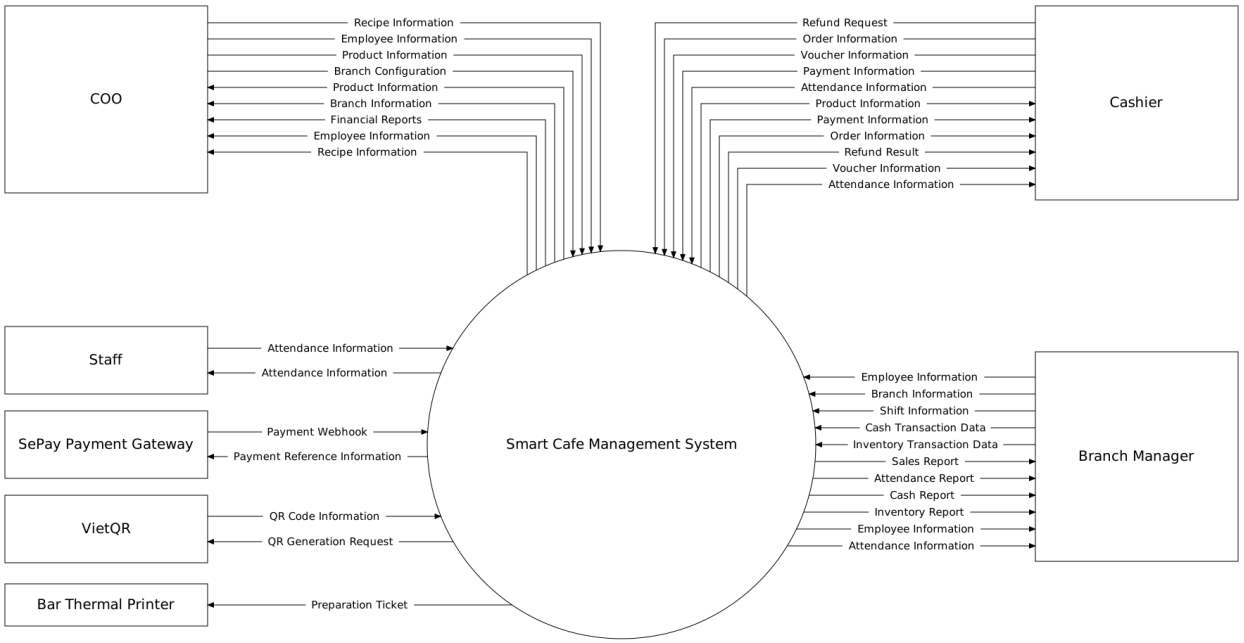
and automated bank transfer reconciliation via SePay. The COO configures settings once and synchronizes them across all branches with a single action, views consolidated financial reports, and controls ingredient shrinkage. Branch Managers schedule shifts, reconcile cash, manage inventory, and view branch-level reports. Cashiers take orders quickly and accept auto-verified QR payments. Staff clock in/out using RFID cards or PINs in less than 0.5 seconds. A standard order is completed in under 5 clicks, the VietQR code is generated in a maximum of 1 second, the order status shifts to "paid" within 1 second after the Webhook triggers, and the system guarantees a minimum availability/uptime of 99.5% during operating hours.

1.4 Key Differentiators

SCMS stands out in four key areas. First, auto-reconciled QR payments: the system matches the order code, amount, and authentication signature from SePay, then automatically updates the order to "paid"; orders with amount discrepancies are held with warnings. Second, cash drawer control by shift lifecycle: shift opening, cash-in/cash-out with required reasons, shift closing via actual physical counts, and mandatory explanations for any discrepancies. Third, centralized management with strict role-based access control (RBAC): Branch Managers only see data for their own branch, managers cannot approve their own actions, paid orders are frozen, and all sensitive actions are recorded in an audit log. Fourth, low cost and ownership autonomy: the system runs on commodity hardware, ESC/POS standard K80 printers, and driverless USB card readers; the enterprise owns the system instead of paying per-store subscription fees.

Version 1.0 focuses on core operations. Out of scope: multi-channel/FoodApp integrations, Kitchen Display Systems (KDS), CRM/loyalty programs, electronic invoicing and S1a/S2a accounting, WebOrder, splitting/merging tables, card/e-wallet payments, offline mode, social insurance (BHXH) and personal income tax (PIT) calculations, and biometrics.

1.5 System Scope and Actors



2. User Requirements

2.1 Actors

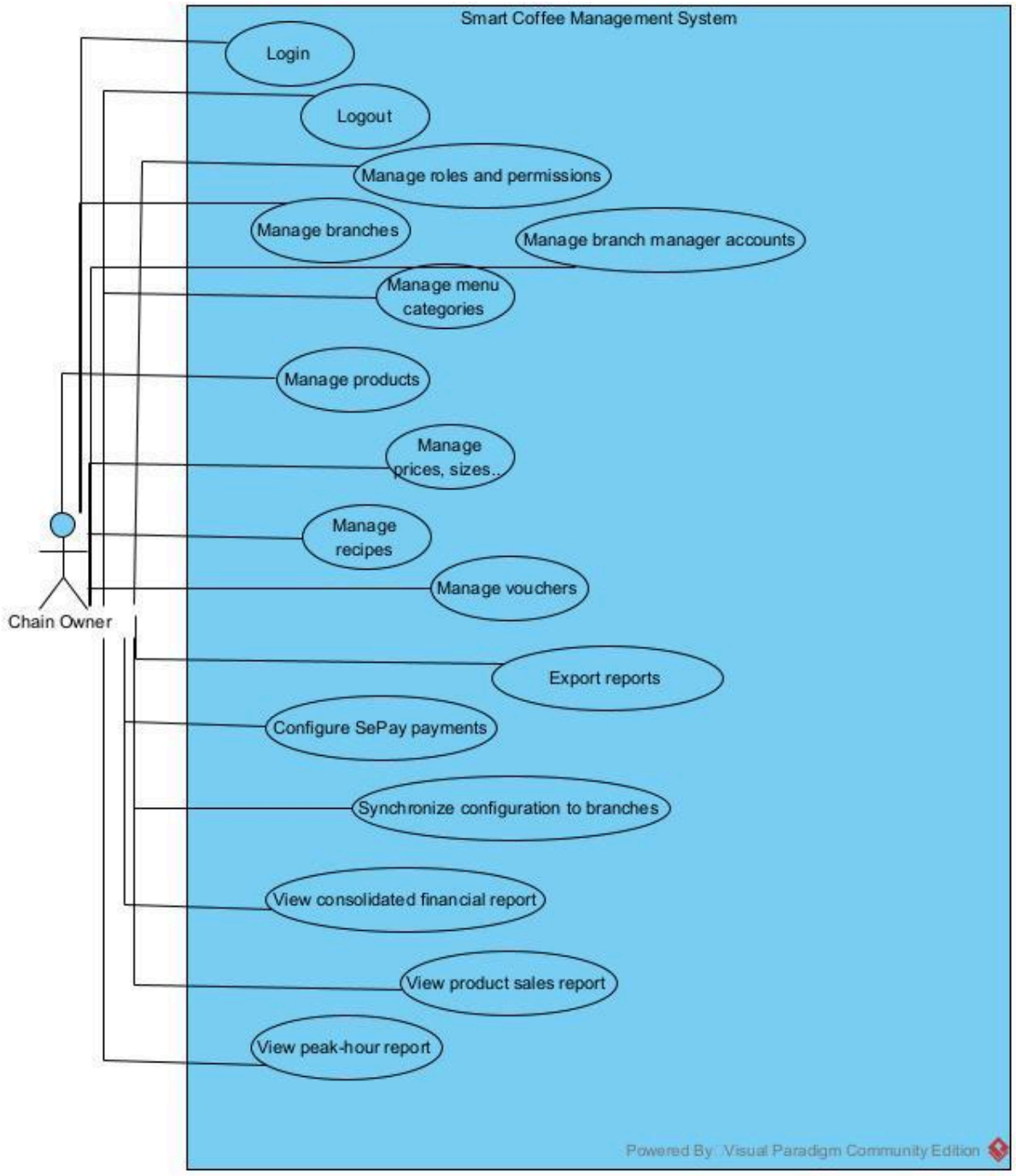
#	Actor	Description
1	COO	The highest-level user with chain-wide scope. Focuses on configuration and monitoring; does not perform sales operations or handle cash. Manages branch and Branch Manager accounts. Centrally manages the master menu, pricing, promotions, standard recipes/portions, and ingredient catalogs, then synchronizes them across all branches with a single action. Configures the SePay receiving bank account and approval thresholds. Approves actions directly related to Branch Managers (timesheets, bonuses/deductions, payroll) and refund requests exceeding standard thresholds. Views consolidated financial reports, monitors chain-wide performance, controls ingredient shrinkage, views the system-wide audit log, and handles data recovery.
2	Branch Manager	The direct manager of a specific branch, restricted to accessing data only for their assigned branch. Organizes shift schedules, reviews shift opening/closing reports, and reconciles cash at the start and end of shifts. Confirms inventory receipts. Manages HR records for Cashiers and Staff, and prepares branch payroll. Views branch-level reports. Strictly prohibited from self-approving their own operational requests.
3	Cashier	Operates the Point of Sale (POS) at the counter of the assigned branch, with access to the orders of the current business day. Creates orders, customizes item options (size, sugar, ice, toppings), collects cash or starts QR payments, and prints preparation slips and customer receipts. Initiates requests for order cancellations (for unpaid orders) and refunds. Performs physical cash drawer counts to finalize shift closures. Records attendance at the Kiosk like every other employee.
4	Staff	Staff represents general store personnel (baristas, servers). Staff have no SCMS account and cannot log in. They only record attendance at the Attendance Kiosk (SCR-02) by tapping an RFID card or by selecting their name and entering a personal 4-digit PIN. Cashiers and Branch Managers record attendance the same way, in addition to their login accounts.
5	Payment Gateway - SePay	An external intermediary payment gateway service that operates automatically via a Webhook mechanism. When a customer transfers funds using a Dynamic VietQR code, SePay detects the bank balance fluctuation and pushes an HTTP POST payload (containing the order code and transaction amount) to the SCMS, allowing the software to automatically update the order status to paid.
6	VietQR	A third-party system providing an API/tool to generate payment QR codes compliant with the NAPAS national standard. The SCMS sends invoice details (transaction

		amount, transfer memo, and bank account number) to VietQR, which returns a dynamic QR code image for the customer to scan and pay at the counter.

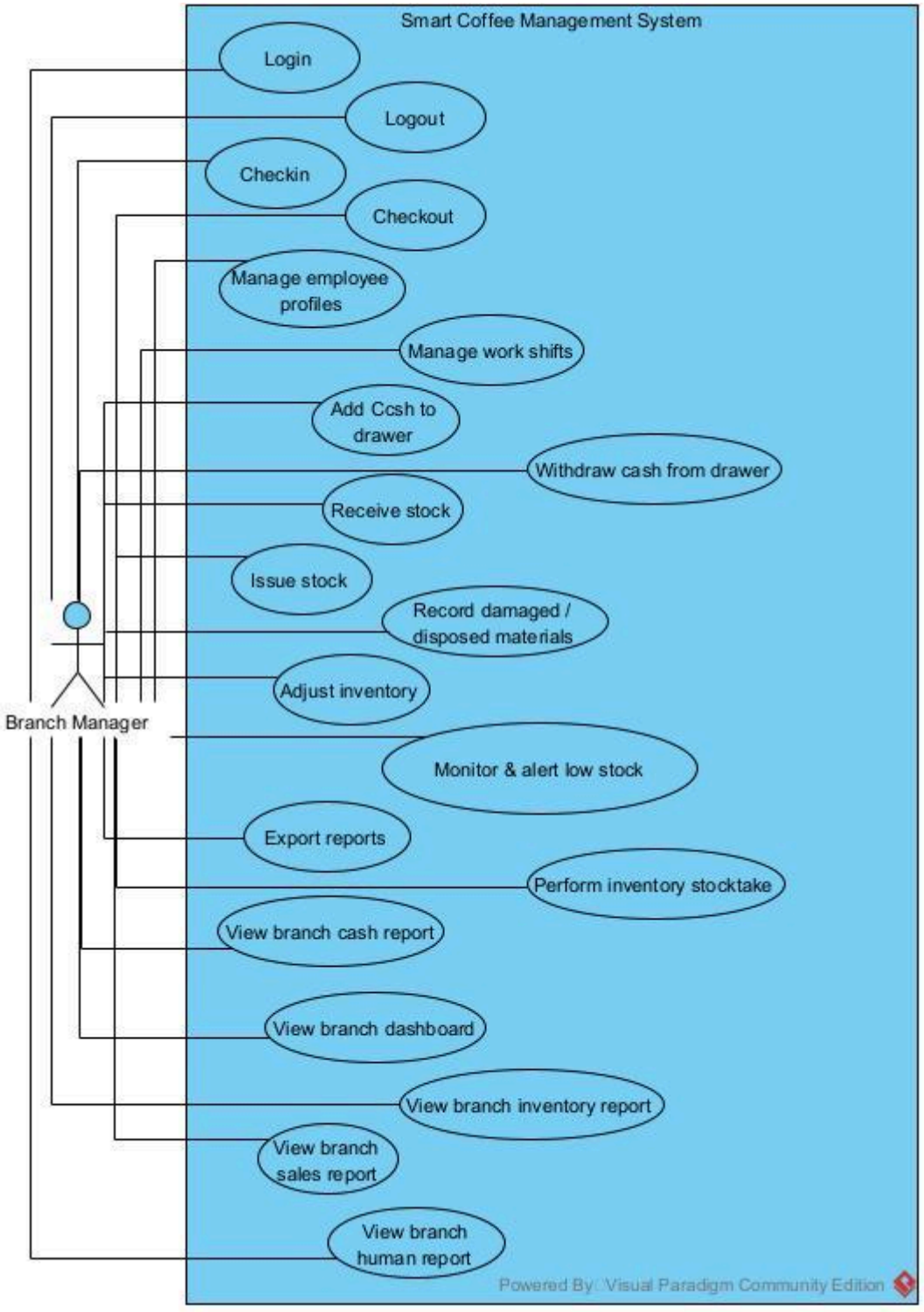
2.2 Use Cases

2.2.1 Diagrams

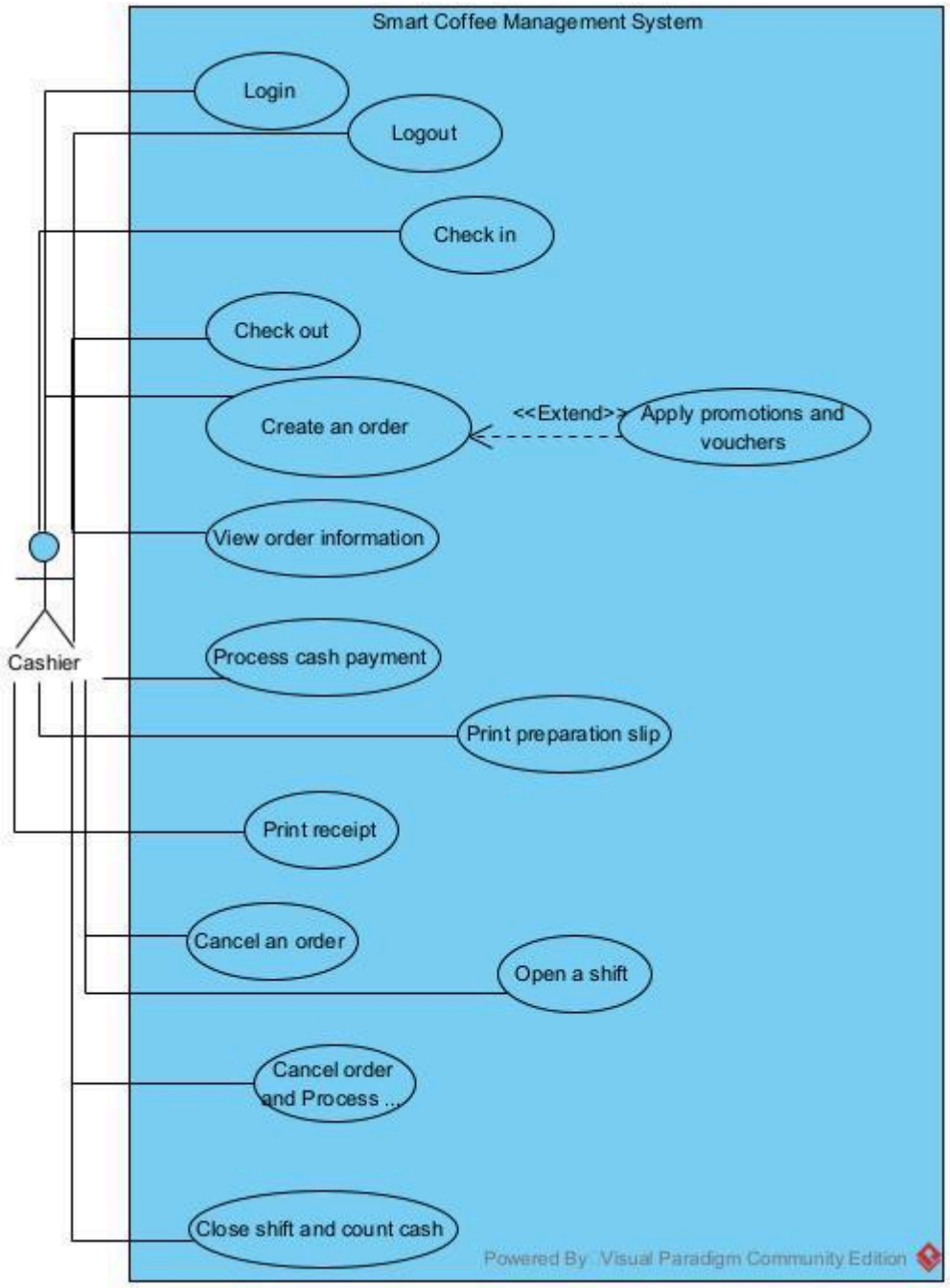
a. Chain owner



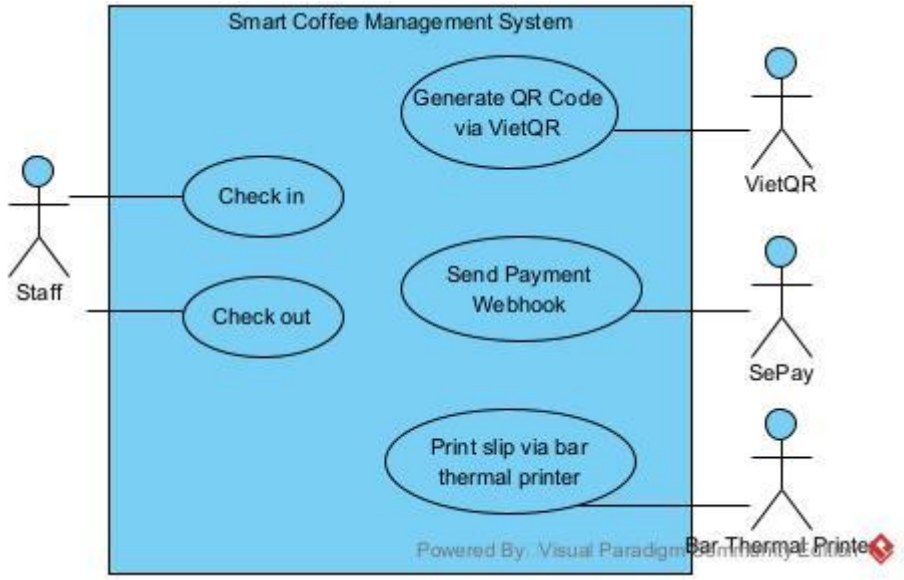
b. Branch Manager



c. Cashier


Smart Coffee Management System
Login
Logout
Check in
Check out
Create an order
<<Extend>>
Apply promotions and vouchers
View order information
Cashier
Process cash payment
Print preparation slip
Print receipt
Cancel an order
Open a shift
Cancel order and Process ...
Close shift and count cash
Powered By Visual Paradigm Community Edition

d. Staff, Sepay, VietQR, Printer



2.2.2 Descriptions

ID	Group Function	Use Case	Actors	Use Case Description
UC01	Authentication & Access Control	Login / Logout	COO, Branch Manager	The user enters login credentials. The system validates the credentials, identifies the user's role, determines the data access scope, and initiates a session.
UC02	Manage Role	Manage Roles	COO	The COO reviews the list of roles and assigns them to individual employees. The system validates these role-based assignments and then saves the access permissions for each employee.
UC03	Attendance Management	Check In	Staff, Cashier	The user enters a PIN to authenticate. The system verifies the working shift status, records the check-in time, and returns a successful result.
UC04	Attendance Management	Check Out	Staff, Cashier	The user requests to end the shift. The system verifies that the shift is active, records the check-out time, and returns a successful result.
UC05	Branch Management	Manage Branches	COO	The COO creates, views, edits, or deactivates branch information. The system receives and saves the changes to the database.
UC06	Branch Manager Account Management	Manage Branch Manager Accounts	COO	The COO creates or updates an account, assigns the account to a specific branch, and configures its permissions. The system stores the account configuration.
UC07	Human Resource Management	Manage Employee Profiles	Branch Manager	The Branch Manager views the employee list and creates, updates, or deactivates employee profiles. The system updates the corresponding information.
UC08	Shift Management	Manage	Branch	The Branch Manager creates shifts, sets working time

		Work Shifts	Manager	frames, and assigns employees. The system checks for schedule conflicts, issues warnings if any, and saves the schedule.
UC09	Menu Management	Manage Menu Categories	COO	The COO creates, modifies, or deactivates menus and assigns products to each menu. The system saves the standard menu configuration.
UC10	Product Management	Manage Products	COO	The COO creates new products, updates product details, or deactivates products. The system saves the information in the product catalog.
UC11	Pricing Management	Manage Prices, Sizes & Toppings	COO	The COO sets or updates selling prices for each product based on size and toppings. The system validates the information and saves the pricing configuration.
UC12	Recipe Management	Manage Recipes	COO	The COO selects a product and defines the required quantities of raw materials for its preparation.
UC13	Promotion Management	Manage Vouchers	COO	The COO creates or updates vouchers and configures usage conditions such as validity period, order value, and applicable products. The system validates and saves the voucher information.
UC14	Chain Management	Synchronize Configuration to Branches	COO	The COO selects configuration data and target branches. The system synchronizes the selected data to the branches and records the synchronization results.
UC15	Payment Management	Configure SePay Payment	COO	The COO configures the receiving account, Payment Reference, and Webhook. The system validates the information and saves the connection configuration.
UC16	POS Order Management	Create New Order	Cashier	The Cashier selects products, enters quantities, chooses sizes/toppings, customizes sugar/ice levels, and adds notes. The system consolidates the information and creates the order.
UC17	Promotion & Voucher Application	Apply Promotion / Voucher	Cashier	The Cashier enters or selects a promotion code. The system verifies the applicable conditions, calculates the discount, and updates the total payment amount.
UC18	Order Management	View Order Information	Cashier	The Cashier selects an order. The system displays the order number, item list, quantities, discount amount, total amount, and current status.
UC19	Cash Payment	Process Cash Payment	Cashier	The Cashier selects the Cash payment method and enters the amount received from the customer. The system calculates the change, the Cashier confirms the payment, and the system updates the order status.
UC20	QR Payment	Confirm QR	SePay	The system receives a Webhook from SePay and automatically reconciles the Payment Reference and

		Payment		payment amount. If the information matches, the system confirms the payment and updates the order status.
UC21	Order Fulfillment	Print Preparation Slip	Cashier, Bar Thermal Printer	Immediately after the order is paid, the system automatically sends the preparation slip to the printer at the bar. The printer prints the slip and returns the printing result.
UC22	Receipt Printing	Print Payment Receipt	Cashier, Bar Thermal Printer	When the transaction is successful, the system automatically generates a receipt and sends a print command to the thermal printer. The printer prints the receipt and returns the printing status.
UC23	Special Order Processing	Process Refund for PAID Order	Cashier	The Cashier selects the order to be cancelled and enters a reason. The system verifies the order status and permissions, processes the cancellation, confirms the refund, and deducts the refunded cash amount from the cash drawer.
UC24	Cash Drawer Management	Open Shift	Cashier	The Cashier selects a working shift and declares the opening cash amount. The system validates the information and saves the shift opening configuration.
UC25	Cash Drawer Management	Add Cash to Drawer	Branch Manager	The Branch Manager enters the cash amount and reason. The system records the cash-in transaction and automatically adds the amount to the cash drawer balance.
UC26	Cash Drawer Management	Withdraw Cash from Drawer	Branch Manager	The Branch Manager enters the cash-out amount and reason. The system records the cash-out transaction and automatically deducts the amount from the cash drawer balance.
UC27	Cash Drawer Management	Close Shift & Count Cash	Cashier	The Cashier requests to close the shift and enters the actual counted cash amount. The system calculates the expected cash amount, reconciles the variance, records the closing report, and locks the shift.
UC28	Branch Inventory Management	Receive Stock	Branch Manager	The Branch Manager creates a goods receipt, selects raw materials, and enters quantities and costs. The system validates the information and increases the corresponding inventory quantities.
UC29	Branch Inventory Management	Issue Stock	Branch Manager	The Branch Manager creates an issue document, selects raw materials, and enters the quantities to be issued. The system checks the current inventory balance and deducts the issued quantities.
UC30	Branch Inventory Management	Record Damaged / Disposed Materials	Branch Manager	The Branch Manager declares the quantity of damaged or disposed raw materials and provides a reason. The system validates the information and deducts the corresponding quantity from inventory.

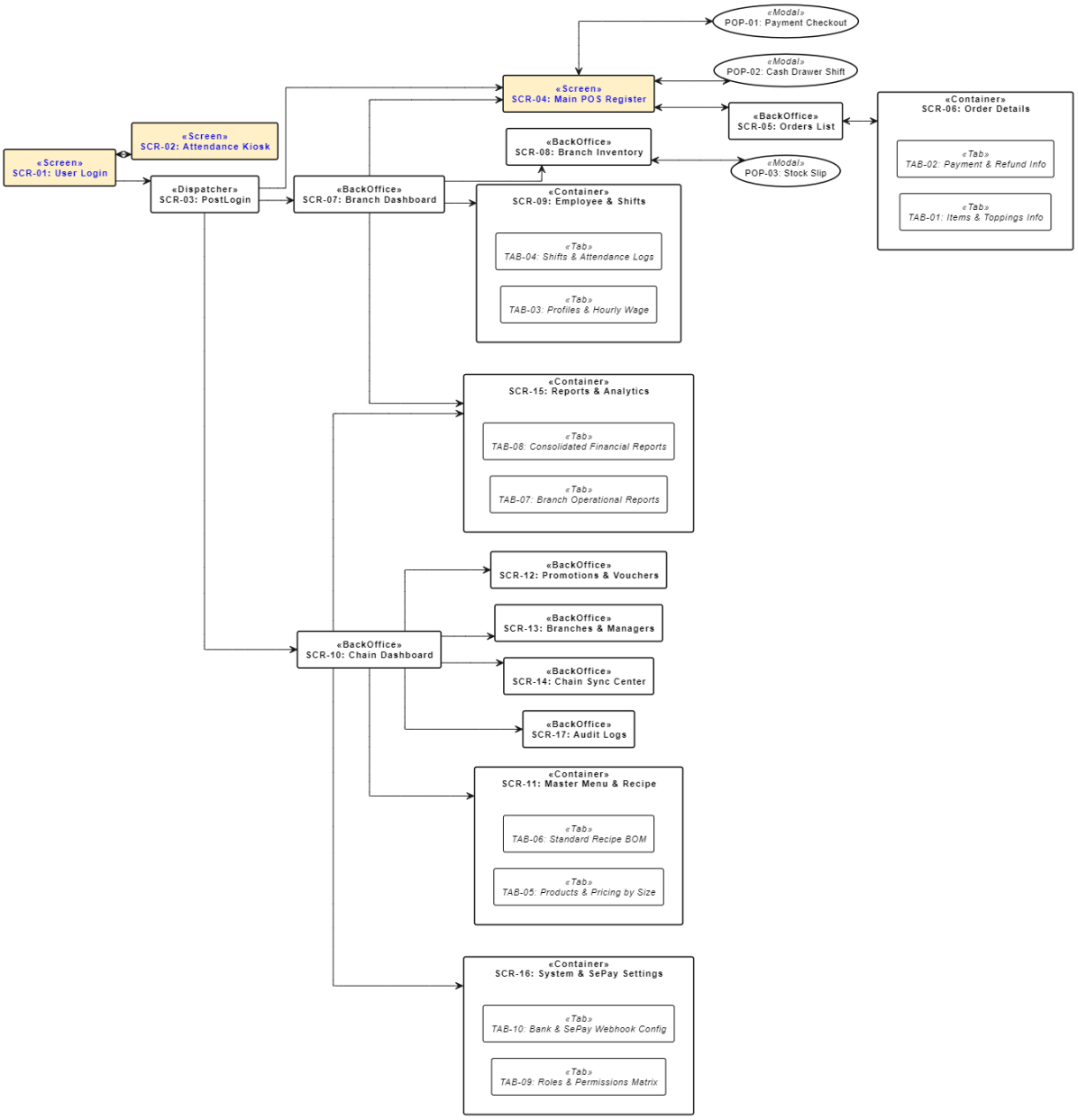
UC31	Branch Inventory Management	Perform Inventory Stocktake	Branch Manager	The Branch Manager creates an inventory stocktake session and enters the actual counted quantities. The system compares the physical count with the system records and records the variance.
UC32	Branch Inventory Management	Adjust Inventory	Branch Manager	Based on the stocktake variance, the Branch Manager enters an adjustment transaction with a reason. The system validates the information and updates the actual inventory balance.
UC33	Branch Inventory Management	Monitor & Alert Low Stock	Branch Manager	The system continuously compares inventory levels with the minimum stock threshold. If the quantity falls below the threshold, the system automatically displays an alert to the Branch Manager.
UC34	Reporting & Statistics	View Branch Dashboard	Branch Manager	The Branch Manager accesses the dashboard. The system aggregates and visually displays key indicators related to revenue, cash flow, inventory, and employee performance.
UC35	Reporting & Statistics	View Branch Sales Report	Branch Manager	The Branch Manager selects a time period. The system retrieves, aggregates, and displays the sales report for the selected branch.
UC36	Reporting & Statistics	Export Reports	Branch Manager, COO	The Branch Manager or COO exports reports as Excel files.
UC37	Reporting & Statistics	View Branch Cash Flow Report	Branch Manager	The Branch Manager requests the report. The system aggregates the opening cash, cash-in/cash-out transactions, revenue, actual cash balance, and displays the cash flow variance.
UC38	Reporting & Statistics	View Branch Inventory Report	Branch Manager	The Branch Manager requests the inventory report. The system aggregates receiving, issuing, disposal, and stocktake data and displays the inventory status and variance.
UC39	Reporting & Statistics	View Branch Human Resources Report	Branch Manager	The Branch Manager requests the report. The system aggregates and displays attendance, work shift, and employee productivity data.
UC40	Reporting & Statistics	View Consolidated Financial Report	COO	The COO requests the consolidated report. The system retrieves data from all branches and generates a consolidated report of revenue, cost of goods sold, and gross profit for the entire chain.
UC41	Reporting & Statistics	View Product	COO	The COO requests the report. The system analyzes sales quantities to identify and display the best-selling

		Sales Report		products.
UC42	Reporting & Statistics	View Peak Hours Report	COO	The COO requests the report. The system analyzes order volumes by time period, identifies peak sales hours, and displays the results.
UC43	External System Integration	Generate QR Code via VietQR	VietQR	The software system sends a request to generate a QR code for an order. VietQR generates the QR code and returns the QR code image together with the payment information.
UC44	External System Integration	Send Payment Webhook via SePay	SePay	SePay records a successful customer bank transfer and sends a Webhook containing the transaction reference, payment amount, and payment status to the software system.

3. Software Features

3.1 Functional Overview

3.1.1 Screens Flow



3.1.2 Screen Descriptions

#	Feature	Screen	Description
SCR-01	Authentication & Timekeeping	User Login	System login screen via Username and Password. Supports quick toggle to the Attendance Kiosk screen (SCR-02). Automatically locks account temporarily after 5 consecutive failed attempts to prevent brute-force attacks. After successful authentication, routes through the System Router (SCR-03).
SCR-02		Attendance Kiosk	Dedicated timekeeping screen placed at counter/store for shift staff. Supports scanning personal QR code or entering Employee ID/PIN to perform shift Check-in / Check-out. Records actual time, cross-checks with assigned shift roster, and saves log to system. Provides return button to login screen (SCR-01).
SCR-03	System Navigation	PostLogin	Intermediate component handling post-login logic (navigation/routing splash screen). Checks account Role, current work shift, and permissions to automatically route workflow: Cashier/Staff straight to POS (SCR-04), Branch Manager to Branch Dashboard (SCR-07), COO to Chain Dashboard (SCR-10).
SCR-04	Point of Sale (POS)	Main POS Register	Primary cashier interface optimized for touchscreens: displays item categories in grid/image card layout, category filters, quick search; current shopping cart area (select size, sugar level, ice level, toppings with extra fee); applies discount codes; displays auto-calculated total including tax/promotions. Allows holding bills temporarily, opening payment checkout popup (POP-01), and opening cash drawer shift management popup (POP-02).
POP-01		Payment Checkout (Modal)	Bill payment popup window: selects payment method (Cash, VietQR/SePay Bank Transfer). Integrates dynamic VietQR code generation containing order code and exact amount, automatically listens to SePay Webhook via WebSocket to confirm "Payment Successful" in real time without requiring manual cashier verification. Supports calculating change for cash payments and auto-printing receipts.
POP-02		Cash Drawer Shift (Modal)	Cash drawer opening and closing workflow popup window: declares opening cash balance (initial float/change), records mid-shift emergency receipts/disbursements (with reason), and conducts end-of-shift cash reconciliation (theoretical total cash revenue vs. actual cash in drawer). Allows printing shift handover reports.

SCR-05	Order Management	Orders List	Branch order list screen. Supports advanced filters by time, status (Pending Payment, Paid, Partially Refunded, Refunded, Cancelled), assigned cashier, and payment method. Displays daily revenue summary and allows clicking to view detailed order information (SCR-06).
SCR-06		Order Details	Detailed view of a specific order, structured as a Container containing 2 functional tabs (TAB-01 and TAB-02). Provides administrative actions: Reprint Receipt, Reprint Prep Label, Cancel Order (requires Branch Manager role or higher), and Refund.
TAB-01		Items & Toppings Info (Tab)	Sub-tab of SCR-06: Displays detailed list of items in the order, cup size, custom measurements (sugar, ice), attached topping list,unit price, quantity, and customer brewing notes.
TAB-02		Payment & Refund Info (Tab)	Sub-tab of SCR-06: Displays payment transaction history: SePay transaction ID, account number, bank, actual payment time, Webhook reconciliation log. Refund processing area: allows partial or full refund, records reason, and updates inventory/bill status.
SCR-07	Branch Operations	Branch Dashboard	Central dashboard for Branch Manager: monitors real-time revenue, hourly order count chart, low-stock inventory status with red alert thresholds, daily staff shift fulfillment rate. Features quick links to POS (SCR-04), Inventory Management (SCR-08), Shift Staff (SCR-09), and Reports (SCR-15).
SCR-08		Branch Inventory	Branch inventory management screen: displays actual on-hand quantity of each raw material (coffee beans, milk, syrup, cups, straws, etc.), minimum safety thresholds, stock shortage alert status. Supports search, category filter, and triggers stock voucher creation via popup (POP-03).
POP-03		Stock Slip (Modal)	Popup window to create and process stock vouchers: Inbound Stock Voucher (from central warehouse or external supplier), Disposal Voucher (due to damage or expiration), Stock Count Voucher (enters actual quantity to balance book discrepancies). Requires Branch Manager confirmation signature.

SCR-09		Employee & Shifts	Branch-level staff and shift management screen, structured as a Container with 2 functional tabs (TAB-03 and TAB-04). Manages branch employee directory, schedules weekly shift rosters, and tracks attendance.
TAB-03		Profiles & Hourly Wage (Tab)	Sub-tab of SCR-09: Manages branch employee profiles (full name, phone number, role, employment status), configures hourly/shift pay rates, attendance PIN codes, and employee activity logs.
TAB-04		Shifts & Attendance Logs (Tab)	Sub-tab of SCR-09: Weekly/monthly shift scheduling (Morning Shift, Afternoon Shift, Evening Shift); detailed attendance log table retrieved from Kiosk (SCR-02): actual check-in/check-out time, minutes late/early departure, cumulative working hours for payroll calculation. Allows Branch Manager to approve attendance time adjustments during exceptions.
SCR-10	Chain Management	Chain Dashboard	Comprehensive dashboard for COO: consolidates daily/monthly revenue across all 6–10 branches, ranks branch sales performance, top-selling items system-wide, abnormal warnings regarding order cancellation costs or inventory discrepancies. Navigates to all centralized administrative modules.
SCR-11		Master Menu & Recipe	System-wide master menu and recipe management screen, structured as a Container containing 2 tabs (TAB-05 and TAB-06). Allows creating item categories, pricing by size, and managing Bill of Materials (BOM) formulas applied synchronously across branches.
TAB-05		Products & Pricing by Size (Tab)	Sub-tab of SCR-11: Configures item details (name, image, category, description), sales pricing policy by size (S, M, L), allowed topping options, and topping unit prices. Supports enabling/disabling items system-wide or allowing individual branches to temporarily pause items when out of stock.
TAB-06		Standard Recipe BOM (Tab)	Sub-tab of SCR-11: Defines raw material consumption standards (Bill of Materials - BOM) per item per size. The system uses this data to calculate theoretical COGS and theoretical ingredient consumption. It does not deduct branch stock per order (BR-41)
SCR-12		Promotions & Vouchers	System-wide promotional campaign management screen: creates discount voucher codes (percentage or fixed amount), usage conditions (minimum order value, happy hour frames, designated branch applications), usage limits, and validity period. Tracks usage count in real time.

SCR-13		Branches & Managers	Branch list management screen: adds new branches, updates branch profile (name, address, hotline, opening hours), assigns responsible Branch Manager account, sets operational status (Open, Temporarily Closed, Maintenance).
SCR-14		Chain Sync Center	Master data publishing hub: publishes a numbered version of menu and prices, standard recipes and promotions to selected branches, shows the publish result per branch and the history of publish versions. It applies new configurations centrally and they are available instantly.
SCR-15	Reports & Analytics	Reports & Analytics	Financial & operational data analysis screen, structured as a Container consisting of 2 in-depth report tabs (TAB-07 and TAB-08). Allows exporting reports to Excel, PDF, CSV formats for accounting purposes.
TAB-07		Branch Operational Reports (Tab)	Sub-tab of SCR-15: Detailed branch-level operational reports: daily/weekly revenue, payment method breakdowns (Cash vs. VietQR SePay), order refund/cancellation rates, inventory movement & material loss reports, and shift staff labor costs.
TAB-08		Consolidated Financial Reports (Tab)	Sub-tab of SCR-15: System-wide consolidated financial reports: P&L (Profit & Loss) comparisons across branches, item profit margins, total cash flow through bank/SePay payment gateways, quarterly/annual revenue growth trend analysis.
SCR-16	Configuration & Security	System & SePay Settings	System & payment gateway configuration screen, structured as a Container with 2 tabs (TAB-09 and TAB-10). Dedicated to COO for configuring security parameters, bank settings, and access control permissions.
TAB-09		Roles & Permissions Matrix (Tab)	Sub-tab of SCR-16: Granular permission matrix for 4 roles in the system (Staff, Cashier, Branch Manager, COO). Allows toggling view, add, edit, delete, approve order, cancel order, refund, and access permissions for each screen/feature.
TAB-10		Bank & SePay Webhook Config (Tab)	Sub-tab of SCR-16: Configures beneficiary bank accounts, transfer syntax templates for dynamic VietQR generation, SePay API Key, Webhook URL, and security Secret Key. Provides connectivity test tools (Test Ping Webhook) and monitors response packet logs from SePay.

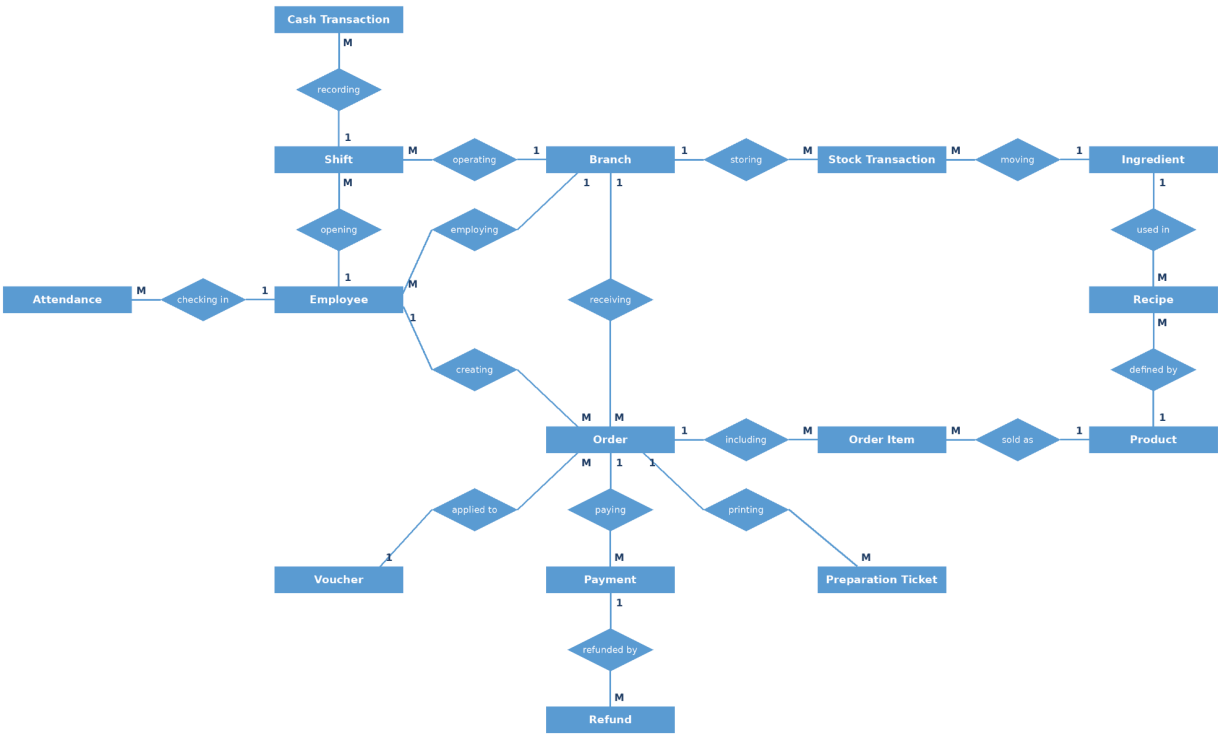
SCR-17	Audit Logs	Audit Logs	System audit log screen: records details of all critical and sensitive operations (Login, Item Price Edit, BOM Standard Edit, Invoice Cancellation, Refund, Manual Inventory Adjustment, Permission Assignment). Displays operator, timestamp, IP address, branch, and data payload details before/after changes (Diff payload).

3.1.3 Screen Authorization

#	Screen	Staff	Cashier	Branch Manager	COO
SCR-01	User Login	x	x	x	x
SCR-02	Attendance Kiosk	x	x	x	
SCR-03	Role Dispatcher	x	x	x	x
SCR-04	Main POS Register		x	x	
POP-01	Item Options		x	x	
POP-02	Payment Checkout		x	x	
POP-03	Cash Drawer Shift		x	x	
SCR-05	Orders List		x	x	x
SCR-06	Order Details	x	x	x	x
TAB-01	Items & Toppings Info	x	x	x	x
TAB-02	Payment & Refund Info		x	x	x
SCR-07	Branch Dashboard			x	x
SCR-08	Branch Inventory			x	x
POP-04	Stock Slip			x	x
SCR-09	Employee & Shifts			x	x
TAB-03	Profiles & Hourly Wage			x	x
TAB-04	Shifts & Attendance Logs	x	x	x	x
SCR-10	Chain Dashboard				x
SCR-11	Master Menu & Recipe			x	x
TAB-05	Products & Pricing by Size			x	x

TAB-06	Standard Recipe BOM			x	x
SCR-12	Promotions & Vouchers			x	x
SCR-13	Branches & Managers				x
SCR-14	Chain Sync Center			x	x
SCR-15	Reports & Analytics			x	x
TAB-07	Branch Operational Reports			x	x
TAB-08	Consolidated Financial Reports				x
SCR-16	System & SePay Settings				x
TAB-09	Roles & Permissions Matrix				x
TAB-10	Bank & SePay Webhook Config				x
SCR-17	Audit Logs				x

3.1.4 Entity Relationship Diagram



Entities Description

#	Entity	Description
1	Branch	A cafe branch within the chain. Stores branch information and operating settings (opening hours, tables, printers, etc.).

2	Employee	An employee of the chain, including personal information, the assigned branch, and the assigned role.
3	Attendance	An employee's attendance record (check-in and check-out time) for a work shift.
4	Shift	A work shift at a branch, opened and closed by an employee. It is the unit for reconciling cash and revenue.
5	Cash Transaction	A cash movement during a shift (cash in, cash out, opening and closing the cash drawer), used for cash reports.
6	Product	A product sold at the cafe (beverage or food), including name, price, and selling status.
7	Recipe	The recipe of a product, defining which ingredients and what quantities are used for each product.
8	Ingredient	A raw material used to prepare products, including its unit of measure and current stock level.
9	Stock Transaction	A stock-in, stock-out, or adjustment record for ingredients, used to track inventory and generate inventory reports.
10	Order	A customer order created by a cashier, including total amount, status, and creation time.
11	Order Item	A single product line within an order, including quantity and the unit price at the time of sale.
12	Voucher	A discount or promotion code applied to an order, including its conditions, value, and validity period.
13	Payment	The payment of an order (cash or bank transfer), storing the method, amount, status, and the QR code information generated by VietQR.
14	Refund	A refund request for a payment, including the reason, amount, and processing result.
15	Preparation Ticket	A ticket printed on the bar thermal printer for each item that needs to be prepared.

3.1.5 Entity Details

a. Branch

#	Attribute name	PK	Type	Mandatory	Description
1	Branch Number	x	String	Yes	Unique identifier of the branch.
2	Name		String	Yes	Name of the branch.
3	Address		String	Yes	Physical address of the branch.
4	Phone		String	No	Contact phone number of the branch.
5	Opening Time		Time	Yes	Opening time of the branch.
6	Closing Time		Time	Yes	Closing time of the branch.
7	Table Count		Number	No	Total number of tables at the branch.
8	Status		String	Yes	Operating status of the branch.

9	Created At		DateTime	Yes	Timestamp when the branch was created.

b. Employee

#	Attribute name	PK	Type	Mandatory	Description
1	Employee Number	x	String	Yes	Unique identifier of the employee.
2	Full Name		String	Yes	Full name of the employee.
3	Phone		String	Yes	Contact phone number.
4	Username		String	No	System login username (applicable for employees with system accounts).
5	PIN		String	No	4-digit PIN code used for kiosk time tracking/check-in.
6	Card Code		String	No	Employee RFID card code used for time tracking.
7	Base Salary		Number	No	Base salary.
8	Status		String	Yes	Employment status of the employee.
9	Created At		DateTime	Yes	Timestamp when the employee profile was created.
10	Role		String	Yes	Role of User in System
11	Hourly Rate		Number	No	Hourly wage rate based on the role policy at the employee's branch.

c. Attendance

#	Attribute name	PK	Type	Mandatory	Description
1	Attendance Number	x	String	Yes	Unique identifier of the attendance record.
2	Work Date		Date	Yes	Working date.
3	Check In Time		DateTime	Yes	Timestamp when the employee checks in to start the shift.
4	Check Out Time		DateTime	No	Check-out timestamp when ending the shift (empty if not yet checked out).
5	Method		String	Yes	Way the employee checked in (PIN code or employee card).
6	Status		String	Yes	Attendance results compared with the assigned working hours (e.g., on time, late, early departure, missing check-out).

d. Shift

#	Attribute name	PK	Type	Mandatory	Description
1	Shift Number	x	String	Yes	Unique identifier of the work shift.
2	Opened At		DateTime	Yes	Timestamp when the shift was opened.
3	Closed At		DateTime	No	Timestamp when the shift was closed.
4	Opening Cash		Number	Yes	Initial cash amount in the cash drawer when opening the shift.
5	Counted Cash		Number	No	Actual physical cash amount counted when closing the shift.
6	Difference Reason		String	No	Explanation/reason when there is a cash drawer discrepancy.
7	Status		String	Yes	Shift status (open, closed).

e. Cash Transaction

#	Attribute name	PK	Type	Mandatory	Description
1	Cash Transaction Number	x	String	Yes	Unique identifier of the cash transaction.
2	Type		String	Yes	Transaction type (cash-in, cash-out, sales collection, refund).
3	Amount		Number	Yes	Transaction cash amount.
4	Reason		String	No	Reason for transaction (mandatory for cash-in and cash-out).
5	Created At		DateTime	Yes	Timestamp when the cash transaction occurred.

f. Product

#	Attribute name	PK	Type	Mandatory	Description
1	Product Number	x	String	Yes	Unique identifier of the product.
2	Name		String	Yes	Product name.
3	Description		String	No	Product description.
4	Price		Number	Yes	Selling price of the product.
5	Status		String	Yes	Business status of the product.

g. Recipe

#	Attribute name	PK	Type	Mandatory	Description
1	Recipe Number	x	String	Yes	Unique identifier of the recipe line item.
2	Quantity		Number	Yes	Standard quantity of ingredient required for one unit of product.
3	Unit		String	Yes	Unit of measurement for the ingredient quantity.

h. Ingredient

#	Attribute name	PK	Type	Mandatory	Description
1	Ingredient Number	x	String	Yes	Unique identifier of the ingredient.
2	Name		String	Yes	Ingredient name.
3	Unit		String	Yes	Dispensing unit of measurement used in preparation/recipes.
4	Pack Unit		String	No	Bulk packaging unit used for inventory receiving.
5	Conversion Ratio		Number	No	Number of base units contained in one bulk packaging unit.
6	Min Stock Level		Number	No	Minimum safe stock threshold for low-stock alerts.
7	Unit Cost		Number	No	Cost price per base unit.

i. Stock Transaction

#	Attribute name	PK	Type	Mandatory	Description
1	Stock Transaction Number	x	String	Yes	Unique identifier of the stock transaction.
2	Type		String	Yes	Transaction type (import, export, disposal, stocktake).
3	Quantity		Number	Yes	Quantity of the ingredient in the ingredient's preparation unit, always a positive value. For import, export and disposal it is the quantity moved; for stocktake it is the counted quantity.

4	Unit Cost		Number	No	Cost price per unit at the time of import. Applicable to import transactions only.
5	Reason		String	No	Reason for disposal or stock difference, required for disposal and stocktake differences.
6	Created At		DateTime	Yes	Timestamp when the stock transaction occurred.

j. Order

#	Attribute name	PK	Type	Mandatory	Description
1	Order Number	x	String	Yes	Unique identifier of the order.
2	Status		String	Yes	Order status (Pending Payment, Paid, Partially Refunded, Refunded, Cancelled)
3	Note		String	No	Order notes or customer special requests.
4	Created At		DateTime	Yes	Timestamp when the order was created.
5	Paid At		DateTime	No	Timestamp when the order was successfully paid.
6	Total Amount		Number	Yes	Order total of order
7	Discount Amount		Number	Yes	Discount amount of order
8	Final Amount		Number	Yes	Accounts receivable of order

k. Order Item

#	Attribute name	PK	Type	Mandatory	Description
1	Order Item Number	x	String	Yes	Unique identifier of the order line item.
2	Quantity		Number	Yes	Quantity ordered.
3	Unit Price		Number	Yes	Selling price of the product, defined by the COO and applied to all branches. It covers the size and toppings options of the product.
4	Sugar Level		String	No	Selected sugar level preference.
5	Ice Level		String	No	Selected ice level preference.
6	Size		String	No	Cup size selected by the customer (S, M, L)
7	Toppings		String	No	List the additional toppings selected by the customer (e.g., tapioca pearls, coconut jelly).

8	Note		String	No	Specific preparation note for this item.

l. Voucher

#	Attribute name	PK	Type	Mandatory	Description
1	Voucher Number	x	String	Yes	Unique identifier of the voucher.
2	Code		String	Yes	Voucher promotional code entered by customer or cashier.
3	Discount Type		String	Yes	Discount type (percentage or fixed amount).
4	Discount Value		Number	Yes	Discount value.
5	Min Order Amount		Number	No	Minimum order total required to apply the voucher.
6	Start Date		Date	Yes	Effective start date of the voucher.
7	End Date		Date	Yes	Expiration date of the voucher.
8	Usage Limit		Number	No	Maximum allowed number of total usages.
9	Status		String	Yes	Voucher status (e.g., active, expired, disabled).

m. Payment

#	Attribute name	PK	Type	Mandatory	Description
1	Payment Number	x	String	Yes	Unique identifier of the payment record.
2	Method		String	Yes	Payment method (cash or QR transfer).
3	Amount		Number	Yes	Amount that has to be paid by this payment.
4	Status		String	Yes	Payment status (pending, completed, failed, amount mismatch).
5	QR Content		String	No	VietQR raw payload string (applicable for QR payment).
6	QR Expired At		DateTime	No	Expiration timestamp of the generated QR code.
7	Reference Code		String	No	Reconciliation reference code provided by SePay.
8	Received Amount		Number	No	Amount reported as received by the payment gateway (QR payments only).

9	Created At		DateTime	Yes	Timestamp when the payment record was created.
10	Paid At		DateTime	No	Timestamp when the payment was confirmed as received.

n. Refund

#	Attribute name	PK	Type	Mandatory	Description
1	Refund Number	x	String	Yes	Unique identifier of the refund request.
2	Amount		Number	Yes	Amount to be refunded to the customer. It cannot exceed the amount paid.
3	Reason		String	Yes	Reason for requesting refund.
4	Status		String	Yes	Refund status (pending approval, approved, rejected, completed).
5	Requested At		DateTime	Yes	Timestamp when the refund was requested.
6	Processed At		DateTime	No	Timestamp when the refund was approved or rejected.

o. Preparation Ticket

#	Attribute name	PK	Type	Mandatory	Description
1	Ticket Number	x	String	Yes	Unique identifier of the kitchen/bar preparation ticket.
2	Print Status		String	Yes	Ticket printing status (printed, error).
3	Printed At		DateTime	No	Timestamp when the ticket was successfully printed.

3.2 Authentication, Navigation & Attendance

3.2.1 User Login

3.2.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Role Selector	Dropdown	Yes	-	Select the intended working role/module (Cashier, Branch Manager, COO, Staff). Default: Cashier.
2	User Name	Text	Yes	30	Enter the username or employee ID (e.g., cashier01, manager_d1).
3	Password	Password	Yes	32	Enter the password. Displays hidden characters (*) instead of actual text. Minimum 6 characters.
4	Remember Me	Checkbox	No	-	Option to save the active working session on the device's browser. Default: Checked.
5	Forgot Password	Link	-	-	Opens instructions to contact the Branch Manager or COO for a password reset.
6	Login	Button	-	-	Submits the authentication request. Upon successful login, redirects to SCR-03: Role Dispatcher.

7	Switch to Kiosk	Link / Button	-	-	Quick switch to the counter attendance screen (SCR-02: Attendance Kiosk).

3.2.1.2 Use Case Description

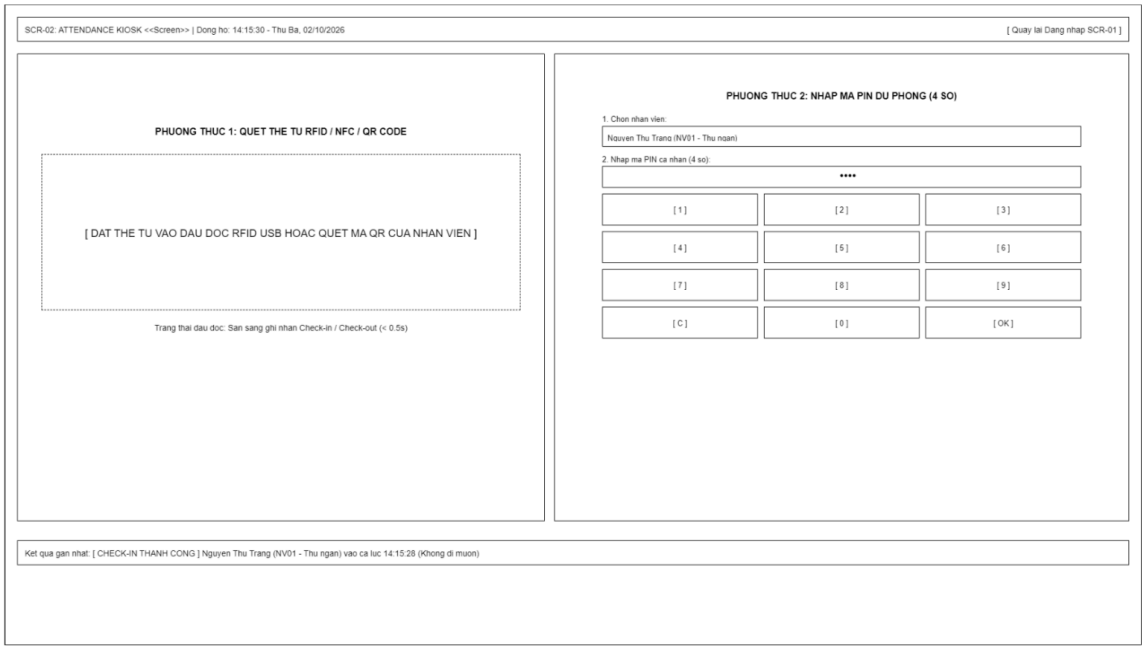
Use Case ID	**UC01**	**Use Case Name**		**Login/Logout**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO, Branch Manager, Cashier, Staff				
Description	The user enters login credentials (username/phone number and password) into the SCMS. The system validates the credentials, identifies the user's role and data access scope, creates an active session, and redirects the user to the appropriate landing dashboard.				
Precondition	PRE-01: The user has an active account provisioned in the SCMS database. PRE-02: The system is connected to the authentication service and database.				
Trigger	TRG-01: The user accesses the SCMS login page and submits their credentials.				
Post-Condition	POS-01: An active login session is created with a valid authentication token. POS-02: The user is logged into the system with permissions and data access corresponding to their assigned role.				
Main flows					
Step	***Actor***	***Action***			
1	User	Navigate to the SCMS login screen (SCR-01) and enter login credentials (username/phone number and password).			
2	User	Click the "Login" button.			
3	SCMS	Validate the input fields to ensure required data is provided.			
4	SCMS	Authenticate credentials against the database and check the user account status (e.g., Active).			
5	SCMS	Identify the user's role (COO, Branch Manager, Cashier, Staff) and determine the corresponding data access scope and branch context.			
6	SCMS	Create an active session, generate an access token, and redirect the user to their designated default screen/dashboard.			

Alternative flows

AT1	Invalid credentials or inactive account (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect invalid username/password or find that the account is deactivated/locked.
2.2	SCMS	Display an error message notifying the user of incorrect credentials or inactive account status.
2.3	SCMS	Keep the user on the login screen and allow re-entry of credentials.

3.2.2 Attendance Kiosk

3.2.2.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	System Clock	Label	-	-	Displays a real-time clock (Hours:Minutes:Seconds - Day of the week, Date/Month/Year) synchronized with the server to reconcile clock-in/out times.
2	Back to Login	Button	-	-	Click to return to the user login screen (SCR-01: User Login).

3	RFID / QR Scan Area	Component	-	-	Receives the card code from the USB RFID reader. A valid card immediately records check-in or check-out (BR-06).
4	Reader Status	Label	-	50	Displays peripheral device connection status (e.g., "Ready for check-in/out" or "Card reader disconnected").
5	Employee Selector	Dropdown	Yes	-	List to select shift-assigned employees (Full Name + Employee ID) used as a backup PIN method when a card is forgotten.
6	PIN Display	Password	Yes	4	Shows masked digits. Exactly 4 digits.
7	Numpad (0 - 9)	Button Grid	-	-	Grid of 10 large touch number keys from 0 to 9 for staff to operate directly on the counter screen.
8	Clear (C)	Button	-	-	Key to clear all recently entered PIN digits in the display field.
9	Submit (OK)	Button	-	-	Submits the PIN code for time tracking authentication reconciled against assigned shifts.
10	Result Banner	Label / Box	-	150	Notification box displaying the latest time tracking result (Full Name, Employee ID, Check-in / Check-out status, actual timestamp, and notes on punctuality/tardiness).

#	Field Name	Type	Mandatory	Max Length	Description
1	Role Selector	Dropdown	Yes	-	Select the intended working role/module (Cashier, Branch Manager, COO, Staff). Default: Cashier.
2	User Name	Text	Yes	30	Enter the username or employee ID (e.g., cashier01, manager_d1).
3	Password	Password	Yes	32	Enter the password. Displays hidden characters (*) instead of actual text. Minimum 6 characters.
4	Remember Me	Checkbox	No	-	Option to save the active working session on the device's browser. Default: Checked.

5	Forgot Password	Link	-	-	Opens instructions to contact the Branch Manager or COO for a password reset.
6	Login	Button	-	-	Submits the authentication request. Upon successful login, redirects to SCR-03: Role Dispatcher.
7	Switch to Kiosk	Link / Button	-	-	Quick switch to the counter attendance screen (SCR-02: Attendance Kiosk).

3.2.2.2 Use Case Description

Use Case ID	**UC03**	**Use Case Name**		**Checkin**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Staff, Cashier, Branch Manager				
Description	The user enters a personal PIN on the branch terminal to authenticate their identity. The system checks the working shift status, records the check-in timestamp, and updates attendance logs.				
Precondition	PRE-01: The user is assigned to the current branch and scheduled for a shift. PRE-02: The POS/Attendance terminal is on the "Check-In" screen (SCR-03).				
Trigger	TRG-01: The user initiates a check-in request at the beginning of their shift.				
Post-Condition	POS-01: A check-in record containing the exact timestamp, user ID, and shift ID is recorded. POS-02: The user's status for the scheduled shift is marked as "Checked-In".				
Main flows					
Step	***Actor***	***Action***			
1	User	Select their name or enter their Employee ID on the attendance terminal.			
2	User	Enter their security PIN and press the "Check In" button.			
3	SCMS	Validate the PIN against user security credentials.			
4	SCMS	Verify current time against scheduled shifts for the employee to associate the check-in with the correct shift.			

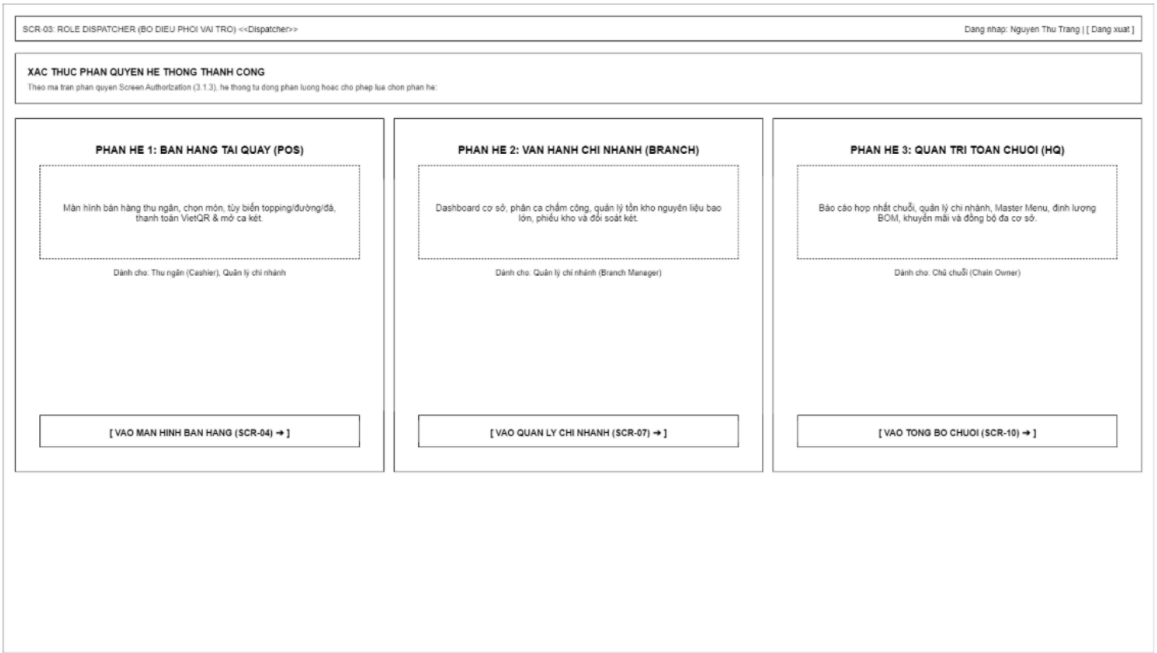
5	SCMS	Record the check-in time, calculate status (e.g., On Time, Late), and update attendance records.
6	SCMS	Display a success message confirming the check-in time.

Alternative flows

AT1	Incorrect PIN entered (At step 3 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that the entered PIN does not match the employee record.
1.2	SCMS	Display an error message "Incorrect PIN. Please try again."
1.3	SCMS	Prompt the user to re-enter their PIN.
AT2	No scheduled shift found (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that no active or upcoming schedule exists for the employee on the current date.
2.2	SCMS	Log an unscheduled attendance record or alert the user.
2.3	SCMS	Display a warning message "No scheduled shift found for today" and request manager override or log as an unscheduled check-in.

3.2.3 Role Dispatcher

3.2.3.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Current User Info	Label	-	80	Displays the name and assigned role of the currently logged-in employee/user (e.g., "Logged in: Nguyen Thu Trang (Cashier - Branch 01)").
2	Logout	Button / Link	-	-	Logs out the current session, clears session/token, and redirects back to the Login screen (SCR-01).
3	Authorization Notice	Label	-	200	Displays a system permission status message and provides instructions for the user to select the appropriate module corresponding to their role.
4	POS Section Card	Container	-	-	Displays information block for Point of Sale (POS) module: item selection, custom sugar/ice/topping adjustments, cash collection, Dynamic VietQR, and cash drawer shift management.
5	Enter POS	Button	-	-	Click to navigate to the main cash register screen (SCR-04: Main POS Register). Visible/active only for Cashier and Branch Manager roles.
6	Branch Section Card	Container	-	-	Displays information block for Branch Management module: monitoring branch revenue, timekeeping, managing bulk raw material inventory, and issuing stock vouchers.
7	Enter Branch Portal	Button	-	-	Click to navigate to the branch dashboard (SCR-07: Branch Dashboard). Active only for Branch Manager and COO roles.
8	HQ Section Card	Container	-	-	Displays information block for Chain HQ Management module: consolidated financial reporting across 6–10 branches, Master Menu setup, BOM formulas, promotions, and data synchronization.
9	Enter HQ Portal	Button	-	-	Click to navigate to the headquarters central dashboard (SCR-10: Chain Dashboard). Visible/active exclusively for the COO role.

3.2.3.1 Use case Description

Use Case ID	**UC02**	**Use Case Name**		**Manage Roles & Permissions**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The COO views the list of system roles and configures detailed functional permissions (e.g., read, create, update, delete) for each role. The system validates and saves the updated access control configuration across the system.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Role & Permission Management" screen (SCR-02).				
Trigger	TRG-01: The COO requests to view, create, or update permissions associated with specific system roles.				
Post-Condition	POS-01: Permission matrices for the specified roles are updated in the system database. POS-02: Access control rules take effect immediately for all user sessions bound to the modified roles.				

Main flows		
Step	***Actor***	***Action***
1	COO	Review the list of existing roles and select a specific role to manage permissions.
2	SCMS	Display the permission matrix associated with the selected role across all system modules.
3	COO	Modify functional permission checkboxes (e.g., Grant/Revoke access to specific features/modules).
4	COO	Click the "Save Permissions" button.
5	SCMS	Validate the permission changes against system security constraints.
6	SCMS	Save the updated permission configuration to the database and display a success notification.

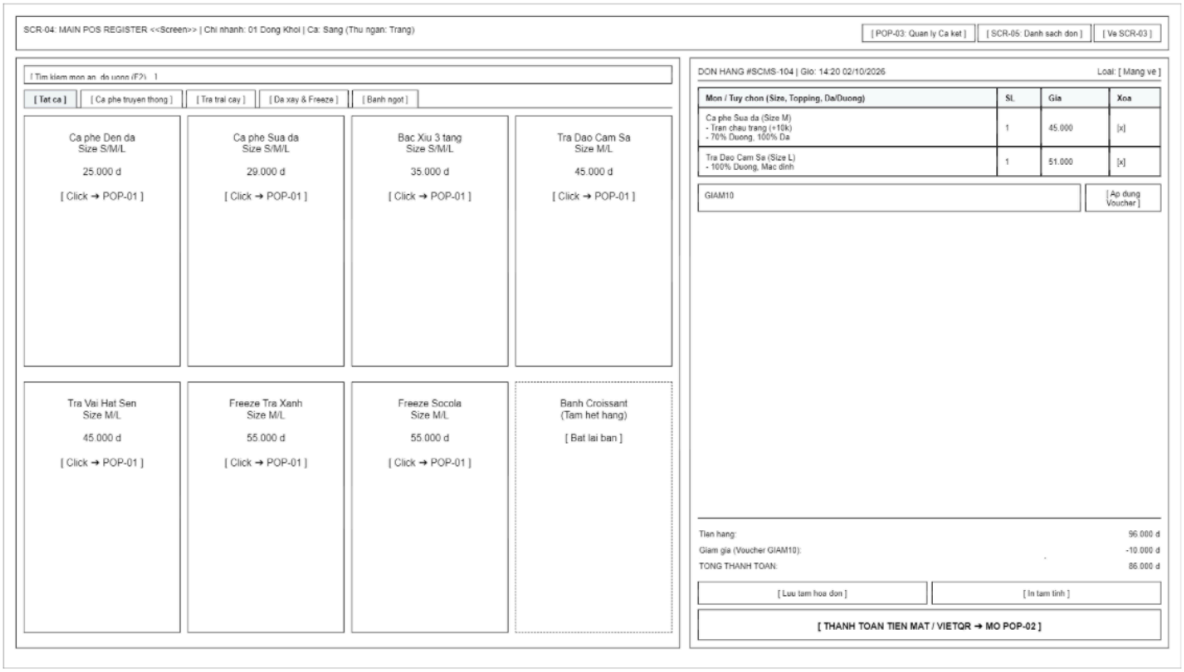
Alternative flows

AT1	Permission configuration conflict or validation failure (At step 5 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect an invalid permission combination (e.g., revoking dependent system admin capabilities).
2.2	SCMS	Display an error message explaining the permission policy restriction.
2.3	SCMS	Revert unapplied permission changes and allow the COO to adjust options.

3.3 Point of Sale (POS) & Checkout

3.3.1 Main POS Register

3.3.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Shift & Branch Info	Label	-	60	Displays current branch name, work shift, and cashier.
2	Cash Drawer Shift	Button	-	-	Opens shift and cash drawer management popup (POP-03).
3	Orders List Nav	Button	-	-	Navigates to the order list screen (SCR-05).

4	Back to Dispatcher	Button	-	-	Returns to the role allocation / module selection screen (SCR-03).
5	Product Search	Text	No	50	Quick search items by name or item code.
6	Category Tabs	Tabs	-	-	Filters items by category: Coffee, Tea, Blended, Pastries.
7	Product Item Card	Component	-	-	Item display card (name, price). Click to open the item customization popup (POP-01).
8	Order Type	Dropdown	Yes	-	Select service type: Takeaway or Dine-in.
9	Cart Table	Table	-	-	Selected item list (Item Name, Size, Topping, Qty, Price).
10	Remove Item	Button	-	-	Removes an item line from the cart.
11	Voucher Code	Text	No	20	Inputs promo code/voucher.
12	Apply Voucher	Button	-	-	Validates and applies a discount to the order.
13	Subtotal Amount	Label	-	15	Displays subtotal (total amount before discount).
14	Discount Amount	Label	-	15	Displays discount amount.
15	Total Amount	Label	-	15	Displays grand total (final amount payable by customer).
16	Hold Order	Button	-	-	Holds current bill temporarily to serve the next customer.
17	Checkout	Button	-	-	Opens payment popup (POP-02).

3.3.1.2 Use Case Description

Use Case ID	**UC04**	**Use Case Name**		**Checkout**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Staff, Cashier, Branch Manager				

Description	The user requests to complete their working shift by authenticating via PIN. The system verifies that an active check-in session exists, records the check-out timestamp, and closes the attendance record for the shift.
Precondition	PRE-01: The user has previously completed a successful check-in (UC03) for the current active shift. PRE-02: The POS/Attendance terminal is on the "Check-Out" screen (SCR-04).
Trigger	TRG-01: The user initiates a check-out request upon ending their work shift.
Post-Condition	POS-01: The check-out timestamp is recorded for the active attendance log. POS-02: Total working hours for the shift are calculated and saved in attendance records.

Main flows

Step	***Actor***	***Action***
1	User	Select their name/Employee ID on the attendance terminal and enter their security PIN.
2	User	Click the "Check Out" button.
3	SCMS	Validate the security PIN against user credentials.
4	SCMS	Verify that an open check-in record exists for the employee.
5	SCMS	Record the check-out timestamp and calculate total working time for the shift.
6	SCMS	Display a confirmation message displaying the check-out time and total shift hours worked.

Alternative flows

AT1	No active check-in record found (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that no open check-in session exists for the user on the current day.
2.2	SCMS	Display an error message "No open check-in record found for this user."
2.3	SCMS	Prompt the user to contact the Branch Manager for manual attendance adjustment.

Use Case ID	UC21	Use Case Name		Print Preparation Slip	
Author	**KhanhPG**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Cashier, Bar Thermal Printer				
Description	The system formats and transmits a drink preparation ticket to the Bar thermal printer immediately after an order transitions to PAID status (BR-31), or upon a manual reprint request by the Cashier from Order Details (SCR-06). The slip contains order number, items, cup sizes, sugar/ice levels, toppings, and barista notes.				
Precondition	PRE-01: An order has successfully transitioned to PAID status (BR-31) via Cash (UC19) or QR (UC20), OR Cashier requests a reprint from Order Details (SCR-06). PRE-02: Bar thermal printer is connected via network/USB and configured in system settings.				
Trigger	TRG-01: Order payment is confirmed and status becomes PAID (BR-31), OR Cashier clicks "Reprint Bar Label" on SCR-06.				
Post-Condition	POS-01: Preparation slip data is formatted and transmitted to the Bar thermal printer. POS-02: Bar thermal printer outputs the preparation ticket / cup labels. POS-03: Print job status is logged in the order audit history.				
Main flows					
Step	***Actor***	***Action***			
1	SCMS	Detect that order status has transitioned to PAID (BR-31) or receive manual reprint command from Cashier.			
2	SCMS	Retrieve order item specifications: product names, sizes, sugar %, ice %, toppings, and barista preparation notes.			
3	SCMS	Format preparation ticket payload with Order Number, Timestamp, Items, and Notes into standard ESC/POS format.			
4	SCMS	Transmit print command to the designated Bar thermal printer over local network/USB.			
5	Bar Thermal Printer	Receive print command, print the preparation ticket / cup stickers, and return completion status.			
6	SCMS	Confirm successful print execution and log print timestamp into order history.			

Alternative flows

AT1	Bar printer offline or out of paper (At step 5 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect printer connection timeout or hardware error signal (paper out/jam).
1.2	SCMS	Display warning toast notification on POS: "Bar printer offline or out of paper. Print job queued."
1.3	SCMS	Save print command to retry queue and allow Cashier to reprint from Order Details (SCR-06).

Use Case ID	**UC22**	**Use Case Name**		**Print Payment Receipt**	
Author	**KhanhPG**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Cashier, Bar Thermal Printer				
Description	The system automatically generates a standard K80 customer receipt and sends a print command to the counter thermal printer once an order is PAID (BR-31), or when Cashier clicks "Reprint Receipt" on Order Details (SCR-06). The receipt displays branch info, order number, items, unit prices, discounts, payment method, and cashier name.				
Precondition	PRE-01: Order has transitioned to PAID status (BR-31), OR Cashier requests a reprint from Order Details (SCR-06). PRE-02: Counter thermal receipt printer is connected and online.				
Trigger	TRG-01: Order transitions to PAID status following cash/QR confirmation, OR Cashier clicks "Reprint Receipt" button on SCR-06.				
Post-Condition	POS-01: Customer payment receipt is printed and handed to the customer. POS-02: Print execution and reprint counter are recorded in system logs.				
Main flows					
Step	***Actor***	***Action***			
1	SCMS	Receive print trigger following payment confirmation or manual reprint request on SCR-06.			
2	SCMS	Aggregate receipt details: Branch Name & Address, Order Number, Date/Time, Cashier, Item List, Subtotal, Discount, Total Due, Payment Method, and Change.			
3	SCMS	Format printable receipt layout adhering to standard K80 80mm thermal roll specifications.			
4	SCMS	Send print command directly to counter thermal receipt printer.			

5	Bar Thermal Printer	Print the paper receipt, perform auto-cut, and return execution status to SCMS.
6	SCMS	Record print completion timestamp in order audit log.

Alternative flows

AT1	Receipt printer paper out or cover open (At step 5 in the main flows)	
Sub step	**Actor**	**Action**
1.1	Bar Thermal Printer	Send error status signal (paper out / cover open) back to SCMS.
1.2	SCMS	Display alert on POS screen: "Receipt printer out of paper. Please reload paper roll."
1.3	Cashier	Reload paper roll and click retry or reprint from Order Details (SCR-06).

3.4.1.2 Use Case Description

Use Case ID	**UC23**	**Use Case Name**		**Process Refund for PAID Order**	
Author	**KhanhPG**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Cashier, Branch Manager				
Description	Allows processing a cash refund for a paid order on Order Details (SCR-06 TAB-02). Cashiers cannot arbitrarily refund orders; refunds require Branch Manager approval and an explicit reason (BR-32). All refunds are paid in cash from the drawer regardless of initial payment method (BR-33), deducted from current shift balance (BR-34), and logged in Audit Log (BR-51).				
Precondition	PRE-01: Cashier or Branch Manager is logged in and viewing a PAID order on Order Details (SCR-06 TAB-02). PRE-02: Order belongs to the current branch and was created during an active sales shift.				
Trigger	TRG-01: Cashier selects refund reason, inputs refund amount, and requests Branch Manager approval (or Branch Manager clicks "Approve Refund").				
Post-Condition	POS-01: Order status remains firmly as PAID (BR-26). A separate Refund record is created and linked to the order. POS-02: Refund amount is deducted in cash from the current shift's cash drawer balance (BR-33, BR-34, BR-39).				

		POS-03: Cancellation reason, manager approval, and refund details are recorded in Audit Log (BR-51).
Main flows		
Step	***Actor***	***Action***
1	Cashier	On Order Details screen (SCR-06), select "Payment & Refund Info" tab (TAB-02) and click "Refund".
2	SCMS	Display refund dialog requesting: Refund Reason (dropdown), Refund Amount, and Branch Manager PIN authorization (BR-32).
3	Cashier	Select refund reason (Customer Changed Mind, Wrong Item, Quality Issue, Other), enter refund amount (maximum: paid amount - already refunded), and request manager approval.
4	Branch Manager	Verify order circumstances, inspect reason, and enter manager credentials/PIN (POP-05) to approve the refund (BR-32).
5	SCMS	Validate manager authorization. Update order status to Partially Refunded or Refunded (original order remains PAID). Create Refund record with status Completed. Deduct refund amount from shift cash drawer balance (BR-33, BR-34), and record in Audit Log (BR-51).
6	Cashier	Refund reason, manager approval, and refund details are recorded in Audit Log (BR-51).

Alternative flows

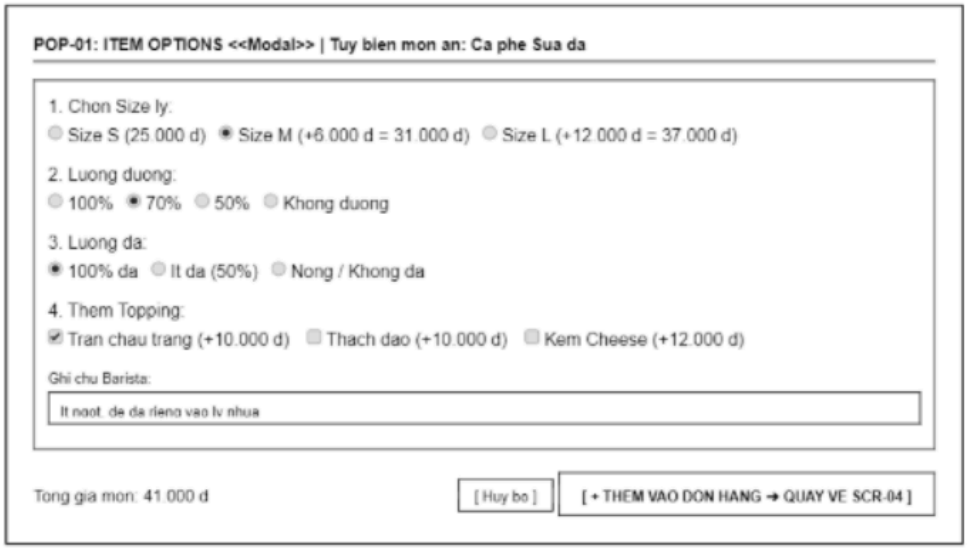
AT1	Refund amount exceeds Branch Manager approval threshold (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that refund amount exceeds Branch Manager approval threshold limit (BR-35).
1.2	SCMS	Require COO approval code or reject operation.
1.3	Branch Manager	Contact COO for override code or reject customer refund request.

AT2	Insufficient cash in drawer for refund (At step 5 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that current cash in drawer is less than required refund amount.
2.2	SCMS	Display alert indicating drawer cash is insufficient and requires Cash In deposit (UC25, BR-37).

2.3	Branch Manager	Perform Cash In transaction (UC25) to supply funds before retrying refund.

3.3.2 Item Options & Apply Promotion

3.3.2.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Modal Header	Label	-	50	Displays popup title and the name of the item being customized.
2	Size Option	Radio	Yes	-	Select cup size: Size S, Size M (+extra fee), Size L (+extra fee). Default: Size M.
3	Sugar Level	Radio	Yes	-	Select sugar level: 100%, 70%, 50%, No Sugar. Default: 70%.
4	Ice Level	Radio	Yes	-	Select ice level: 100% Ice, Less Ice (50%), Hot/No Ice. Default: 100% Ice.
5	Topping List	Checkbox	No	-	Select add-on toppings (White Boba, Peach Jelly, Cheese Foam, etc., with surcharges).

6	Barista Note	Text	No	100	Inputs custom brewing/preparation notes from customer (e.g., "Less sweet, ice on the side").
7	Item Total Price	Label	-	15	Displays item total price after adding size and topping adjustments.
8	Cancel	Button	-	-	Closes popup and cancels item selection.
9	Add to Cart	Button	-	-	Adds item with customized options to cart (SCR-04) and closes popup.

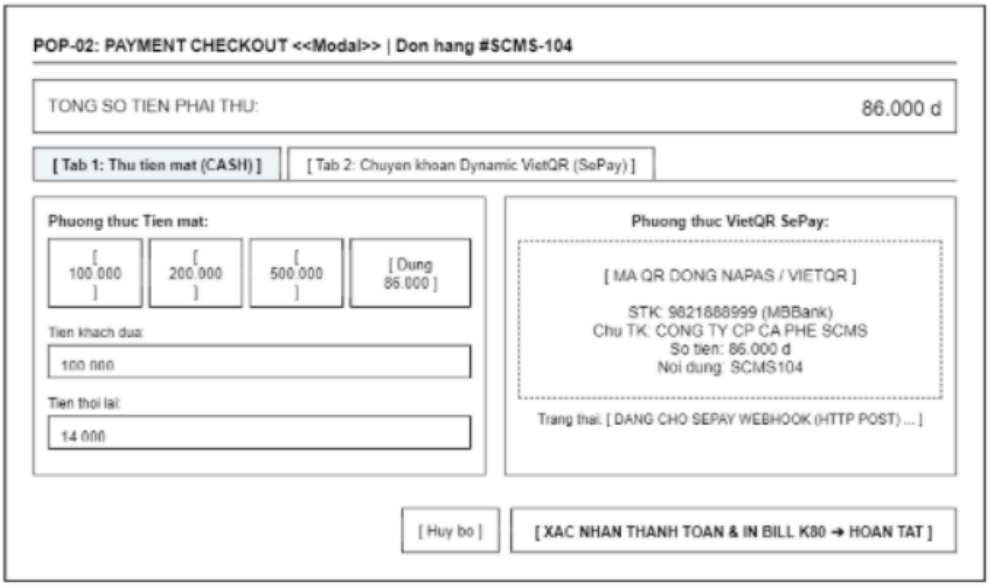
3.3.2.2 Use Case Description

Use Case ID		**UC17**	**Use Case Name**		**Apply Promotion / Voucher**	
Author		**VanNT**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor		Cashier				
Description		The Cashier enters or selects a promotion code. The system verifies the applicable conditions, calculates the discount, and updates the total payment amount.				
Precondition		PRE-01: The Cashier is logged into the SCMS (UC01) and is currently operating the "Main POS Register" screen (SCR-04). PRE-02: There is an active, unpaid order currently being processed (Business Rule BR-01 states promotions cannot be applied to PAID orders).				
Trigger		TRG-01: The Cashier inputs a promotion code or selects an available voucher to apply to the customer's current order.				
Post-Condition		POS-01: The system successfully calculates the discount and updates the total payment amount, ensuring the total payable amount is not less than VND 0 (BR-05).				
Main flows						
Step	***Actor***	***Action***				
1	Cashier	Enter the voucher code or select an applicable promotion on the POS interface.				
2	SCMS	Verify the voucher's validity by checking its Status, Start Date, End Date, Min Order Amount, and Usage Limit against the current order details.				
3	SCMS	Verify that no other discount or promotion has already been applied to this order, enforcing the limit of one discount per order (BR-05).				
4	SCMS	Calculate the discount amount based on the voucher's Discount Type (percentage or fixed amount) and Discount Value.				
5	SCMS	Apply the discount, update the order's total payment amount (ensuring it is				

		≥ VND 0), and display the new total on the POS screen.
Alternative flows		
AT1	Order is already PAID (At step 2 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that the current order has already been changed to PAID status.
1.2	SCMS	Display system error message and return to Orders List.
AT2	Voucher conditions are not met (At step 2 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that the voucher is invalid (e.g., expired End Date, order value is below Min Order Amount, or Usage Limit exceeded).
2.2	SCMS	Reject the voucher application, do not apply any discount, and display a warning message indicating why the voucher is invalid.
AT3	A promotion is already applied (At step 3 in the main flows)	
Sub step	**Actor**	**Action**
3.1	SCMS	Detect that the order already has an existing discount or voucher applied.
3.2	SCMS	Reject the new voucher to enforce the rule that each order may have a maximum of one discount/promotion applied (BR-05). Display a warning message to the Cashier.

3.3.3 Payment Checkout

3.3.3.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Modal Header	Label	-	50	Displays popup title and order code currently being paid.
2	Total Amount Due	Label	-	15	Displays total amount payable by the customer.
3	Payment Method Tabs	Tabs	-	-	Toggles between two payment methods: Cash and VietQR Transfer.
4	Quick Cash Buttons	Button Grid	-	-	Quick-select options for cash tendered (100k, 200k, 500k, Exact Amount).
5	Cash Received	Text	Yes	15	Amount tendered by customer (manual input or quick-select option).
6	Change Return	Label	-	15	Displays change due to customer (automatically calculated).
7	Dynamic VietQR Display	Component	-	-	Displays dynamic VietQR code containing exact order amount and transaction reference code.
8	Webhook Status	Label	-	50	Displays payment status received via SePay integration (Pending / Paid).
9	Cancel	Button	-	-	Closes popup and returns to the sales register screen (SCR-04).
10	Confirm & Print Bill	Button	-	-	Confirms payment completion and triggers 80mm receipt printing.

3.3.3.2 Use Case Description

UC43 – Generate VietQR Dynamic QR Code

Use Case ID	**UC43**	**Use Case Name**		**Generate VietQR Dynamic QR Code**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	VietQR				
Description	SCMS submits a QR generation request for an order. VietQR generates a dynamic QR code and returns the QR image along with payment transfer details.				
Precondition	PRE-01: Order is in pending payment status with total payable amount greater than 0. PRE-02: COO has configured the recipient bank account. PRE-03: Connection to VietQR service is operational.				

Trigger		TRG-01: Cashier selects QR payment method for the order.
Post-Condition		POS-01: QR code information is stored in the order payment record. POS-02: QR code is rendered for Cashier and customer to scan.
Main flows		
Step	**Actor**	**Action**
1	SCMS	Sends QR generation request to VietQR including payment amount, transfer narration (containing order code), and recipient account number.
2	VietQR	Validates payload and generates dynamic QR code according to NAPAS standards.
3	VietQR	Returns QR code image and payment details to SCMS.
4	SCMS	Saves QR information into order payment record and displays QR code on Cashier screen.

Alternative flows

AT1	At step 2 or 3, if VietQR does not respond or returns an error:	
Sub step	**Actor**	**Action**
2.1	SCMS	Notifies Cashier of QR generation failure, allowing retry or switching to cash payment.
2.2	SCMS	Preserves current order state without modification.
AT2	At step 2, if VietQR rejects due to missing or invalid recipient account configuration:	
Sub step	**Actor**	**Action**
2.1	SCMS	Displays receiving account configuration error and advises contacting COO.

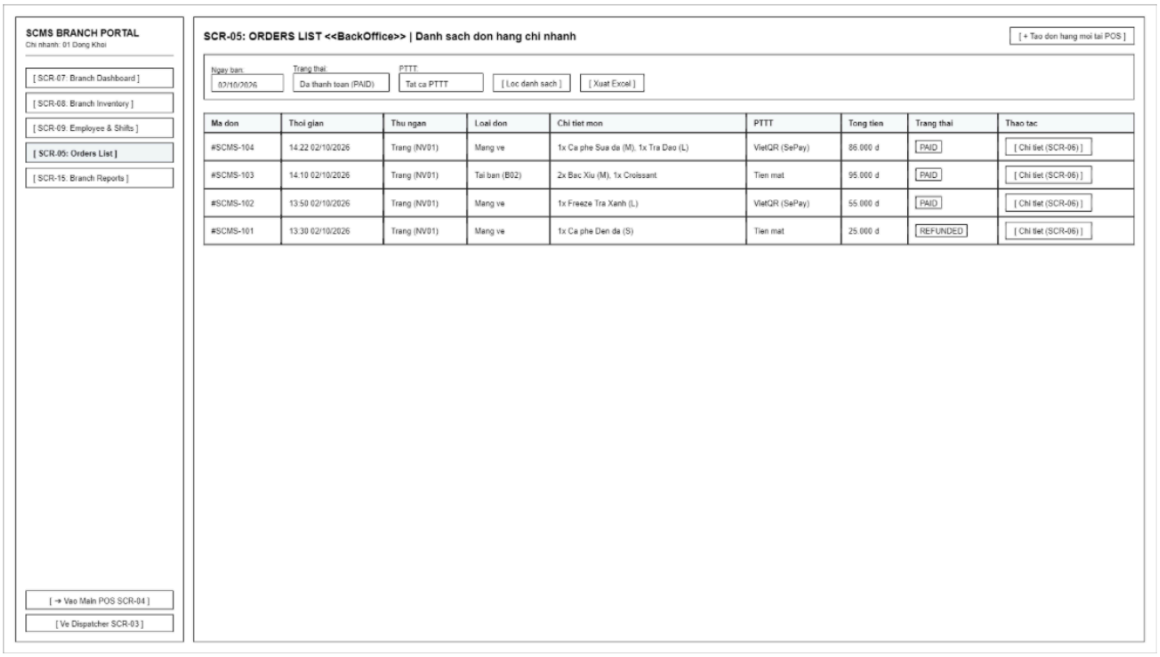
Business Rules

#	Rule Description
BR-01	Amount encoded in QR code equals the exact order payable amount and cannot be edited (BR-24).
BR-02	Transfer narration must contain the unique order code for reconciliation when SCMS receives SePay webhook (UC44) (BR-28).
BR-03	Beneficiary bank account is configured exclusively by COO (BR-28).
BR-04	QR code has an expiration duration. Once expired, a new QR code must be generated (BR-28).
BR-05	QR code must be displayed within a maximum of 3 seconds.

3.4 Order & Cash Drawer Management

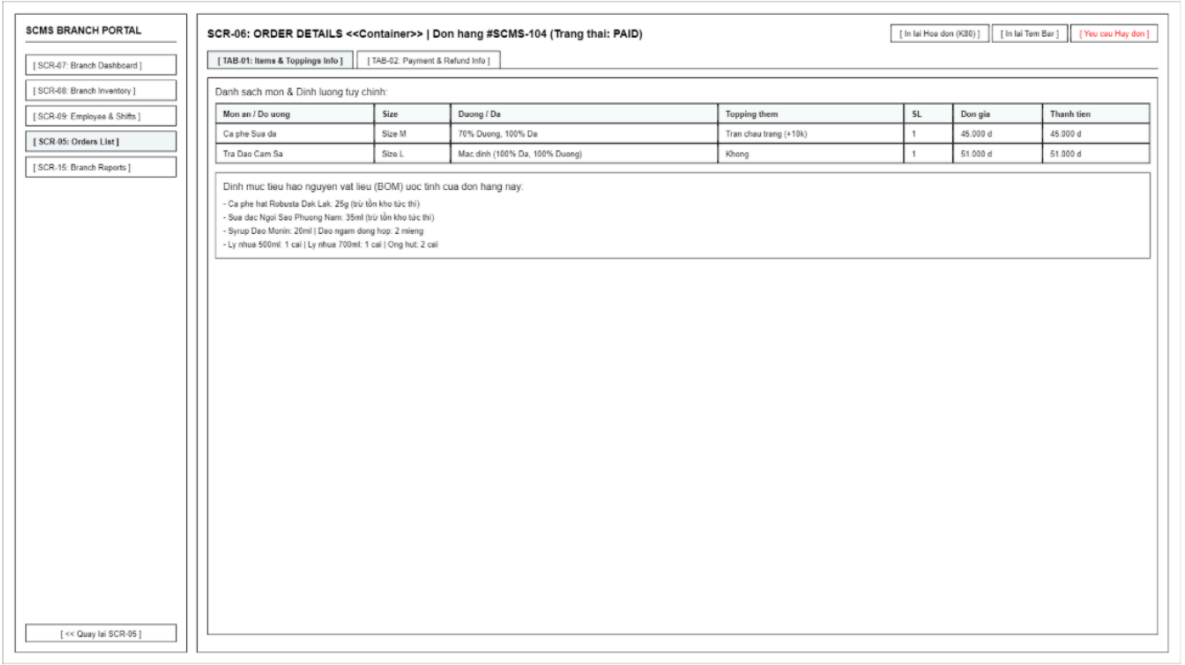
3.4.1 Order Details & Refund

3.4.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Order Date Filter	Text / Date	No	10	Filter orders by sales date (Default: Current date).
2	Status Filter	Dropdown	No	-	Filter by order status: All, Pending Payment, Paid, Partially Refunded, Refunded, Cancelled.
3	Payment Method Filter	Dropdown	No	-	Filter by payment method: All, Cash, VietQR.
4	Filter Action	Button	-	-	Apply filters and reload the order list.
5	Export Excel	Button	-	-	Export the order list to an Excel file.
6	Orders Table	Table	-	-	List of orders (Order Code, Time, Cashier, Item Details, Payment Method, Total Amount, Status).
7	View Order Details	Button	-	-	Navigate to the order details screen (SCR-06).
8	Nav to POS	Button	-	-	Return to the POS/sales screen (SCR-04).



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Order Header Info	Label	-	80	Display order code, creation time, and order status.
2	Sub-Tab Switcher	Tabs	-	-	Switch between TAB-01 (Items & Toppings) and TAB-02 (Payment & Refund).
3	Reprint Receipt	Button	-	-	Send command to reprint the K80 receipt.
4	Reprint Bar Label	Button	-	-	Send command to reprint cup labels for the bar/preparation counter.
5	Cancel Order	Button	-	-	Request order cancellation (Requires Manager approval).
6	Back to Orders List	Button	-	-	Return to the orders list screen (SCR-05).
7	(TAB-01) Order Items Table	Table	-	-	Table listing items, size, sugar, ice, toppings, quantity, and total amount.
8	(TAB-01) BOM Usage Box	Box / Label	-	200	Display estimated raw material consumption quota based on the order's recipe.
9	(TAB-02) Transaction Details	Table	-	-	Payment details: SePay transaction ID, account number, bank, and Webhook matching time.

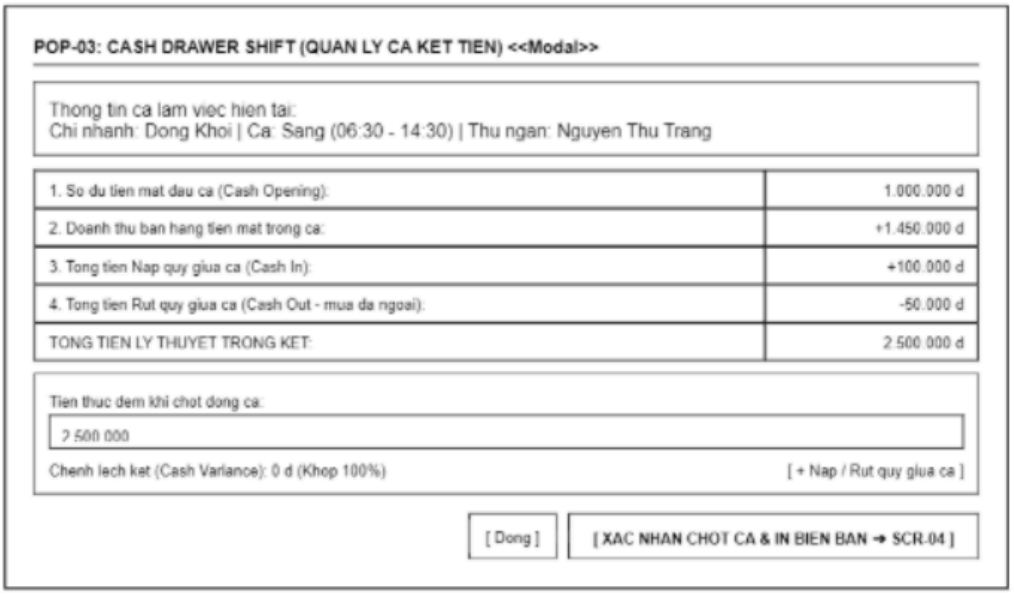
10	(TAB-02) Refund Reason	Dropdown	Yes	-	Select refund reason (Customer changed mind, Wrong item, Quality issue).

3.4.1.2 Use Case Description

Use Case ID	**UC18**	**Use Case Name**		**View Order Information**	
Author	**VanNT**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Cashier				
Description	The Cashier selects an order. The system displays the order number, item list, quantities, discount amount, total amount, and current status.				
Precondition	PRE-01: The Cashier has successfully logged into the SCMS (UC01). PRE-02: The Cashier is currently on the "Main POS Register" screen (SCR-04). PRE-03: A sales shift is currently active (BR-09).				
Trigger	TRG-01: The Cashier selects a specific order from the list/queue on the POS interface to view its details.				
Post-Condition	POS-01: The system successfully retrieves and displays all detailed information related to the selected order on the screen.				
Main flows					
Step	***Actor***	***Action***			
1	Cashier	Select an order from the active or historical order list on the POS screen.			
2	SCMS	Retrieve the order record from the database, including data from the Order entity (Order Number, Status), Order Item entity (Quantity, Unit Price, Sugar/Ice Levels, Note), and any applied Voucher or Payment records.			
3	SCMS	Aggregate the data to calculate the total amount and any applicable discount amount.			
4	SCMS	Display the detailed order information to the Cashier, including: order number, item list, quantities, discount amount, total amount, and current status.			
Alternative flows					
AT1	System fails to retrieve order details (At step 2 in the main flows)				
Sub step	**Actor**	**Action**			
1.1	SCMS	Attempt to retrieve the order data but encounter an error (e.g., network timeout or order was deleted/not found).			
1.2	SCMS	Strictly prohibit applying the promotion (BR-01) and display an error message to the Cashier.			
1.3	SCMS	Return the Cashier to the main order list view.			

3.4.2 Cash Drawer Shift

3.4.2.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Shift Header	Label	-	60	Displays branch information, work shift, and assigned cashier.
2	Opening Cash	Label	-	15	Displays opening cash balance handed over at the start of the shift.
3	Cash Sales	Label	-	15	Displays total cash sales revenue generated during the shift.
4	Cash In Total	Label	-	15	Displays total mid-shift cash top-ups added to the cash drawer.
5	Cash Out Total	Label	-	15	Displays total emergency cash withdrawals made from the cash drawer during the shift.
6	Theoretical Cash	Label	-	15	Displays calculated theoretical cash balance that should be in the drawer.
7	Actual Cash Counted	Text	Yes	15	Inputs actual cash counted upon shift closure.

8	Cash Variance	Label	-	30	Displays variance between actual and theoretical cash balance (Balanced / Surplus / Shortage).
9	Cash Adjustment Action	Button	-	-	Opens form to record emergency mid-shift cash top-ups or withdrawals along with reason.
10	Close Modal	Button	-	-	Closes popup and returns to the POS screen.
11	Confirm Shift Close	Button	-	-	Confirms shift closing, locks cash drawer, and prints handover report.

3.4.2.2 Use Case Description

Use Case ID	**UC19**	**Use Case Name**		**Process Cash Payment**	
Author	**KhanhPG**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Cashier				
Description	The Cashier processes a cash payment for an active order on the Payment Checkout modal (POP-02). The Cashier enters the cash amount received from the customer or selects a quick-cash shortcut. The system validates the amount, calculates the change return, updates order status to PAID, freezes order data (BR-26), records cash into the current shift's cash drawer balance (BR-27, BR-39), and triggers automatic printing of receipts and bar slips (UC21, UC22, BR-31).				
Precondition	PRE-01: The Cashier has successfully logged into SCMS (UC01) and has an active sales shift (UC24, BR-36). PRE-02: The Cashier is on the "Main POS Register" screen (SCR-04) with an active order in PENDING status. PRE-03: The "Payment Checkout" modal (POP-02) is open displaying the order Total Amount Due.				
Trigger	TRG-01: The Cashier selects the "Cash" payment tab on modal POP-02, enters cash received, and clicks "Confirm & Print Bill".				
Post-Condition	POS-01: Order status transitions to PAID and order details are permanently frozen against modifications (BR-26). POS-02: The cash amount is recorded into sales revenue and added to the active shift's cash drawer balance (BR-27, BR-39). POS-03: Change return amount is accurately calculated and displayed. POS-04: Print commands are sent to print Bar Preparation Slip (UC21) and Payment Receipt (UC22) (BR-31). POS-05: Modal POP-02 closes and POS register (SCR-04) resets for the next order.				
Main flows					

Step	*Actor*	*Action*
1	Cashier	On the "Payment Checkout" modal (POP-02), verify Total Amount Due and select the "Cash" payment tab (default active).
2	SCMS	Display cash payment UI with Total Amount Due, quick-cash shortcut buttons (100k, 200k, 500k, Exact), Cash Received input, and Change Return field.
3	Cashier	Input the amount of cash received from customer into "Cash Received" field or click a quick-cash button.
4	SCMS	Validate that cash received is >= Total Amount Due, calculate change return amount (Change Return = Cash Received - Total Amount Due), and display it in real time.
5	Cashier	Verify the displayed change return, collect cash, return change if applicable, and click "Confirm & Print Bill".
6	SCMS	Save payment with method CASH, transition order to PAID, freeze order data (BR-26), add cash to shift drawer balance (BR-27, BR-39), trigger printing of preparation slip (UC21) and customer receipt (UC22) (BR-31), close modal POP-02, and reset POS register.

Alternative flows

AT1	Cash received is less than total amount due (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that Cash Received is less than Total Amount Due.
1.2	SCMS	Display an inline error message notifying that cash received is insufficient and disable "Confirm & Print Bill".
1.3	SCMS	Keep the Cashier on modal POP-02 to re-enter an adequate cash amount.

Alternative flows

AT2	Cashier cancels cash payment (At step 1, 3, or 5 in the main flows)	
Sub step	**Actor**	**Action**
2.1	Cashier	Click the "Cancel" ("Hủy bỏ") button on modal POP-02.
2.2	SCMS	Close modal POP-02 without creating any payment record or modifying order status (order remains PENDING).
2.3	SCMS	Return Cashier to Main POS Register (SCR-04) with current order cart preserved.

Use Case ID	UC20	Use Case Name		Confirm QR Payment	
Author	**KhanhPG**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	SePay				
Description	The system receives an automated payment Webhook from SePay after the customer completes a bank transfer via dynamic VietQR. The system validates the Webhook signature and reconciles the Order ID and payment amount (BR-29). Upon a match, the system transitions order status to PAID, freezes order data (BR-26), notifies POS in real time, and triggers automatic printing of receipts and bar tickets (BR-31).				
Precondition	PRE-01: Cashier opened Payment Checkout modal (POP-02) and selected "Dynamic VietQR" tab (Tab 2). PRE-02: System generated and displayed a dynamic VietQR code with chain account, order amount, and reference prefix CAFE (BR-24, BR-28, UC43). PRE-03: Customer completed bank transfer via mobile banking app.				
Trigger	TRG-01: SePay payment gateway sends an HTTP POST Webhook payload containing transaction details to the SCMS webhook endpoint (UC44).				
Post-Condition	POS-01: Webhook authentication signature and transaction details are successfully verified (BR-29). POS-02: Order status updates from PENDING to PAID and is frozen against edits (BR-26, BR-29). POS-03: Payment record is saved with method VIETQR, referencing SePay transaction ID. POS-04: Modal POP-02 displays payment success, triggers receipt printing (UC21, UC22, BR-31), and resets POS cart.				

Main flows

Step	***Actor***	***Action***
1	SePay	Send an HTTP POST Webhook payload to SCMS containing Transaction ID, Order Reference, Transfer Amount, and Authentication Signature.
2	SCMS	Verify the Webhook authentication signature against the configured SePay API Secret Key (BR-29, SCR-16 TAB-10).
3	SCMS	Reconcile Order ID and verify that the transferred amount exactly equals the order Total Amount Due (BR-29).
4	SCMS	Save payment transaction with method VIETQR, update order status to PAID, and freeze order data (BR-26).
5	SCMS	Push real-time notification via WebSocket to POS register (SCR-04 / POP-02) indicating payment success.

6	SCMS	Trigger automatic printing of Bar Preparation Slip (UC21) and Payment Receipt (UC22) (BR-31), close modal POP-02, and reset POS register for next order.

Alternative flows

AT1	Transferred amount does not match total amount due (At step 3 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that transferred amount does not match order total amount due.
1.2	SCMS	Retain order in PENDING status, log discrepancy, and push reconciliation alert to POS screen (BR-30).
1.3	SCMS	Require Cashier / Branch Manager to verify bank transaction details manually.

AT2	Invalid signature or unauthorized webhook (At step 2 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect invalid authentication signature in Webhook payload.
2.2	SCMS	Reject Webhook with HTTP 401/403 status and log unauthorized security event.
2.3	SCMS	Keep order in PENDING status without updating.

3.5 Branch Operations

3.5.1 Branch Dashboard

3.5.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Branch Sidebar Menu	Menu	-	-	Navigation bar for internal branch modules (Dashboard, Inventory, HR, Orders, Reports).
2	Daily Revenue KPI	Card	-	20	Displays accumulated total sales revenue for the day.
3	Active Shift KPI	Card	-	20	Displays the number of active sales shifts at the counter.
4	Low Stock Alert KPI	Card	-	20	Displays the number of raw materials reaching critical low-stock alert thresholds.
5	Cash Variance KPI	Card	-	20	Displays cash variance from the most recent shift.
6	Peak Hours Chart	Chart	-	-	Bar chart depicting order count and revenue by hourly time slots.
7	Active Staff List	Table	-	-	List of staff members currently on shift along with their check-in status.

8	Quick Nav to POS	Button	-	-	Quick access to the sales register screen (SCR-04).
9	Quick Nav to Inventory	Button	-	-	Quick access to the inventory management screen (SCR-08).
10	Quick Nav to Staff	Button	-	-	Quick access to the staff shift roster management screen (SCR-09).

3.5.1.2 Use Case Description

UC42 – View Peak Hours Report

Use Case ID	**UC42**	**Use Case Name**		**View Peak Hours Report**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	COO				
Description	COO requests peak hours reporting. The system analyzes order volume by hourly time slots, identifies peak sales periods, and displays traffic distribution.				
Precondition	PRE-01: COO is logged in. PRE-02: Orders exist within the selected viewing period.				
Trigger	TRG-01: COO selects the "Peak Hours Report" function.				
Post-Condition	POS-01: Order volume by hourly slot and identified peak hours are displayed.				

Main flows		
Step	**Actor**	**Action**
1	COO	Selects the peak hours report function.
2	SCMS	Displays time range filter and scope selection (chain-wide or specific branch).
3	COO	Selects filter parameters and requests to view report.
4	SCMS	Groups paid orders by hourly slots within store operating hours.
5	SCMS	Identifies time slots with the highest order volume.
6	SCMS	Displays traffic volume by hourly slot and highlights peak hours.

Alternative flows

AT1	At step 4, if no orders exist within the selected time range:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays a message indicating no data available for analysis.

Business Rules

#	**Rule Description**
BR-01	Time slots fall within chain operating hours (06:00 to 23:00).
BR-02	Each time slot has a duration of 1 hour (*).

BR-03	Orders are grouped into time slots based on order creation timestamp (*).
BR-04	Peak hours are defined as slots with order counts belonging to the highest tier during the period (*).

Use Case ID	**UC34**	**Use Case Name**		**View Branch Dashboard**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager accesses the dashboard. The system aggregates and visually displays key indicators related to revenue, cash flow, inventory, and employee performance.				
Precondition	PRE-01: Successfully logged in with Branch Manager role.				
Trigger	TRG-01: Branch Manager accesses home page or clicks "Dashboard" menu.				
Post-Condition	POS-01: Overview charts and key performance indicators (KPIs) of the assigned branch are displayed completely and accurately.				

Main flows

Step	***Actor***	***Action***
1	Branch Manager	Clicks "Dashboard" item on navigation bar.
2	SCMS	Verifies assigned branch scope and retrieves real-time aggregated data: sales revenue, order counts, cash drawer flow (opening cash, cash in/out, refunds), and low stock alerts.
3	SCMS	Displays information as visual charts, KPI indicator cards, and action alert lists.
4	Branch Manager	Views overview metrics or clicks widgets to drill down into corresponding detailed modules.

Alternative flows

AT1	At step 2 of the main flow, system fails to retrieve aggregation data due to connection issues:	
Sub step	**Actor**	**Action**
2.1	SCMS	Displays empty state widgets with message: "Failed to load dashboard data. Please refresh."
2.2	SCMS	Provides a "Retry" button for the user.

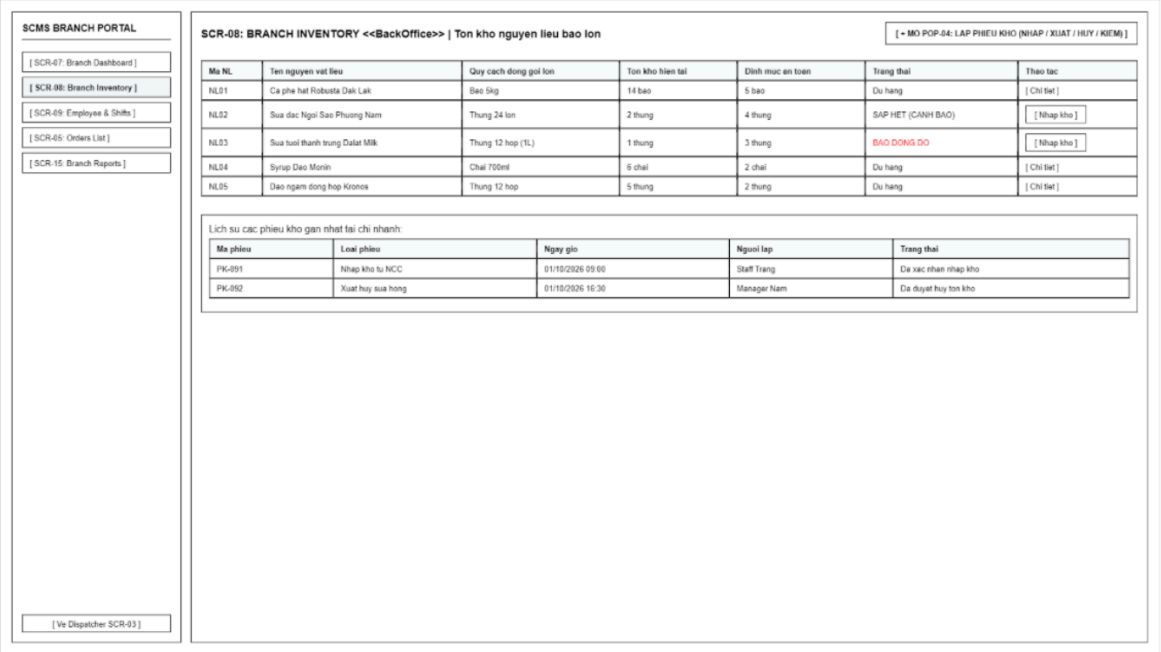
Business Rules

#	**Rule Description**

BR-01	Operational data must be strictly isolated by branch: Branch Managers are only allowed to view Dashboard metrics of their assigned branch (Rule BR-04).
BR-02	The system must provide a Dashboard and detailed Sales Reports over time for the Branch Manager to track revenue and order volumes (Rule BR-46).
BR-03	The system must provide in-depth branch-level tracking: Cash Drawer Flow (over/short reconciliation) and Inventory In-Out-Balance status (Rule BR-47).

3.5.2 Branch Inventory & Stock Slip

3.5.2.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Open Stock Slip	Button	-	-	Opens the popup to create a new stock voucher (POP-04).
2	Inventory Table	Table	-	-	Table listing bulk inventory items (Code, Name, Packaging Specification, On-hand Stock, Threshold, Status).
3	Stock Item Detail	Button	-	-	Views the inbound/outbound transaction history of a specific raw material.

4	Quick Restock Button	Button	-	-	Quickly opens an inbound stock voucher for raw materials currently triggering low-stock alerts.
5	Recent Stock Slips	Table	-	-	Table displaying the list of most recently created stock vouchers at the branch.

3.5.2.2 Use Case Description

Use Case ID	**UC28**	**Use Case Name**		**Receive Stock**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager creates a goods receipt, selects raw materials, and enters quantities and costs. The system validates the information and increases the corresponding inventory quantities.				
Precondition	PRE-01: Branch Manager has successfully logged into the system. PRE-02: Raw materials catalog and standard packaging units have been configured in the system.				
Trigger	TRG-01: Branch Manager needs to import additional raw materials into branch stock.				
Post-Condition	POS-01: The system records and stores a new inventory import receipt. POS-02: Inventory quantity in the branch warehouse is updated and added successfully.				
Main flows					
Step	***Actor***	***Action***			
1	Branch Manager	Selects the "Create inventory import receipt" feature.			
2	SCMS	Displays the import receipt creation screen with fields: Raw Material, Quantity, Import Price, Packaging Unit, Notes, and a "Confirm" button.			
3	Branch Manager	Selects raw material, enters quantity and price, and clicks "Confirm".			
4	SCMS	Validates information (quantity > 0, price >= 0, valid packaging unit).			
5	SCMS	Saves the import receipt, adds quantity to branch inventory stock, and logs the operation in the audit trail.			
6	SCMS	Displays a success notification.			

Alternative flows

AT1	At step 4 of the main flow, if entered information is missing or incorrectly formatted:	
Sub step	**Actor**	**Action**
4.1	SCMS	Highlights empty mandatory fields and displays message: "The * field is required and must be a valid positive number."
4.2	SCMS	Halts the save operation and remains on the import receipt screen.
AT2	At step 4 of the main flow, if the selected unit is not a standard large packaging unit:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays error message: "Goods received must be managed by large packaging units (cartons, bags, boxes, kg)."
4.2	SCMS	Prompts user to select a valid packaging unit before proceeding.

Business Rules

#	Rule Description
BR-01	Branch Managers are only allowed to create import receipts and manage inventory data belonging to their assigned branch (Rule BR-04).
BR-02	Goods received at the branch must be managed by large packaging units (cartons, bags, boxes, kg) and fully record the quantity and import cost to increase inventory (Rule BR-40).
BR-03	All sensitive inventory import operations must be automatically recorded in the Audit Log and cannot be modified or deleted (Rule BR-51).

Use Case ID	**UC29**	**Use Case Name**		**Issue Stock**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager creates an issue document, selects raw materials, and enters the quantities to be issued. The system checks the current inventory balance and deducts the issued quantities.				
Precondition	PRE-01: Branch Manager has successfully logged into the system. PRE-02: Branch warehouse contains raw materials with available stock balance > 0.				
Trigger	TRG-01: Need to retrieve raw materials from inventory for Bar counter usage or transfer.				

Post-Condition	POS-01: Inventory export receipt is created and stored in the system. POS-02: Corresponding inventory quantity is deducted from the branch warehouse.

Main flows		
Step	***Actor***	***Action***
1	Branch Manager	Selects the "Create inventory export receipt" feature.
2	SCMS	Displays export receipt creation screen with raw materials list, available balance, export quantity, and purpose.
3	Branch Manager	Selects raw materials to export, enters quantity in intact packaging units, and clicks "Confirm".
4	SCMS	Checks available inventory balance and validates packaging specifications.
5	SCMS	Saves export receipt, deducts quantity from branch inventory stock, and logs the operation in the audit trail.
6	SCMS	Displays success notification.

Alternative flows

AT1	At step 4 of the main flow, if exported quantity exceeds current inventory stock balance:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays error message: "Export quantity exceeds available stock balance."
4.2	SCMS	Halts the save operation and remains on the export receipt creation page.
AT2	At step 4 of the main flow, if quantity is not in intact large packaging units:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays warning: "Raw materials are only deducted from branch inventory when issued to the Bar counter in intact large packaging units."
4.2	SCMS	Requires user to adjust the quantity to match standard packaging units.

Business Rules

#	**Rule Description**
BR-01	Operational data must be strictly isolated by branch: Branch Managers are only allowed to issue stock belonging to their assigned branch (Rule BR-04).

BR-02	Raw materials are only deducted from branch inventory when issued to the Bar counter in intact large packaging units. The system does not deduct branch inventory in fractional amounts per cup sold (Rule BR-41).
BR-03	The quantity of raw materials issued to the bar must not exceed the currently available inventory quantity in the branch warehouse (Rule BR-42).
BR-04	All stock issue transactions must be automatically recorded in the Audit Log and cannot be modified or deleted (Rule BR-51).

Use Case ID	**UC30**	**Use Case Name**		**Record Damaged / Disposed Materials**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager declares the quantity of damaged or disposed raw materials and provides a reason. The system validates the information and deducts the corresponding quantity from inventory.				
Precondition	PRE-01: Branch Manager has successfully logged into the system. PRE-02: Materials to be disposed of exist in the branch inventory with positive available stock.				
Trigger	TRG-01: Discovered damaged, expired, or broken raw materials requiring disposal in warehouse or bar.				
Post-Condition	POS-01: Disposal receipt is recorded and stored with attached explanation reason. POS-02: Branch inventory stock is reduced accordingly.				
Main flows					
Step	***Actor***	***Action***			
1	Branch Manager	Selects the "Record material disposal" feature.			
2	SCMS	Displays material disposal declaration form (Raw Material, Quantity, Reason for Disposal, Photo evidence).			
3	Branch Manager	Selects material, enters quantity, enters complete disposal reason, and clicks "Confirm Disposal".			
4	SCMS	Validates data: disposal quantity <= available stock, reason is mandatory.			
5	SCMS	Validates value against Branch Manager inventory adjustment limit (TAB-11). If below limit, saves data, deducts stock and logs in audit trail.			

		If above limit, saves request in 'Pending' status awaiting COO approval (UC49) before deducting stock (BR-51).
6	SCMS	Displays success notification.

Alternative flows

AT1	At step 4 of the main flow, if Reason for Disposal is left empty:	
Sub step	**Actor**	**Action**
4.1	SCMS	Highlights the reason field and displays message: "Reason for disposal is mandatory."
4.2	SCMS	Halts the save operation until a valid reason is provided.
AT2	At step 4 of the main flow, if disposal quantity exceeds current stock balance:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays error message: "Disposal quantity cannot exceed currently available inventory quantity."
4.2	SCMS	Keeps the form for user adjustment.
AT3	At step 2 in the main flow, an internal system error occurs:	
Sub step	**Actor**	**Action**
2.1	SCMS	Displays "Error" page with message: "Internal System Error, please contact administrator."

Business Rules

#	Rule Description
BR-01	Branch Managers are only authorized to declare disposal for materials belonging to their assigned branch (Rule BR-04).
BR-02	The quantity of raw materials declared for disposal must not exceed the currently available inventory quantity in the branch warehouse (Rule BR-42).
BR-03	All disposal transactions for damaged, expired, or broken raw materials must include a specific disposal reason (Rule BR-43).
BR-04	All disposal transactions that reduce inventory must be automatically recorded in the Audit Log and cannot be modified or deleted (Rule BR-51).

Use Case ID	**UC31**	**Use Case Name**	**Perform Inventory Stocktake**

Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager creates an inventory stocktake session and enters the actual counted quantities. The system compares the physical count with the system records and records the variance.				
Precondition	PRE-01: Branch Manager has successfully logged into the system. PRE-02: Branch raw materials catalog has book inventory data in the system.				
Trigger	TRG-01: Periodic inventory count period reached (end-of-day/week/month) or urgent stock check needed.				
Post-Condition	POS-01: Audit session is saved with system book balance, physical count, and actual variance data. POS-02: Variance reconciliation report is generated, ready for the adjustment phase.				

Main flows		
Step	***Actor***	***Action***
1	Branch Manager	Selects "Create inventory audit session" feature.
2	SCMS	Generates audit session ID and displays list of raw materials with current system book levels, "Actual Quantity" input fields, and "Complete Count" button.
3	Branch Manager	Conducts physical count, enters counted quantities for items, and clicks "Complete Count".
4	SCMS	Validates entered quantities (must be non-negative numbers).
5	SCMS	Automatically calculates variance (Actual - Book), displays variance report, and saves audit session in "Pending Adjustment" status.
6	SCMS	Displays audit session completion notification.

Alternative flows

AT1	At step 4 of the main flow, if an entered count value is non-numeric or negative:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays error message: "Quantity must be a non-negative number (>= 0)."
4.2	SCMS	Highlights invalid input cell and prevents completing the audit session.
AT2	At step 3 of the main flow, if some items have not been counted:	

Sub step	Actor	Action
3.1	SCMS	Displays warning prompt: "Some raw materials have not had their actual count entered."
3.2	Branch Manager	Either fills in missing counts or confirms uncounted items have 0 actual quantity.

Business Rules

#	Rule Description
BR-01	Inventory stocktake operations are strictly isolated by branch; Branch Managers can only audit their assigned branch (Rule BR-04).
BR-02	After completing a periodic inventory count, the system must log the inventory discrepancy record showing book balance, physical count, and variances (Rule BR-44).
BR-03	All completed audit session records must be automatically captured in the Audit Log for transparency and verification (Rule BR-51).

Use Case ID	**UC32**	**Use Case Name**		**Adjust Inventory**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	Based on the stocktake variance, the Branch Manager enters an adjustment transaction with a reason. The system validates the information and updates the actual inventory balance.				
Precondition	PRE-01: An audit session with variances has been completed (UC31) and is in pending adjustment status.				
Trigger	TRG-01: Branch Manager needs to synchronize system stock quantity with actual count post-audit.				
Post-Condition	POS-01: System book inventory quantity is adjusted to match physical count. POS-02: Inventory adjustment voucher is stored with mandatory reason and the audit session is closed.				
Main flows					
Step	***Actor***	***Action***			
1	Branch Manager	Opens completed audit session and selects "Adjust Inventory".			

2	SCMS	Displays adjustment entry form showing discrepancy items, adjustment quantities, mandatory "Reason for Adjustment" field, and "Confirm Adjustment" button.
3	Branch Manager	Enters specific justification reason and clicks "Confirm Adjustment".
4	SCMS	Validates reason field (mandatory, cannot be blank).
5	SCMS	Validates value against Branch Manager inventory adjustment limit (TAB-11). If below limit, updates branch inventory balance according to actual audit session quantities. If above limit, saves adjustment request in 'Pending' status awaiting COO approval (UC49) before updating stock (BR-51).
6	SCMS	Closes audit session, logs operation in Audit Log, and displays success notification.

Alternative flows

AT1	At step 4 of the main flow, if Reason for Adjustment is left empty:	
Sub step	**Actor**	**Action**
4.1	SCMS	Highlights the reason field and displays message: "Reason for adjustment is mandatory."
4.2	SCMS	Remains on the form and blocks confirmation until a valid reason is provided.
AT2	At step 1 of the main flow, if all items in the audit session have zero variance:	
Sub step	**Actor**	**Action**
1.1	SCMS	Displays notification: "All physical counts match system book records; no adjustment is required."
1.2	SCMS	Allows closing the audit session directly without creating an adjustment voucher.

Business Rules

#	Rule Description
BR-01	Branch Managers are only authorized to adjust inventory for their assigned branch (Rule BR-04).
BR-02	After completing a periodic inventory count, the system must automatically update the book inventory balance according to the physical count and log the inventory discrepancy record (Rule BR-44).
BR-03	All sensitive business operations, including inventory adjustments, must be automatically recorded in the Audit Log and cannot be modified or deleted (Rule BR-51).

Use Case ID	UC33	Use Case Name		Monitor & Alert Low Stock	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager, SCMS				
Description	The system continuously compares inventory levels with the minimum stock threshold. If the quantity falls below the threshold, the system automatically displays an alert to the Branch Manager.				
Precondition	PRE-01: Minimum inventory stock / Safety Stock thresholds have been configured for raw materials at the branch.				
Trigger	TRG-01: Export, disposal, or adjustment transaction causes stock quantity to drop to or below minimum threshold.				
Post-Condition	POS-01: Low stock alert is generated and displayed to Branch Manager via notification bell and Dashboard widgets.				

Main flows		
Step	***Actor***	***Action***
1	SCMS	Continuously tracks stock quantities of all items after every inventory transaction (export, disposal, adjustment).
2	SCMS	Detects items with quantity <= minimum Safety Stock threshold.
3	SCMS	Sends and displays "Low stock" alert via notification bell and Dashboard widgets to the assigned Branch Manager.
4	Branch Manager	Clicks on alert to view details of nearly depleted materials (current balance, safety stock, recommended order quantity).

Alternative flows

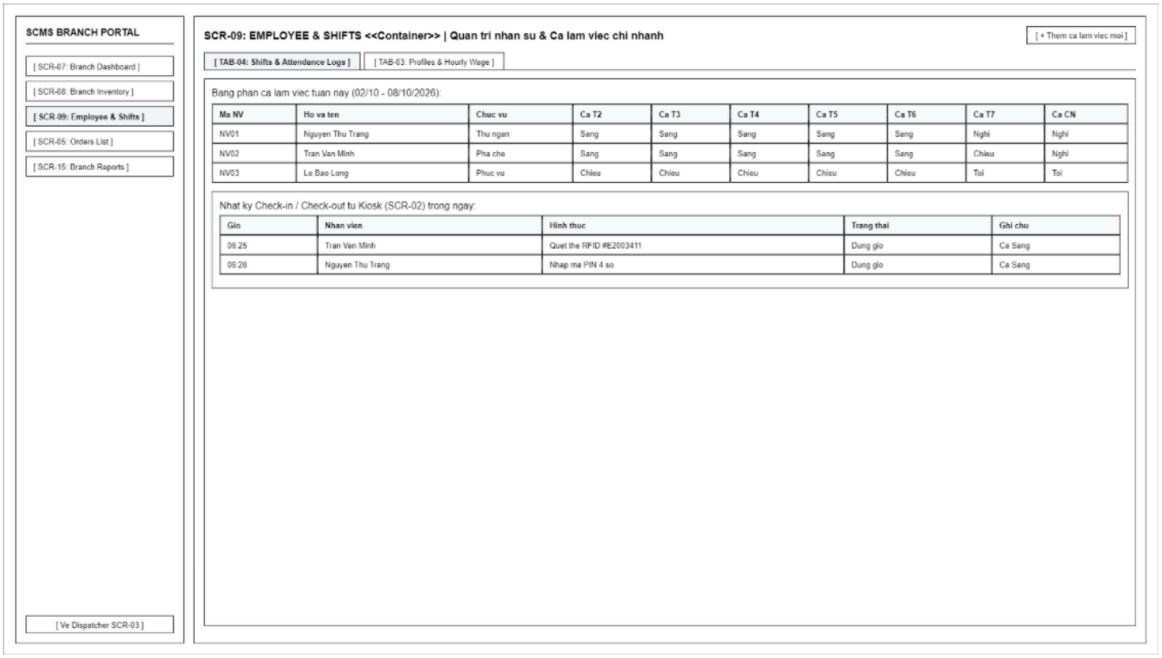
AT1	At step 2 of the main flow, if stock quantity drops to 0 (completely depleted):	
Sub step	**Actor**	**Action**
2.1	SCMS	Upgrades alert level to "Critical / Out of Stock" (highlighted in red) on Dashboard.
2.2	SCMS	Prompts a quick action button to create an import receipt (Receive Stock).
AT2	If real-time notification service is temporarily disconnected:	
Sub step	**Actor**	**Action**
3.1	SCMS	Queues the alert and delivers it immediately once the Branch Manager opens or refreshes Dashboard.

Business Rules

#	Rule Description
BR-01	Stock alerts are strictly partitioned by branch; alerts are only sent to the Branch Manager of the affected branch (Rule BR-04).
BR-02	When the inventory quantity of a raw material drops to or below the Safety Stock level, the system must automatically issue an out-of-stock alert to the Branch Manager (Rule BR-45).

3.5.3 Employee & Shifts

3.5.3.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Add Shift / Staff	Button	-	-	Opens form to schedule a new shift or add a employee profile.
2	Sub-Tab Switcher	Tabs	-	-	Toggles between TAB-04 (Shift Roster & Timekeeping) and TAB-03 (Profiles & Hourly Wage).
3	*(TAB-04)* Weekly Schedule	Table	-	-	Shift assignment matrix for branch staff from Monday to Sunday.

4	*(TAB-04)* Kiosk Logs Table	Table	-	-	Actual Check-in / Check-out time logs recorded from Kiosk (SCR-02).
5	*(TAB-04)* Adjust Attendance	Button	-	-	Manages approvals for employee attendance time adjustments during system/issue exceptions.
6	*(TAB-03)* Profiles Table	Table	-	-	Employee directory table (Employee ID, Full Name, Phone Number, Position, Employment Status).
7	*(TAB-03)* Hourly Wage Rate	Text	Yes	10	Configures base hourly wage rates for employees.
8	*(TAB-03)* RFID / PIN Config	Text	Yes	15	Configures RFID card ID or 4-digit timekeeping PIN code for employees.
9	*(TAB-03)* Edit Profile	Button	-	-	Opens window to edit employee profile information.

3.5.3.2 Use Case Description

Use Case ID	**UC07**	**Use Case Name**		**Manage Employee Profiles**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager manages store staff profiles within their assigned branch, including creating, updating employee details (e.g., full name, role, phone, PIN code), or deactivating profiles upon termination.				
Precondition	PRE-01: The Branch Manager has successfully logged into the SCMS (UC01). PRE-02: The Branch Manager has navigated to the "Employee Profiles" screen (SCR-07).				
Trigger	TRG-01: The Branch Manager chooses to add a new employee, edit employee details, or change an employee's employment status.				
Post-Condition	POS-01: Employee records for the specific branch are updated in the central database. POS-02: Updated employee details and PIN codes are immediately applicable for attendance and operations.				
Main flows					

Step	*Actor*	*Action*
1	Branch Manager	View the employee list for the managed branch and select "Add Employee" or choose an existing employee to edit/deactivate.
2	Branch Manager	Fill in or update employee profile data (Full Name, Phone, Role/Position, Attendance PIN, Status).
3	Branch Manager	Click the "Save" button.
4	SCMS	Validate employee profile inputs (e.g., required fields, unique attendance PIN within the branch).
5	SCMS	Save the employee record associated with the current branch context.
6	SCMS	Display a success confirmation message and update the employee directory display.

Alternative flows

AT1	Duplicate PIN code within the same branch (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that the entered PIN code is already assigned to another employee in the branch.
2.2	SCMS	Display a validation error message "PIN code already in use by another staff member."
2.3	SCMS	Prompt the Branch Manager to provide a unique PIN code.

Use Case ID	**UC08**	**Use Case Name**		**Manage Work Shifts**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager defines work shift timeframes, creates shift schedules for upcoming periods, and assigns employees to specific shifts. The system checks for scheduling conflicts and saves the work schedule.				
Precondition	PRE-01: The Branch Manager has successfully logged into the SCMS (UC01).				

	PRE-02: Active employees exist in the branch directory (UC07). PRE-03: The Branch Manager has navigated to the "Shift Management" screen (SCR-08).
Trigger	TRG-01: The Branch Manager creates a new shift schedule or assigns employees to working shifts for a calendar period.
Post-Condition	POS-01: Work shift schedules and employee shift assignments are persisted in the database. POS-02: The shift schedule becomes active and is available for employee check-in verification (UC03)

Main flows		
Step	***Actor***	***Action***
1	Branch Manager	Select shift configuration (e.g., Morning Shift, Evening Shift) or pick a date on the shift schedule calendar.
2	Branch Manager	Define/adjust shift time frames (Start Time, End Time) and select target employees to assign to the shift.
3	Branch Manager	Click the "Save Schedule" button.
4	SCMS	Validate shift configurations and check for schedule conflicts (e.g., employee assigned to overlapping shifts on the same day).
5	SCMS	Save shift schedules and employee assignment data to the database.
6	SCMS	Display a confirmation message and refresh the schedule view.

Alternative flows

AT1	Schedule conflict detected (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
2.1	SCMS	Detect that one or more assigned employees are already scheduled for an overlapping shift time.
2.2	SCMS	Display a warning message detailing the conflicting employees and shift times.
2.3	SCMS	Highlight conflicting assignments and allow the Branch Manager to resolve or reassign staff.

UC39 – View Branch Personnel Report

Use Case ID	UC39	Use Case Name		View Branch Personnel Report	
Author	VuNL	Version	1.0	Date	04/10/2026
Actor	Branch Manager				
Description	The Branch Manager views the personnel report of their assigned branch. The system aggregates and displays data regarding attendance, work shifts, and productivity indicators for employees.				
Precondition	PRE-01: Branch Manager is logged in. PRE-02: The branch has employees and timekeeping data available.				
Trigger	TRG-01: Branch Manager selects the "Personnel Report" feature.				
Post-Condition	POS-01: The branch personnel report is displayed.				

Main flows		
Step	**Actor**	**Action**
1	Branch Manager	Selects the personnel report feature.
2	SCMS	Displays the time period filter.
3	Branch Manager	Selects the filter criteria and requests to view the report.
4	SCMS	Aggregates timekeeping data for each employee: check-in time, check-out time, total working hours, late arrivals, early departures, and missing check-outs.
5	SCMS	Compares assigned shifts with actual worked shifts.
6	SCMS	Calculates productivity indicators for each employee (*).
7	SCMS	Displays the report detailed by individual employee and aggregated for the entire branch.

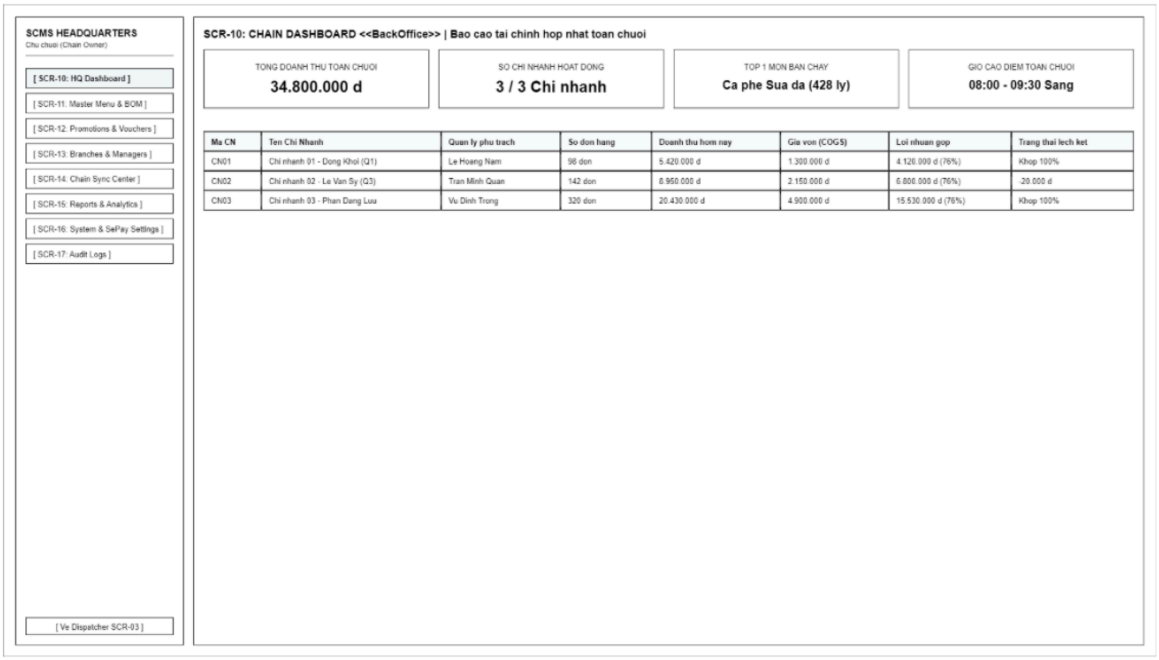
Alternative flows

AT1	No timekeeping data found.At step 4, if there is no timekeeping data within the selected time period:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays a notification indicating that no personnel data is available.
AT2	Missing attendance record for scheduled shift. At step 5, if an employee is assigned to a shift but has no timekeeping record:	
Sub step	**Actor**	**Action**
5.1	SCMS	Marks that shift as "Absent" in the report.

3.6 Chain Management & Master Data

3.6.1 Chain Dashboard

3.6.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	HQ Sidebar Menu	Menu	-	-	Navigation bar for high-level management modules of the Chain Headquarters.
2	Chain Total Revenue	Card	-	25	Display the total daily revenue of all 6-10 branches.
3	Active Branches KPI	Card	-	15	Display the number of open and operating branches.
4	Top 1 Best Seller	Card	-	50	Display the name of the food/beverage item with the highest consumption across the chain.
5	Peak Hours KPI	Card	-	30	Display the peak hour timeframe with the highest number of orders across the chain.
6	Branch Performance Table	Table	-	-	Table comparing revenue, COGS (Cost of Goods Sold), gross profit, and cash drawer discrepancy status of each facility.

3.6.1.2 Use Case Description

UC40 – View Consolidated Financial Report

Use Case ID	UC40	Use Case Name		View Consolidated Financial Report	
Author	VuNL	Version	1.0	Date	04/10/2026
Actor	COO				
Description	COO requests consolidated reporting. The system retrieves data from all branches to compute and display total Revenue, Cost of Goods Sold (COGS), and Gross Profit across the entire chain.				
Precondition	PRE-01: COO is logged in. PRE-02: The chain has at least one branch that has processed orders.				
Trigger	TRG-01: COO selects the "Consolidated Financial Report" function.				
Post-Condition	POS-01: Revenue, COGS, and Gross Profit reports across the chain and per branch are displayed.				

Main flows		
Step	**Actor**	**Action**
1	COO	Selects the consolidated financial report function.
2	SCMS	Displays time range filter and branch selection list (defaults to all branches).
3	COO	Selects filter parameters and requests to view report.
4	SCMS	Retrieves paid order data from all selected branches to calculate revenue.
5	SCMS	Calculates COGS based on recipe ingredient quantities, sales volumes, and ingredient unit costs.
6	SCMS	Calculates gross profit.
7	SCMS	Displays consolidated chain-wide figures and detailed breakdown per branch.

Alternative flows

AT1	At step 4, if a branch has not generated any orders during the period:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays zero values for that branch and still includes it in the consolidated report.
AT2	At step 5, if products lack recipes or ingredients lack unit cost prices:	
Sub step	**Actor**	**Action**
5.1	SCMS	Displays an incomplete cost data warning while still presenting available calculated figures.

Business Rules

#	Rule Description
BR-01	Only COO is permitted to view the chain-wide consolidated report (BR-02, BR-04).
BR-02	Revenue only accounts for paid orders, excluding cancelled or refunded orders (BR-26, BR-48).
BR-03	COGS = Σ (recipe ingredient quantity × product sales volume × ingredient unit cost) (BR-19, BR-48).
BR-04	Gross Profit = Revenue − COGS (BR-48).

UC41 – View Sales Report by Product

Use Case ID	**UC41**	**Use Case Name**		**View Sales Report by Product**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	COO				
Description	COO requests product sales report. The system calculates sales volumes, ranks top-selling products in descending order, and displays sales revenue per product.				
Precondition	PRE-01: COO is logged in. PRE-02: Paid orders exist within the selected viewing period.				
Trigger	TRG-01: COO selects the "Sales Report by Product" function.				
Post-Condition	POS-01: List of top-selling products is displayed in descending order.				

Main flows		
Step	**Actor**	**Action**
1	COO	Selects the sales report by product function.
2	SCMS	Displays time range filter and scope selection (chain-wide or specific branch).
3	COO	Selects filter parameters and requests to view report.
4	SCMS	Aggregates sales quantity for each product from paid orders.
5	SCMS	Sorts products in descending order by quantity sold.
6	SCMS	Displays list of best-selling products along with quantities sold and generated revenue.

Alternative flows

AT1	At step 4, if no paid orders exist within the selected time range:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays a message indicating no sales data available.

Business Rules

#	Rule Description
BR-01	Only paid orders are counted; cancelled orders are excluded (BR-26).
BR-02	Products are ranked by sales volume (quantity sold), not by revenue (BR-49).
BR-03	The list defaults to displaying the top 10 products (BR-49).

UC42 – View Peak Hours Report

Use Case ID	UC42	Use Case Name		View Peak Hours Report	
Author	VuNL	Version	1.0	Date	04/10/2026
Actor	COO				
Description	COO requests peak hours reporting. The system analyzes order volume by hourly time slots, identifies peak sales periods, and displays traffic distribution.				
Precondition	PRE-01: COO is logged in. PRE-02: Orders exist within the selected viewing period.				
Trigger	TRG-01: COO selects the "Peak Hours Report" function.				
Post-Condition	POS-01: Order volume by hourly slot and identified peak hours are displayed.				

Main flows		
Step	**Actor**	**Action**
1	COO	Selects the peak hours report function.
2	SCMS	Displays time range filter and scope selection (chain-wide or specific branch).
3	COO	Selects filter parameters and requests to view report.
4	SCMS	Groups paid orders by hourly slots within store operating hours.
5	SCMS	Identifies time slots with the highest order volume.
6	SCMS	Displays traffic volume by hourly slot and highlights peak hours.

Alternative flows

AT1	At step 4, if no orders exist within the selected time range:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays a message indicating no data available for analysis.

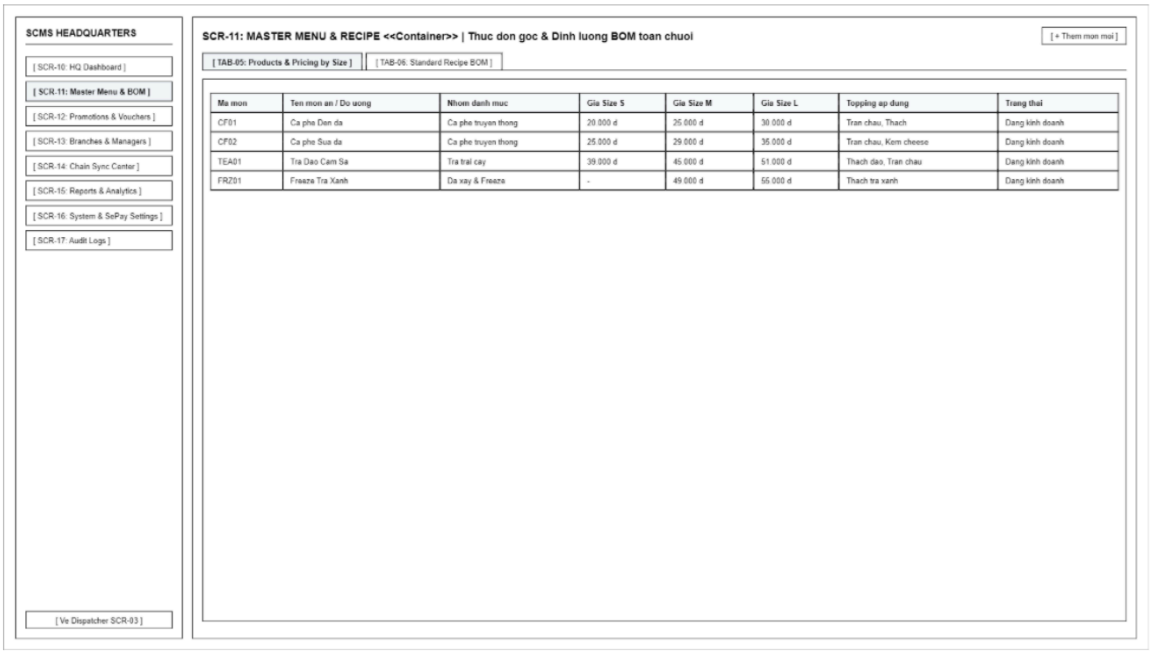
Business Rules

#	Rule Description
BR-01	Time slots fall within chain operating hours (06:00 to 23:00) (BR-49).
BR-02	Each time slot has a duration of 1 hour (BR-49).
BR-03	Orders are grouped into time slots based on order creation timestamp (BR-25, BR-49).

BR-04	Peak hours are defined as slots with order counts belonging to the highest tier during the period (BR-49).

3.6.2 Master Menu & Recipe

3.6.2.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description
1	Add New Product	Button	-	-	Open the form to create a new item in the chain menu.
2	Sub-Tab Switcher	Tabs	-	-	Switch between TAB-05 (Products & Prices) and TAB-06 (BOM Recipe).
3	(TAB-05) Products Table	Table	-	-	List of items, categories, selling price table by S/M/L sizes, and applicable toppings.
4	(TAB-05) Active Status	Toggle	-	-	Enable/Disable the active business status of the item across the entire chain system.
5	(TAB-06) Recipe BOM Table	Table	-	-	Detailed raw material consumption quota table for each item by size.
6	(TAB-06) Estimated COGS	Label	-	15	Display the theoretical cost of goods sold (COGS) automatically calculated from the BOM formula.

7	(TAB-06) Edit BOM Formula	Button	-	-	Open the form to edit the ingredient quantities for the item.

3.6.2.2 Use Case Description

Use Case ID	**UC09**	**Use Case Name**		**Manage Menu Categories**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The COO creates, modifies, or deactivates standard menu categories and master menus, assigning products to each menu structure. The system records the standardized menu configuration centrally.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Menu Management" screen (SCR-09).				
Trigger	TRG-01: The COO initiates creation or modification of master menu categories or item groupings.				
Post-Condition	POS-01: Master menu structures and category-product mappings are updated in the database. POS-02: The updated menu hierarchy becomes available for store synchronization (UC14).				
Main flows					
Step	***Actor***	***Action***			
1	COO	Select "Create Category" or choose an existing menu category to edit/deactivate.			
2	COO	Enter/update category details (Category Name, Display Sequence, Description, Active Status).			
3	COO	Assign/reorder products associated with the menu category.			
4	COO	Click the "Save Menu Category" button.			
5	SCMS	Validate category inputs and verify structural integrity (e.g., mandatory names, valid display order).			

6	SCMS	Save standard menu structure to the central database and display a success notification.

Alternative flows

AT1	Category contains assigned products upon deletion request (At step 1 in the main flows)	
Sub step	**Actor**	**Action**
2.1	COO	Attempt to delete/deactivate a menu category that currently contains active products.
2.2	SCMS	Detect linked products within the target category.
2.3	SCMS	Display a warning message "Category contains active products. Please reassign or remove products before deleting."
2.4	SCMS	Abort deletion and maintain current category status.

Use Case ID	**UC10**	**Use Case Name**		**Manage Products**	
Author	**VanNT**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The function allows the COO to create new products, update product details, or deactivate products. The system saves the information in the product catalog.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Master Menu & Recipe" screen (SCR-11).				
Trigger	TRG-01: The COO clicks on the "Add Product" button or selects an existing product from the list to view/edit details.				
Post-Condition	POS-01: The product information is successfully created or updated in the system database. POS-02: If a product is deactivated, its status is changed to "Unavailable" and it is disabled on the POS interface (BR-04).				
Main flows					
Step	***Actor***	***Action***			
1	COO	Click the "Add Product" button on the Master Menu & Recipe screen.			
2	SCMS	Display the Product Information form with the following fields: - Name (Required) - Description - Price (Required) - Status (Active/Unavailable)			

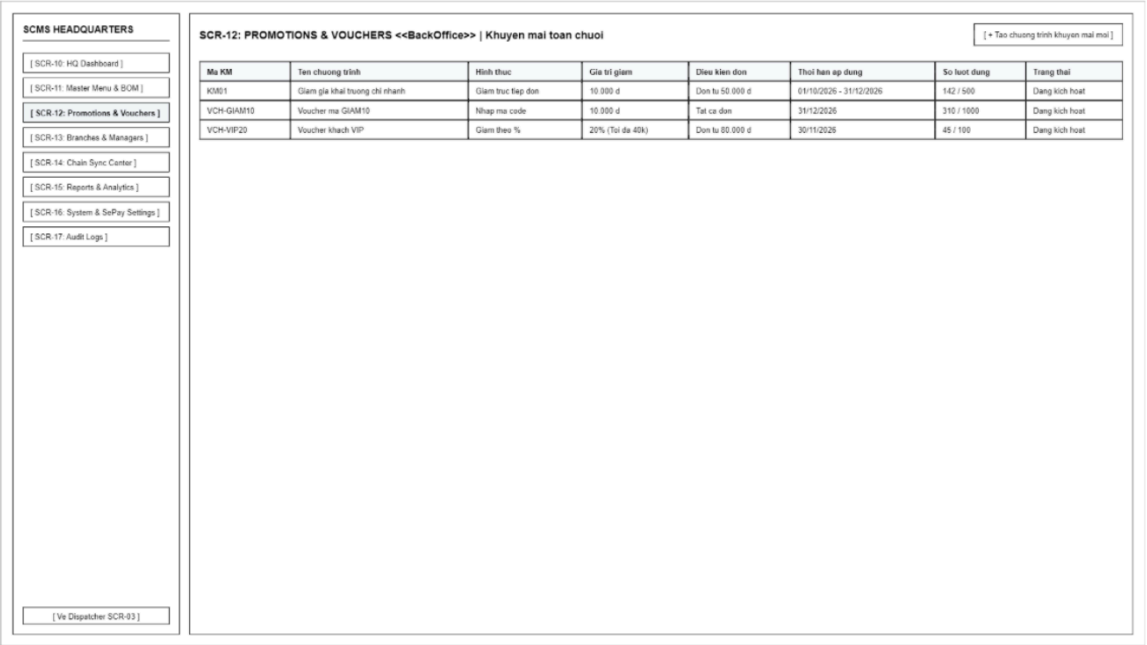
3	COO	Enter the required product details and click the "Save" button.
4	SCMS	Validate the entered data (check for empty required fields and data types).
5	SCMS	Save the new product into the database and display a success toast message.
Alternative flows		
AT1	At step 1 in the main flows, update an existing product	
Sub step	**Actor**	**Action**
1.1	COO	Select an existing product from the list and click the "Edit" icon.
1.2	SCMS	Display the Product Information form pre-filled with the current data.
1.3	COO	Modify the product details (e.g., Name, Description) and click the "Save" button.
1.4	SCMS	Validate, update the information in the database, and display a success toast message.
AT2	At step 1 in the main flows: Deactivate a product.	
Sub step	**Actor**	**Action**
2.1	COO	Select an existing product, change its Status field to "Unavailable", and click "Save".
2.2	SCMS	Update the status in the database. Apply Business Rule BR-04: The system strictly disables this item on the POS interface so it cannot be added to any new order.

Use Case ID	**UC11**	**Use Case Name**		**Manage Prices, Sizes & Toppings**	
Author	**VanNT**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The function allows the COO to set or update selling prices for each product based on size and toppings. The system validates the information and saves the pricing configuration.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Master Menu & Recipe" screen (SCR-11) and accessed the "Products & Pricing by Size" sub-tab.				
Trigger	TRG-01: The COO selects a product to configure its size options and topping prices.				
Post-Condition	POS-01: The pricing configuration (sizes and toppings) is successfully saved in the database and applied to the product.				

Main flows		
Step	***Actor***	***Action***
1	COO	Select a product from the catalog and click the "Configure Price/Size" button.
2	SCMS	Display the Pricing Configuration form, showing available sizes (Small/Medium/Large) and applicable toppings.
3	COO	Input or update the numeric selling prices for each active size and topping, then click "Save".
4	SCMS	Validate the entered data (ensure price fields are filled and contain valid non-negative numbers).
5	SCMS	Save the pricing configuration into the database and display a success toast message (e.g., MSG03).
Alternative flows		
AT1	At step 1 in the main flows, update an existing product	
Sub step	**Actor**	**Action**
1.1	SCMS	Detects that a required price field is left blank.
1.2	SCMS	Highlight the empty mandatory field(s) and display message MSG02: "The * field is required." in red under the text box
1.3	SCMS	The system halts the save operation and remains on the configuration form page.

3.6.3 Promotions & Vouchers

3.6.3.1 Screen Mock-up

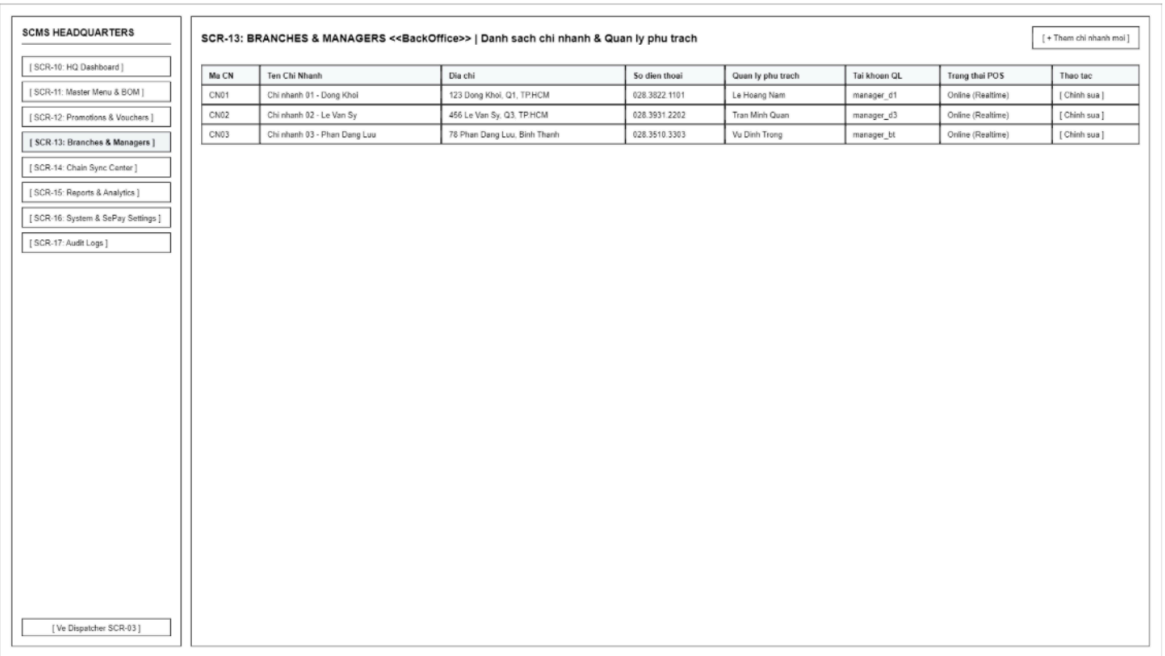


Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Create Promotion	Button	-	-	Open the form to create a new promotion campaign or voucher code.
2	Promotion Code	Text	Yes	15	Voucher code applied during POS checkout (e.g., GIAM10).
3	Discount Type & Value	Dropdown / Text	Yes	15	Set the discount type (Percentage discount or Fixed amount discount).
4	Min Order Value	Text	No	15	Minimum order value condition to apply the code.
5	Usage Limit / Count	Label	-	20	Display the number of times used over the maximum issued limit.
6	Validity Period	Text / Date	Yes	25	Start and end validity period of the program.
7	Status Toggle	Toggle	-	-	Activate or suspend the validity of the promotion.

3.6.4 Branches & Managers

3.6.4.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Add Branch	Button	-	-	Open the form to add a new branch to the chain system.
2	Branches Table	Table	-	-	Branch information table (Branch Code, Name, Address, Phone Number, Assigned Manager).
3	Assigned Manager	Dropdown	Yes	-	Assign the Branch Manager account in charge of the facility.
4	POS Connection Status	Label	-	20	Display the network connection status of the POS system at the branch (Online/Offline).
5	Edit Branch Action	Button	-	-	Open the form to edit information or temporarily close the branch.

3.6.4.2 Use Case Description

Use Case ID	**UC05**	**Use Case Name**		**Manage Branch**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The COO creates, views, edits, or deactivates store branch profiles (including address, contact details, operating status, and tax info). The system validates and saves the branch configuration in the central database.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Branch Management" screen (SCR-05).				
Trigger	TRG-01: The COO chooses to add a new branch, update existing branch info, or change a branch operational status.				
Post-Condition	POS-01: Branch information is updated or created in the central system database. POS-02: Updated branch profile data becomes available across chain-wide modules.				
Main flows					
Step	***Actor***	***Action***			

1	COO	Select an action: "Create New Branch" or choose an existing branch from the list to "Edit" or "Deactivate".
2	COO	Input or modify branch details (Branch Name, Code, Address, Phone Number, Operating Hours, Status).
3	COO	Click the "Save" button.
4	SCMS	Validate the entered information (e.g., check for unique Branch Code and required fields).
5	SCMS	Save the branch record to the system database.
6	SCMS	Display a success message and update the branch list table on screen.

Alternative flows

AT1	Duplicate branch code or missing required fields (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect missing mandatory fields or a duplicate Branch Code in the database.
1.2	SCMS	Highlight the erroneous input fields and display appropriate validation error messages.
1.3	SCMS	Allow the COO to correct the information and re-submit.

Use Case ID	**UC06**	**Use Case Name**		**Manage Branch Manager**	
Author	**TamDHM**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				
Description	The COO creates or updates Branch Manager user accounts, assigns specific branch ownership/data scopes, and sets login credentials and permissions. The system updates the manager account database.				
Precondition	PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: Target branches exist within the system (UC05). PRE-03: The COO has navigated to the "Branch Manager Account Management" screen (SCR-06).				

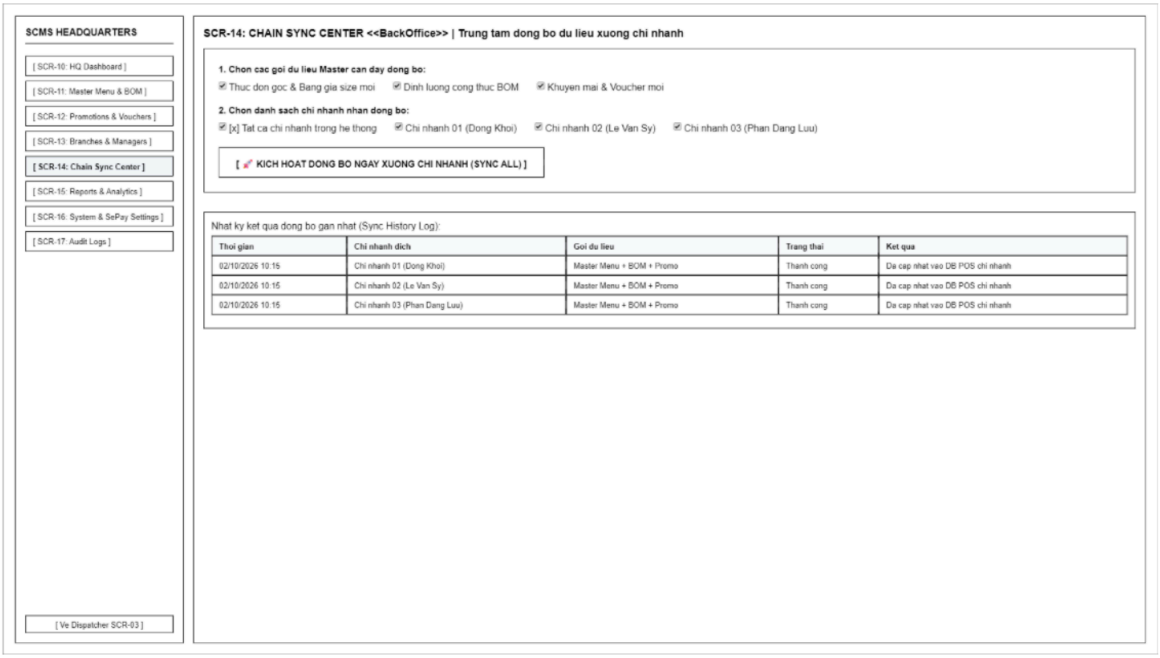
Trigger		TRG-01: The COO selects the option to create a new Branch Manager account or modify an existing manager's assigned branch/credentials.
Post-Condition		POS-01: Branch Manager account details, assigned branch scope, and credentials are saved in the system. POS-02: The assigned Branch Manager can log into the system with target branch permissions.
Main flows		
Step	***Actor***	***Action***
1	COO	Select "Create Manager Account" or select an existing Branch Manager account to edit.
2	COO	Enter/modify manager details (Full Name, Phone Number, Email, Username, Password) and select the target branch to assign.
3	COO	Click the "Save Account" button.
4	SCMS	Validate account details (e.g., unique username/email, strong password, valid branch assignment).
5	SCMS	Store account details and bind the manager profile to the specified branch scope.
6	SCMS	Display a success message confirming account setup/update.

Alternative flows

AT1	Username or email already exists (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Detect that the username or email is already registered to another account.
1.2	SCMS	Display an error message "Username/Email already exists in the system."
1.3	SCMS	Retain the form data and allow the COO to enter unique credentials.

3.6.5 Chain Sync Center

3.6.5.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Data Packages Select	Checkbox Group	Yes	-	Select data packages to synchronize (Menu & Prices, BOM Recipes, Promotions).
2	Target Branches Select	Checkbox Group	Yes	-	Select the list of branches to receive the synchronization (All or specific branches).
3	Trigger Sync Button	Button	-	-	Trigger the process of instantly pushing Master data to the POS DB of the branches.
4	Sync History Table	Table	-	-	Table tracking the results of synchronization batches (Time, Target Branch, Data Package, Status).

3.6.5.2 Use Case Description

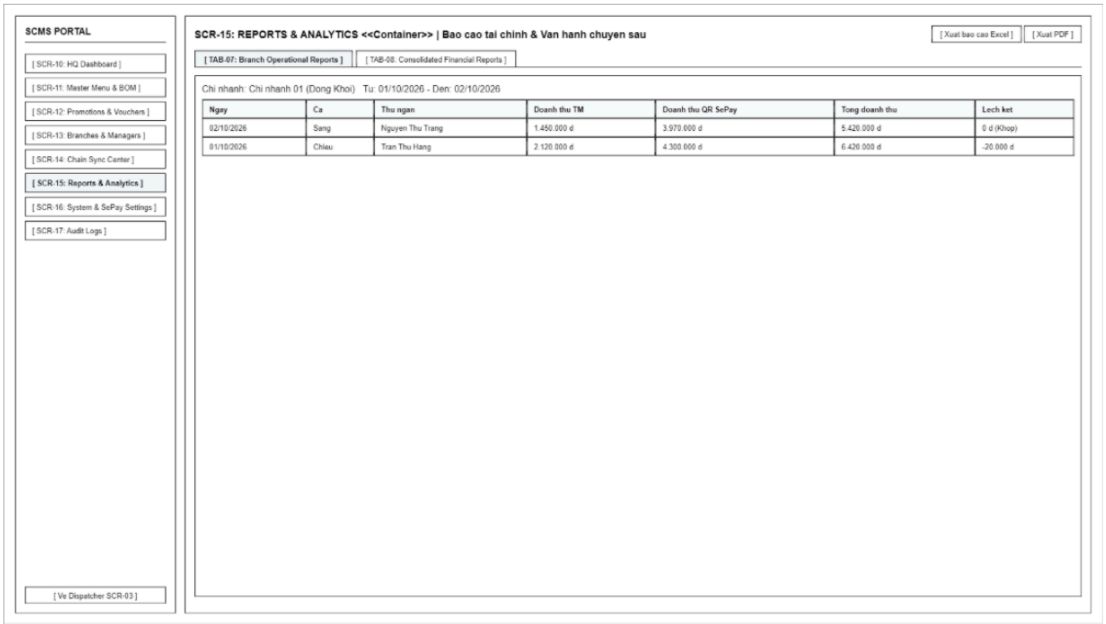
Use Case ID	**UC14**	**Use Case Name**		**Synchronize Configuration to Branches**	
Author	**VanNT**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	COO				

Description		The COO selects configuration data and target branches. The system synchronizes the selected data to the branches and records the synchronization results.
Precondition		PRE-01: The COO has successfully logged into the SCMS (UC01). PRE-02: The COO has navigated to the "Chain Sync Center" screen (SCR-14).
Trigger		TRG-01: The COO initiates the synchronization process to push updated master data down to the branch level.
Post-Condition		POS-01: Configuration data (including master menu items, pricing tiers, BOM recipes, and active promotions) is successfully updated in the selected branches' databases. POS-02: The system records the synchronization log/results.
Main flows		
Step	***Actor***	***Action***
1	COO	Review the configuration data packages available for syncing, which encompass master menu items, pricing tiers, BOM recipes, and active promotions.
2	COO	Select the specific target branches to receive the update.
3	COO	Click the synchronization button to trigger the one-click sync process.
4	SCMS	Validate the selected data packages and establish a connection to the target branches' databases.
5	SCMS	Push and synchronize the selected configuration data to the databases of the chosen branches.
6	SCMS	Record the synchronization results and display a success message confirming the operation is complete.
Alternative flows		
AT1	Synchronization fails for one or more branches (At step 4 in the main flows)	
Sub step	**Actor**	**Action**
1.1	SCMS	Attempt to connect to the target branch's database.
1.2	SCMS	Detect a connection failure (e.g., the branch network is offline).
1.3	SCMS	Log the synchronization failure for the specific unreachable branch, flag it in the synchronization results, and proceed to sync the remaining available branches.
1.4	SCMS	Display a warning message summarizing which branches succeeded and which failed.

3.7 Reports & Analytics

3.7.1 Reports & Analytics

3.7.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Date Range Filter	Text / Date	Yes	25	Select the time range to view the report (From date - To date).
2	Export Excel	Button	-	-	Export the entire current report table to an Excel file.
3	Export PDF	Button	-	-	Export the report in PDF format for printing and archiving.
4	Sub-Tab Switcher	Tabs	-	-	Switch between TAB-07 (Branch Report) and TAB-08 (Chain Consolidated Report).
5	(TAB-07) Payment Mix Table	Table	-	-	Statistics on revenue proportion between Cash and SePay QR Transfer by shift.
6	(TAB-07) Cash Variance Log	Table	-	-	Detailed log of cash overage/shortage reconciliation in the cash register for work shifts.
7	(TAB-08) Chain P&L Table	Table	-	-	Consolidated report of net revenue, COGS (Cost of Goods Sold), and gross profit for the entire system.
8	(TAB-08) Gross Margin KPI	Label	-	15	Display the average gross profit margin of the entire chain.

3.7.1.2 Use Case Description

Use Case ID	**UC35**	**Use Case Name**		**View Branch Sales Report**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager				
Description	The Branch Manager selects a time period. The system retrieves, aggregates, and displays the sales report for the selected branch.				
Precondition	PRE-01: Successfully logged in with Branch Manager role.				
Trigger	TRG-01: Need to view detailed sales figures and performance metrics for a specific timeframe.				
Post-Condition	POS-01: Sales report is displayed matching applied filters in table and chart formats.				

Main flows		
Step	***Actor***	***Action***
1	Branch Manager	Navigates to "Reports" module > "Branch Sales".
2	SCMS	Displays filter options: From date - To date, Shift, Payment method (Cash / VietQR), and "Search" button.
3	Branch Manager	Selects date range and filter criteria, then clicks "Search".
4	SCMS	Validates date range (From date <= To date).
5	SCMS	Retrieves and aggregates sales data from orders with PAID status within selected timeframe.
6	SCMS	Presents results in visual charts, revenue summary cards, and itemized order/product sales tables.

Alternative flows

AT1	At step 3 of the main flow, user selects 'From date' that is greater than 'To date':	
Sub step	**Actor**	**Action**
3.1	SCMS	Displays error message: "'From date' cannot be greater than 'To date'."
3.2	SCMS	Halts query execution and keeps previous report view.
AT2	At step 5 of the main flow, if no sales transactions exist in the selected period:	

Sub step	Actor	Action
5.1	SCMS	Displays empty state notification: "No sales data available within the selected timeframe."

Business Rules

#	Rule Description
BR-01	Branch Managers are only allowed to access and view sales report data belonging to their assigned branch (Rule BR-04).
BR-02	Sales revenue is only recognized from orders in PAID status; cash collected is booked into the respective cashier shift (Rule BR-26, BR-27).
BR-03	The system must provide detailed Sales Reports over time for the Branch Manager to track revenue and order volumes (Rule BR-46).
BR-04	All sales report views must support exporting data to Excel format (.xlsx) (Rule BR-50).

Use Case ID	**UC36**	**Use Case Name**		**Export Reports**	
Author	**AnhTV**	**Version**	**1.0**	**Date**	**04/10/2026**
Actor	Branch Manager, COO				
Description	The Branch Manager or COO exports reports as Excel files.				
Precondition	PRE-01: User is authenticated with Branch Manager or COO role. PRE-02: Selected report contains data within the specified timeframe.				
Trigger	TRG-01: User clicks "Export Excel" button on current report screen.				
Post-Condition	POS-01: An Excel file (.xlsx) containing report data is generated and downloaded to user's computer. POS-02: Export operation is recorded in the system audit log (*).				
Main flows					
Step	***Actor***	***Action***			
1	User	Selects report type to export (Sales, Inventory In-Out-Balance, Cash Drawer Flow, Employee Hours).			
2	SCMS	Displays filter options based on role: date range, and branch selector (COO only).			
3	User	Selects filters and clicks "Export Excel".			

4	SCMS	Verifies data access permissions according to user role and branch scope.
5	SCMS	Aggregates data according to filters and generates formatted Excel file (.xlsx).
6	SCMS	Returns file for browser download and records operation in Audit Log.

Alternative flows

AT1	At step 4, if Branch Manager selects a branch outside their assigned scope:	
Sub step	**Actor**	**Action**
4.1	SCMS	Rejects operation and displays message: 'Access denied: You do not have permission to access data outside your assigned branch.'
AT2	At step 5, if the current report dataset is completely empty:	
Sub step	**Actor**	**Action**
5.1	SCMS	Displays toast message: 'No data available to export.' and aborts file generation.
AT3	At step 5, if an error occurs during file generation:	
Sub step	**Actor**	**Action**
5.1	SCMS	Displays error message: 'File generation failed. Please try again.'

Business Rules

#	Rule Description
BR-01	Branch Manager can only export data belonging to their assigned branch (Rule BR-04).
BR-02	Excel file contents must strictly match the report data displayed on screen under the same filter criteria (Rule BR-50).
BR-03	File export format must be Excel (.xlsx) (Rule BR-50).
BR-04	All report export actions must be automatically recorded in the system Audit Log (*) (Rule BR-51).

UC37 – View Branch Cash Flow Report

Use Case ID	**UC37**	**Use Case Name**		**View Branch Cash Flow Report**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026

Actor		Branch Manager
Description		Branch Manager views the cash flow report of their assigned branch. The system aggregates shift cash, sales revenue, cash-in/cash-out, and calculates expected cash versus counted cash to report drawer discrepancies.
Precondition		PRE-01: Branch Manager is logged in. PRE-02: The branch has opened work shifts and sales transactions.
Trigger		TRG-01: Branch Manager selects the "Cash Flow Report" function.
Post-Condition		POS-01: Cash flow report and cash drawer discrepancy details are displayed.
Main flows		
Step	**Actor**	**Action**
1	Branch Manager	Selects the cash flow report function.
2	SCMS	Displays filters by date or shift (defaults to current date).
3	Branch Manager	Selects filter parameters and requests to view report.
4	SCMS	Aggregates data for each shift at the branch: opening cash, cash-in/cash-out with reasons, cash sales revenue, cash refunds.
5	SCMS	Calculates expected cash, reconciles with counted cash upon shift closing, and computes discrepancy.
6	SCMS	Displays summary and shift breakdown report with discrepancy explanations, highlighting shifts with variances.

Alternative flows

AT1	At step 4, if no shifts exist within the selected time range:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays a message indicating no cash flow data available.
AT2	At step 5, if shift is not yet closed and physical count has not occurred:	
Sub step	**Actor**	**Action**
5.1	SCMS	Displays "Unreconciled" status for that shift and does not calculate discrepancy.

Business Rules

#	Rule Description
BR-01	Branch Manager can only view data of their assigned branch.(BR-04)
BR-02	Expected Cash = Opening Cash + Cash Sales Revenue + Cash In − Cash Out − Cash Refunds (BR-39).
BR-03	Discrepancy = Counted Cash − Expected Cash. Shifts with discrepancies must provide an explanation (BR-38).

BR-04	Revenue in cash flow report only includes orders paid in cash (BR-27).

UC38 – View Branch Inventory Report

Use Case ID	**UC38**	**Use Case Name**		**View Branch Inventory Report**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	Branch Manager				
Description	Branch Manager views the inventory report for their assigned branch. The system aggregates stock receipts, issues, disposals, physical counts, and displays inventory status along with discrepancies.				
Precondition	PRE-01: Branch Manager is logged in. PRE-02: The branch has an active ingredient catalog.				
Trigger	TRG-01: Branch Manager selects the "Inventory Report" function.				
Post-Condition	POS-01: Inventory report, stock status, and stock discrepancies are displayed.				

Main flows		
Step	**Actor**	**Action**
1	Branch Manager	Selects the inventory report function.
2	SCMS	Displays time range filter.
3	Branch Manager	Selects filter parameters and requests to view report.
4	SCMS	Aggregates stock transactions in the period: stock receipts, consumption, disposals, physical inventory counts.
5	SCMS	Calculates opening inventory, closing inventory, and compares system stock against physical count to determine variance.
6	SCMS	Determines stock status for each ingredient and displays report, highlighting items with discrepancies.

Alternative flows

AT1	At step 4, if no stock transactions occurred during the period:	
Sub step	**Actor**	**Action**
4.1	SCMS	Displays current stock on hand for each ingredient with a notification that no transactions occurred during the period.
AT2	At step 5, if an ingredient has never been physically counted:	
Sub step	**Actor**	**Action**
5.1	SCMS	Displays discrepancy status as "Uncounted" for that ingredient.

Business Rules

#	Rule Description
BR-01	Branch Manager can only view inventory data of their assigned branch (BR-04).
BR-02	Discrepancy = Physical Count Quantity − System Inventory at the time of count (BR-44).
BR-03	Ingredients with stock less than or equal to minimum stock level are flagged as "Low Stock" (BR-45).
BR-04	Quantities are displayed in bulk packaging units along with conversion to dispensing units used in recipes (BR-40, BR-41).

UC39 – View Branch HR & Shifts Report

Use Case ID	**UC39**	**Use Case Name**		**View Branch HR & Shifts Report**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	Branch Manager				
Description	Branch Manager views the human resources and shift report of their branch. The system aggregates and displays attendance data, work shifts, and employee productivity indicators.				
Precondition	PRE-01: Branch Manager is logged in. PRE-02: The branch has registered employees and time attendance records.				
Trigger	TRG-01: Branch Manager selects the "HR & Shifts Report" function.				
Post-Condition	POS-01: Branch HR and attendance report is displayed.				
Main flows					
Step	**Actor**	**Action**			
1	Branch Manager	Selects the HR & shifts report function.			
2	SCMS	Displays time range filter.			
3	Branch Manager	Selects filter parameters and requests to view report.			
4	SCMS	Aggregates attendance data for each employee: check-in time, check-out time, hours worked, late arrivals, early departures, missing check-outs.			
5	SCMS	Reconciles scheduled shifts against actual worked shifts.			
6	SCMS	Calculates productivity metrics for each employee (*).			
7	SCMS	Displays report broken down by employee and aggregated for the entire branch.			

Alternative flows

AT1	At step 4, if no attendance data exists within the selected time range:

Sub step	Actor	Action
4.1	SCMS	Displays a message indicating no HR data available.
AT2	At step 5, if an employee was scheduled for a shift but has no attendance record:	
Sub step	**Actor**	**Action**
5.1	SCMS	Marks that shift as absent in the report.

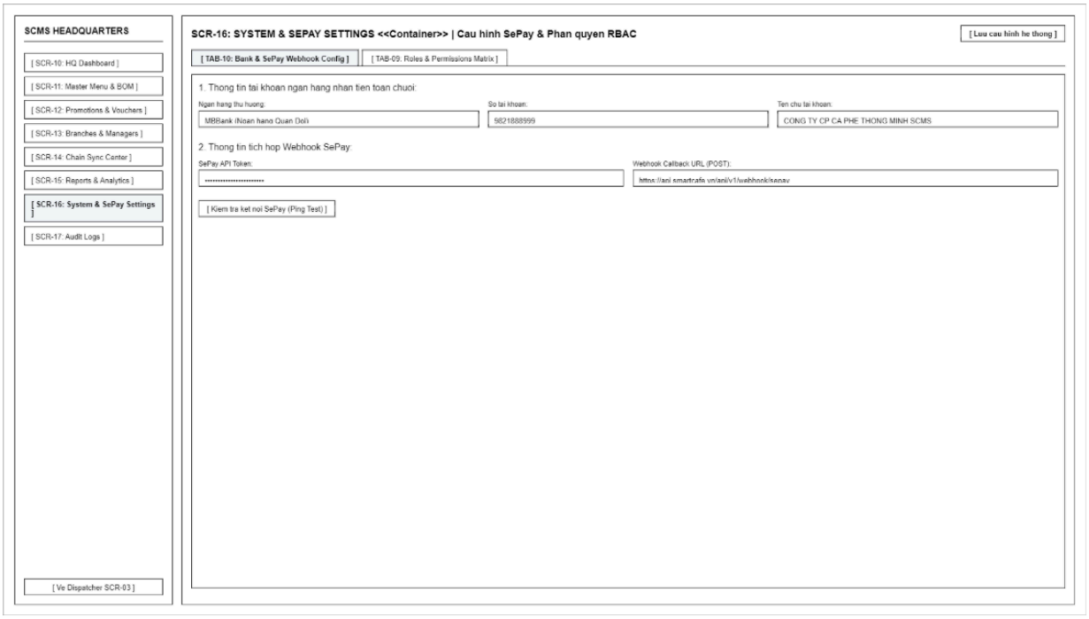
Business Rules

#	Rule Description
BR-01	Branch Manager can only view employees of their assigned branch (BR-04).
BR-02	Working hours are calculated from check-in time to check-out time(BR-06, BR-07).
BR-03	Productivity metric for Cashiers is calculated based on orders processed and average revenue generated per working hour (BR-09, BR-10).
BR-04	This report does not include payroll or salary information (BR-47).

3.8 System Configuration & Security

3.8.1 System & SePay Settings

3.8.1.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Save Settings	Button	-	-	Save all configuration changes to the system.

2	Sub-Tab Switcher	Tabs	-	-	Switch between TAB-10 (SePay Configuration) and TAB-09 (RBAC Authorization Matrix).
3	(TAB-10) Bank Name	Text	Yes	50	Beneficiary bank name receiving money for the entire chain (e.g., MBBank).
4	(TAB-10) Account Number	Text	Yes	20	Beneficiary bank account number.
5	(TAB-10) Account Holder	Text	Yes	50	Beneficiary bank account holder name.
6	(TAB-10) SePay API Token	Password	Yes	60	API Token security key from the SePay account.
7	(TAB-10) Webhook URL	Text	Yes	120	API Endpoint URL to receive IPN data from the SePay gateway.
8	(TAB-10) Ping Test	Button	-	-	Send a mock packet to test the connection with the SePay Webhook.
9	(TAB-09) RBAC Matrix Table	Table	-	-	View/Create/Edit/Delete/Approve authorization matrix table for 4 roles (Staff, Cashier, Manager, Owner).

3.8.1.2 Use Case Description

Use Case ID	**UC44**	**Use Case Name**		**Send payment webhook via SePay**	
Author	VuNL	**Version**	1.0	**Date**	04/10/2026
Actor	SePay				
Description	The SePay system records a successful customer bank transfer transaction, generates a webhook, and sends the transaction code, amount, and status to the software. SCMS authenticates and automatically confirms the order as paid.				
Precondition	PRE-01: An order with a pending QR payment exists and the QR code has been generated (UC43). PRE-02: SePay has been configured with the webhook receiving endpoint and SCMS authentication key.				
Trigger	TRG-01: SePay records a balance fluctuation due to a successful customer bank transfer.				
Post-Condition	POS-01: The payment status changes to completed and the order status changes to paid. POS-02: The webhook is logged, even if rejected.				
Main flows					
Step	**Actor**	**Action**			

1	SePay	Records the transaction and sends a webhook (HTTP POST) containing the transaction ID, amount, transfer description, and status to SCMS.
2	SCMS	Authenticates the webhook signature.
3	SCMS	Checks that the transaction ID has not been processed previously.
4	SCMS	Extracts the order code from the transfer description and finds the corresponding pending payment.
5	SCMS	Reconciles the received amount with the required payment amount.
6	SCMS	Changes the payment status to completed, updates the order to paid (maximum 1 second after receiving the webhook), and logs the webhook.
7	SCMS	Responds to SePay with a success acknowledgment and updates the Cashier screen.

Alternative flows

AT1	At step 2, if the signature is invalid	
Sub step	**Actor**	**Action**
2.1	SCMS	Rejects the webhook, logs the event, and does not change the order status.
AT2	At step 3, if the transaction ID has already been processed	
Sub step	**Actor**	**Action**
3.1	SCMS	Skips processing, logs it as a duplicate transaction, and responds to SePay with an acknowledgment.
AT3	At step 4, if no corresponding order is found or the order is already paid/cancelled:	
Sub step	**Actor**	**Action**
4.1	SCMS	Logs the event for manual handling and alerts the Branch Manager (*).
AT4	At step 5, if the amount does not match	
Sub step	**Actor**	**Action**
5.1	SCMS	Keeps the order in pending status and does not change it to paid.
5.2	SCMS	Logs the amount discrepancy and sends an alert to the Cashier and Branch Manager.

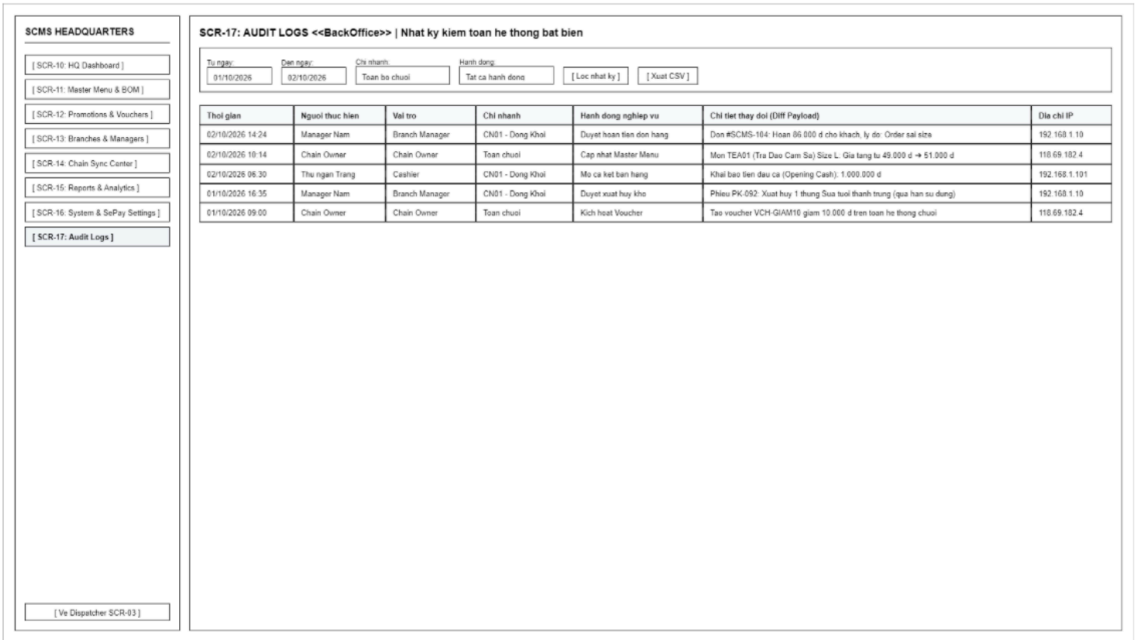
Business Rules

#	Rule Description
BR-01	Only automatically confirm payment when the order code, amount, and authentication signature from SePay all match (BR-29).
BR-02	Each SePay transaction ID can only be processed once (BR-26).
BR-03	A paid order is frozen and cannot be edited (BR-30).

BR-04	All received webhooks must be logged, including rejected or duplicate webhooks (BR-31).

3.8.2 Audit Logs

3.8.2.1 Screen Mock-up



Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Date Filter	Text / Date	No	25	Filter audit logs by the time range the action occurred.
2	Branch Filter	Dropdown	No	-	Filter logs by the executing branch or the entire chain.
3	Action Type Filter	Dropdown	No	-	Filter by sensitive action type (Price adjustment, Refund approval, Order cancellation, Inventory disposal).
4	Filter Action	Button	-	-	Apply filters to search for log records.
5	Export CSV	Button	-	-	Export all audit log data to a CSV file.
6	Audit Logs Table	Table	-	-	Immutable log table (Time, Actor, Role, Branch, Action, Diff Payload, IP).

4. Non-Functional Requirements

4.1 External Interfaces

This section defines the communication requirements between the Smart Cafe Management System and external hardware or service platforms.

- SePay Integration: The system must connect to the SePay Payment Gateway to receive automated payment notifications via a Payment Webhook.
- Payment Reconciliation: The system must receive and store Payment Reference Information from SePay to enable automated transaction matching.
- VietQR Integration: The system must connect to the VietQR platform to send a QR Generation Request for individual billing.
- QR Code Display: The system must successfully receive the Generated QR Code from VietQR and display it on the customer-facing screen for payment.
- Hardware Printer Communication: The system must connect to and transmit Preparation Slips directly to the Bar Thermal Printer immediately after an order is confirmed by the cashier.

4.2 Quality Attributes

4.2.1 Usability

- Cashier Interface Efficiency: The interface designed for the Cashier must be optimized for rapid operations, ensuring that the processing of Order Information, Voucher Information, and Payment Information can be completed in under 5 clicks or taps.
- Training Time: The system must be highly intuitive, ensuring that the training time for a new Staff member or Cashier to understand Login Credentials and basic operations does not exceed 2 hours.
- Management Dashboard: The interfaces for the COO and Branch Manager must present the Sales Report and Operational & Financial Reports in highly visual, easily interpretable charts.

4.2.2 Reliability

- Availability: The system must maintain a 99.5% uptime guarantee during all active shifts (Shift Information) across all cafe locations.
- NFR-REL-02 Offline operation is out of scope for v1.0. When the internet connection or the SCMS server is unavailable, the POS cannot create or pay orders. Branches follow a manual fallback procedure (using paper order slips and cash only); the Branch Manager enters the manual sales records after service is restored.

4.2.3 Performance

- Business Response Time: Core functions that calculate Price Information and retrieve Product Information on the Cashier's screen must respond in under 1 second to prevent customer service delays.
- Integration Latency: The round-trip time for a QR Generation Request sent to VietQR and its subsequent display on the screen must be completed within a maximum of 3 seconds. The internal processing within SCMS before sending the request must be under 300ms.
- Concurrent Processing Capability: The system must handle the simultaneous load of multiple Branch Managers querying the Inventory Report or Attendance Report at the end of the day without degrading the performance of active cashier counters.

4.2.4 Security & Authorization

- Transmission Security: All sensitive data flows—especially Login Credentials, Payment Information, and Operational & Financial Reports—must be strictly encrypted over the network.
- Role-Based Access Control: The system must strictly enforce information access rights through configured Permission Information.
- Chain-Level Data Restrictions: Only the designated COO is authorized to view or modify the Branch Configuration and access aggregate data across all branches.
- Internal Branch Authority: The Branch Manager is restricted to managing Cash In / Out Information and viewing Employee Information strictly within their assigned branch.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Users must be authenticated with valid credentials before accessing the system; sessions must be strictly limited based on the assigned role and data scope.
BR-02	The COO is the only role authorized to set up the role matrix and configure functional permissions across the entire system.
BR-03	Branch Managers are strictly prohibited from self-approving their own attendance data, working hours adjustments, bonuses/deductions, or payrolls (All such approvals must be performed by the COO).
BR-04	Operational data must be strictly isolated by branch: Branch Managers and Cashiers are only allowed to access data belonging to their assigned branch. Cross-branch data access is exclusively reserved for the COO role.
BR-05	Each RFID/NFC card can only be linked to one active employee at a time; the system strictly prohibits assigning the same card to multiple employees.
BR-06	A valid card swipe at the reader must automatically record a Check-in if it is the first attendance of the shift/day, or automatically record a Check-out if a Check-in already exists for that shift.
BR-07	The time interval between a valid Check-in and Check-out for the same employee must be at least 15 minutes to prevent accidental duplicate card swipes.
BR-08	In case of a forgotten/lost card or a broken reader, employees are allowed a fallback attendance method by selecting their name and entering a 4-digit personal PIN on the Kiosk screen.
BR-09	If an employee has Checked-in but fails to Check-out after 14 hours, the system must automatically flag a "Forgot Check-out" warning and suspend payroll calculation for that shift until verified by the Branch Manager.
BR-10	Only Branch Managers have the authority to adjust the Check-in/Check-out times of their subordinate employees, and a reason must be recorded for every adjustment.

BR-11	The COO is the only role authorized to create, edit information, or deactivate branches within the chain.
BR-12	A Branch Manager account belongs to one branch. Only the COO can reassign it; the change is written to the Audit Log and historical data stays with the original branch.
BR-13	Hard deleting employee profiles is strictly prohibited. Accounts of terminated employees must only be switched to an Inactive state to preserve attendance, order, and payroll history.
BR-14	When a Branch Manager updates the hourly wage rate, the new rate will only take effect for work shifts occurring after the modification time.
BR-15	The system must prevent the assignment of overlapping work schedules or duplicate shifts for the same employee on the same day.
BR-16	The menu categories and product information across the entire system must be centrally managed by the COO; branches are not allowed to arbitrarily add or remove menu items.
BR-17	Any product with an Unavailable status must be automatically disabled on the POS interface and cannot be added to new orders.
BR-18	Selling prices based on specifications (size, toppings) must be centrally managed by the COO; branches are not allowed to arbitrarily change selling prices.
BR-19	The standard bill of materials (BOM/Recipe) for each item must be centrally issued by the COO to serve as a basis for calculating Cost of Goods Sold (COGS) and reconciling loss.
BR-20	Promotion policies and vouchers (duration, minimum order value, applicable categories) must be centrally managed by the COO.
BR-21	A maximum of one discount program/voucher can be applied per order. The final total payment amount of the order must not be less than 0 VND.
BR-22	Cashiers are only allowed to apply discounts/vouchers to an order when the order fully satisfies all constraint conditions of the program.
BR-23	Changes to the master menu, selling prices, standard recipes and promotions take effect at branches only after the COO publishes them (UC14). Each publish creates a numbered version and a publish record per target branch. Orders created before the effective time keep the prices stored on the order.
BR-24	All bank transfer payments must be routed directly to the official chain bank account configured by the COO (not to personal accounts).
BR-25	Cashiers creating orders at the counter must record all attributes: product, quantity, size, toppings, sugar/ice levels, and preparation notes.
BR-26	Once an order has transitioned to the PAID status, the system strictly prohibits any operations to edit items, change size/toppings, apply promotions, or cancel the order (Order data freeze).
BR-27	All cash collected from customers at the counter must be instantly recorded in the sales revenue of the current Cashier's shift.

BR-28	The generated dynamic VietQR code must contain the exact chain bank account number, the order amount, and the payment description formatted as CAFE.
BR-29	The system will only automatically transition an order to the PAID status upon receiving a SePay Webhook that exactly matches all of the following: Order ID, Transaction Amount, and Authentication Signature.
BR-30	If the amount transferred by the customer via SePay does not match the total order amount, the system must retain the Pending Payment status and issue a reconciliation alert.
BR-31	Bar preparation slips and customer payment receipts are only allowed to print automatically after the order has successfully transitioned to the PAID status.
BR-32	Sensitive counter operations (refund, cash-in, cash-out, shift-close variance above tolerance) follow a two-person rule: the Cashier initiates the request with a mandatory reason, and a Branch Manager approves it by entering their own PIN on the same terminal (POP-05). If the amount exceeds the threshold configured for the Branch Manager (BR-35), the request also requires COO approval in the Approvals Inbox (SCR-18).
BR-33	All refunds are always paid in cash from the drawer (even if the customer originally paid via QR code/bank transfer).
BR-34	The refund amount to the customer must be automatically deducted from the cash drawer balance during the current work shift.
BR-35	Refund amounts exceeding the Branch Manager's approval threshold limit must be approved by the COO.
BR-36	A sales shift is only allowed to start after a Cashier or Branch Manager has performed the Open Shift operation and declared the starting cash amount in the drawer.
BR-37	Cash-in and cash-out transactions during a shift are initiated by the Cashier, require a business reason, and are approved by the Branch Manager (BR-32). A transaction initiated by a Branch Manager is recorded immediately, flagged "self-initiated", and listed for COO review.
BR-38	During Close Shift, the responsible personnel must enter the physically counted cash amount. If there is a discrepancy with the theoretical balance, an explanatory reason must be provided.
BR-39	The theoretical end-of-shift cash balance must be calculated using the formula: Starting Cash + Cash Revenue from PAID orders + Cash In - Cash Out - Customer Refunds.
BR-40	Goods received at the branch must be managed by large packaging units (cartons, bags, boxes, kg) and fully record the quantity and import cost to increase inventory.
BR-41	Raw materials are only deducted from branch inventory when issued to the Bar counter in intact large packaging units. The system does not deduct branch inventory in fractional amounts per cup sold.
BR-42	The quantity of raw materials issued to the bar or declared for disposal must not exceed the currently available inventory quantity in the branch warehouse.
BR-43	All disposal transactions for damaged, expired, or broken raw materials must include a specific disposal reason.

BR-44	After completing a periodic inventory count, the system must automatically update the book inventory balance according to the physical count and log the inventory discrepancy record.
BR-45	When the inventory quantity of a raw material drops to or below the Safety Stock level, the system must automatically issue an out-of-stock alert to the Branch Manager.
BR-46	The system must provide a Dashboard and detailed Sales Reports over time for the Branch Manager to track revenue and order volumes.
BR-47	The system must provide in-depth branch-level reports: Cash Drawer Flow Report (over/short reconciliation), Inventory In-Out-Balance Report, and Employee Working Hours Report.
BR-48	Revenue data, Cost of Goods Sold (COGS), and gross profit must be automatically consolidated from all branches to provide real-time chain-wide Financial Reports for the COO.
BR-49	The system must analyze sales data to provide the COO with a list of Best-sellers and order density during Peak hours.
BR-50	All financial and operational report forms must allow export to Excel format for independent storage and reconciliation.
BR-51	All sensitive business operations (price edits, order cancellations, refunds, cash in/out, working hours adjustments, inventory adjustments) must be automatically recorded in the Audit Log and cannot be modified or deleted.
BR-52	Ingredient shrinkage for a period = Quantity issued to bar (base units) - Theoretical consumption (SUM of BOM quantity x quantity sold of PAID orders, excluding fully refunded orders). A positive result is reported as shrinkage.

5.2 Common Requirements

5.2.1 General User Interface (UI/UX) Requirements

1. UI Consistency:
    - All screens must adhere to the system's color palette and standard Design System:
        - Navigation Bar (Sidebar/Header): Dark Primary theme containing user profile, current branch information, and logout button.
        - Primary Action Buttons: Coffee Orange/Brown (Accent Color) used for confirmation, payment, and data saving actions.
        - Status Indicators: Red (Danger/Cancel/Out of Stock/Failed), Green (Success/Paid/In Stock), Yellow/Orange (Warning/Pending).
    - Standard Font: Clear, sans-serif font with full support for Vietnamese diacritics.
2. Device & Role-Specific Layouts:
    - POS & Kiosk Interfaces (SCR-01, SCR-02, SCR-04): Optimized for 10-inch to 15.6-inch touchscreens. Item selection buttons, size/sugar/ice options, and the numeric keypad (Numpad) must have large touch targets (minimum 44x44 px) for fast manual operation.

    - Admin Interface (Dashboard & Portal): Optimized for desktop web browsers (Desktop Screen Resolution 1920x1080 or higher) with detailed tables, charts, and permission matrices.
3. Quick Operations & Shortcuts at POS:
    - Support physical keyboard shortcuts for Cashiers at the sales counter (SCR-04):
        - F1: Quick search for items/beverages.
        - F8: Apply Discount Code / Voucher.
        - F9: Open payment window (POP-01).
        - F10: Confirm cash payment.
        - Esc: Close/Cancel currently open Pop-ups/Modals.

5.2.2 Data Formatting & Localization Standards

1. Currency Formatting:
    - Default currency is Vietnamese Dong (VND).
    - Monetary amounts must use a dot (.) as thousands separators and include the currency symbol *(e.g., 45.000 VNĐ)*.
    - No decimal places are used for VND amounts.
2. Date & Time Formatting:
    - Date: DD/MM/YYYY *(e.g., 04/10/2026)*.
    - Time: HH:mm:ss (24-hour format) *(e.g., 14:30:00)*.
    - Combined Date & Time: DD/MM/YYYY HH:mm:ss *(e.g., 04/10/2026 14:30:00)*.
3. Quantity & Unit Formatting:
    - Retail item quantity/orders: Positive integers *(e.g., 1, 2, 5)*.
    - Raw materials/Inventory (BOM): Up to 2–3 decimal places allowed using a comma (,) or dot (.) consistently *(e.g., 0.02 kg Arabica, 0.03 Liters milk)*.

5.2.3 Common Validation & Error Handling

1. Mandatory Fields:
    - All data fields marked as Mandatory: Yes in the Screen Definition table must display a red asterisk (*).
    - If left blank upon submission, the system must focus on the first missing field and immediately display an error message below it: [Field Name] cannot be left blank.

2. Input Validation:
    - Phone Number: Only numeric digits allowed, 10–11 characters long, starting with 0.
    - Timekeeping PIN: exactly 4 digits, unique within the branch, stored hashed. After 5 consecutive wrong PIN entries for the same employee the kiosk blocks that employee for 5 minutes.
    - Amount / Price / Inventory: Only positive integers/floats allowed (≥ 0). Negative numbers are not permitted except for valid inventory write-off/adjustment cases.
3. Confirmation Modals for Critical Actions:
    - Actions involving cancellations or sensitive data modifications (Order Cancellation, Refunds, Employee Deletion, Inventory Write-offs, Price/BOM Updates) must trigger a Confirmation Pop-up with a "Cancel" button and a red "Confirm" button.

5.2.4 Table, Pagination & Search Requirements

1. Grid/Table View:

    - Support fixed table headers (Fixed Table Header) when scrolling through long lists.
    - Display a clean "Empty State" with an appropriate placeholder image when a list is empty or search filters yield no results.
2. Pagination:
    - Apply pagination across all list screens (Order History, Timekeeping Logs, Inventory, Audit Logs).
    - Default items per page: 10, 25, 50, or 100 rows.
    - Display total record count and current page position *(e.g., Showing 1-10 of 245 records)*.
3. Filter & Search Behavior:
    - Live Search: Automatically filter results in real-time (with a 300ms debounce delay) as the user types into the search box.
    - Filter Persistence: Retain selected filter parameters (Time range, Status, Branch) when users navigate away and return to the previous screen.

5.2.5 Audit Logging & Security Baseline

1. Automatic Audit Trail:
    - Every Create, Edit, Delete, Cancel, Refund, Approval, Configuration Sync, and Login/Logout action must be automatically logged into the system's Audit Log table (SCR-17).
    - Required captured details: Timestamp, User ID, Role, IP Address, Branch, Action, and Data Before/After Change (Diff payload).
2. Session Control & RBAC Baseline:
    - Automatic session timeout after 30 minutes of inactivity on Manager/COO screens.
    - If a user attempts to access a URL or function outside their role permissions, the system blocks access and redirects to the Role Dispatcher screen (SCR-03) with the message: You do not have permission to access this function.

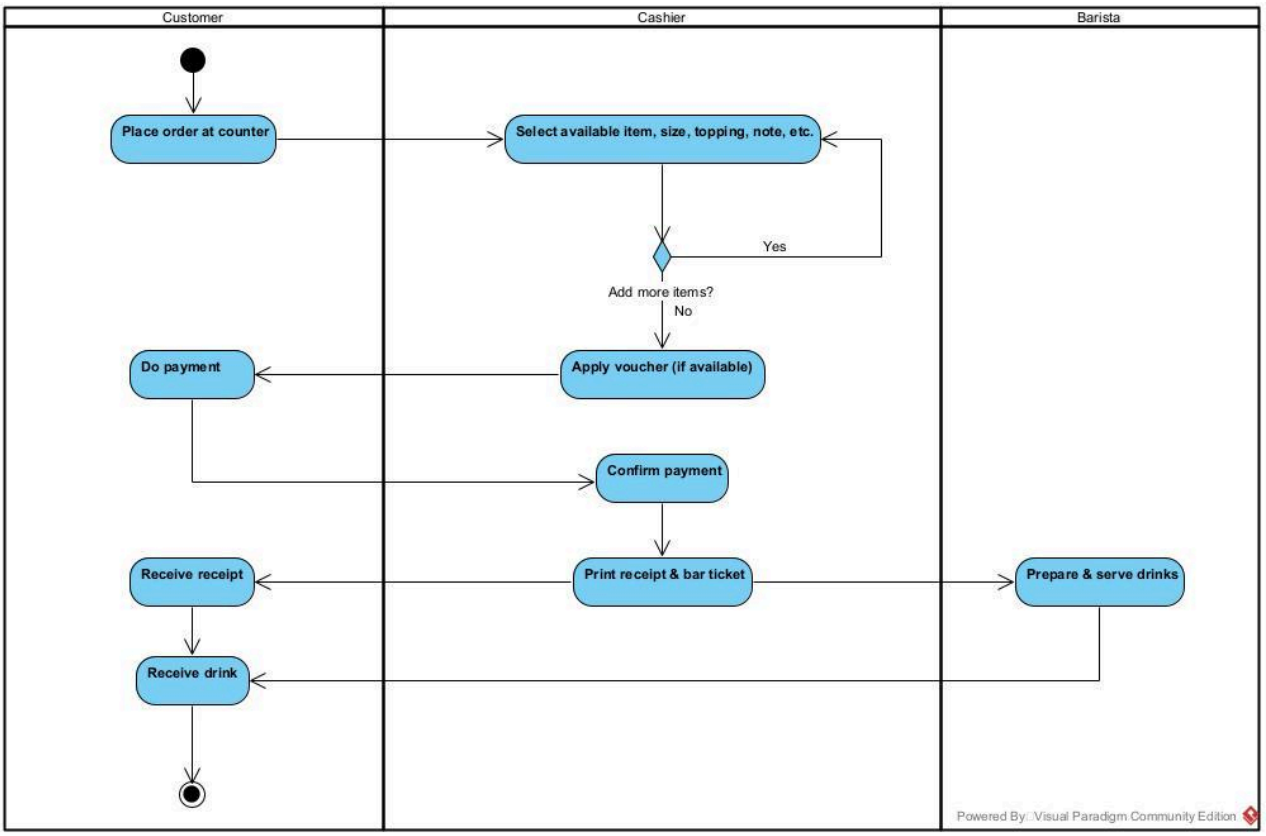
5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In line	There is not any search result	No search results.
2	MSG02	In red, under the text box	Input-required fields are empty	The * field is required.
3	MSG03	Toast message	Updating asset(s) information successfully	Update asset(s) successfully.
4	MSG04	Toast message	Adding new asset successfully	Add asset successfully.
5	MSG05	Toast message	Confirming email of asset hand-over is sent successfully	A confirmation email has been sent to {email_address}.

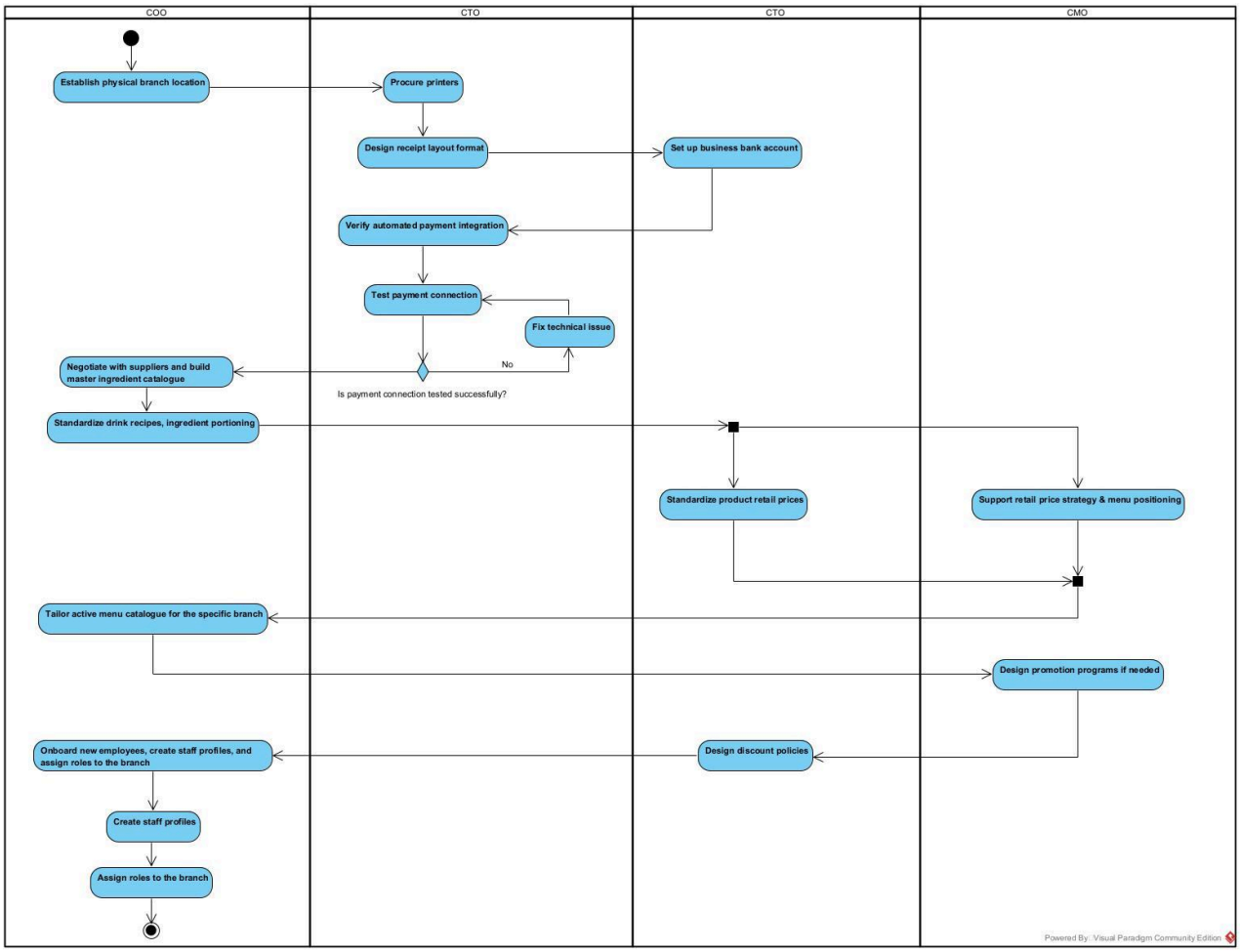
6	MSG06	Toast message	Resetting asset information successfully	Return asset(s) successfully.
7	MSG07	Toast message	Deleting asset information successfully	Delete asset(s) successfully.
8	MSG08	In red, under the text box	Input value length > max length	Exceed max length of {max_length}.
9	MSG09	In line	Username or password is not correct when clicking sign-in	Incorrrect user name or password. Please check again.

6. Business Process

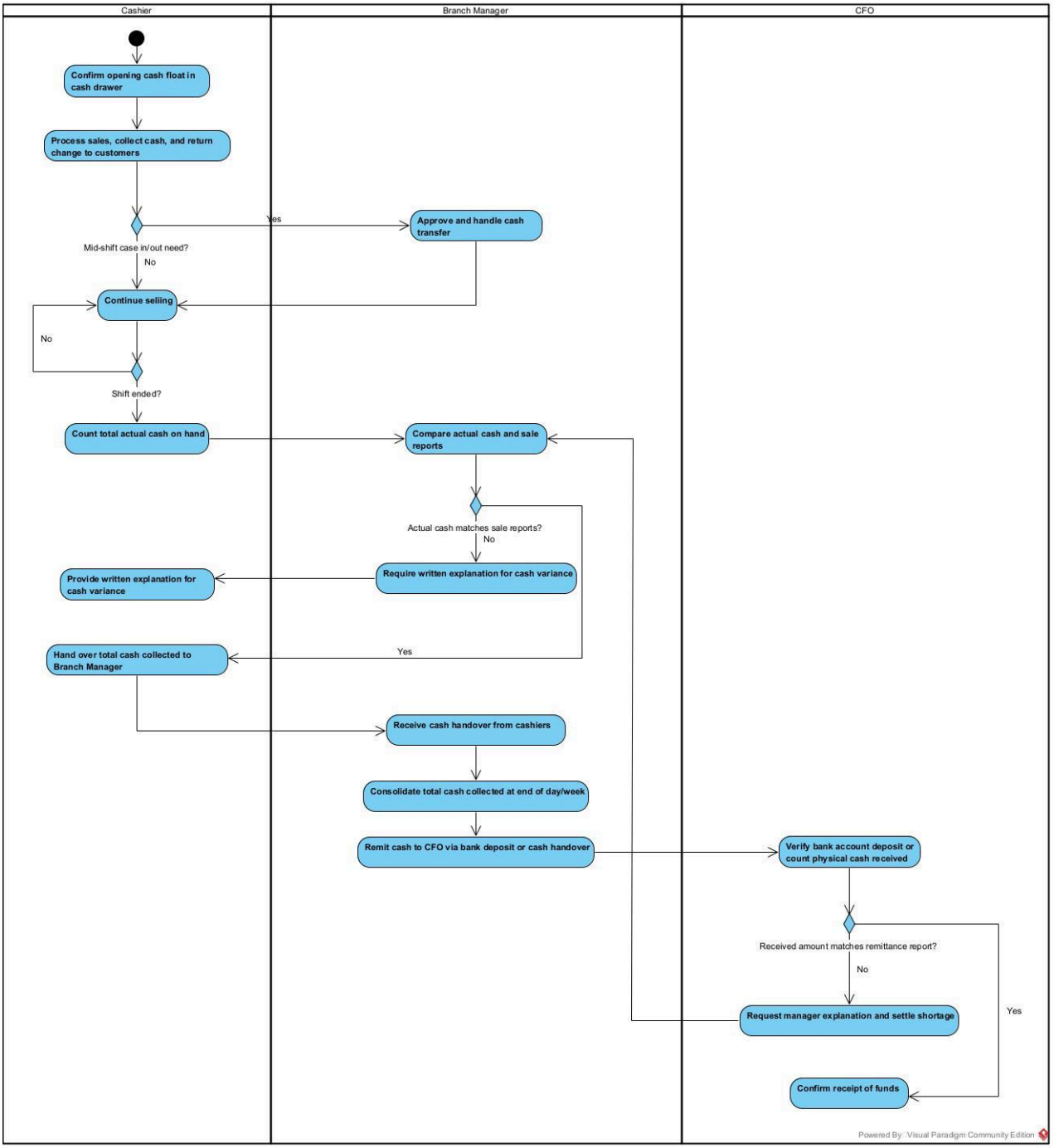
6.1. Order process



6.2. Chain set up process


COO
CTO
CTO
CMO
Establish physical branch location
Procure printers
Design receipt layout format
Set up business bank account
Verify automated payment integration
Test payment connection
Fix technical issue
No
Is payment connection tested successfully?
Negotiate with suppliers and build master ingredient catalogue
Standardize drink recipes, ingredient portioning
Standardize product retail prices
Support retail price strategy & menu positioning
Tailor active menu catalogue for the specific branch
Design promotion programs if needed
Onboard new employees, create staff profiles, and assign roles to the branch
Design discount policies
Create staff profiles
Assign roles to the branch
Powered By: Visual Paradigm Community Edition

6.3. Cash shift handover



6.4. Attendance to payroll

